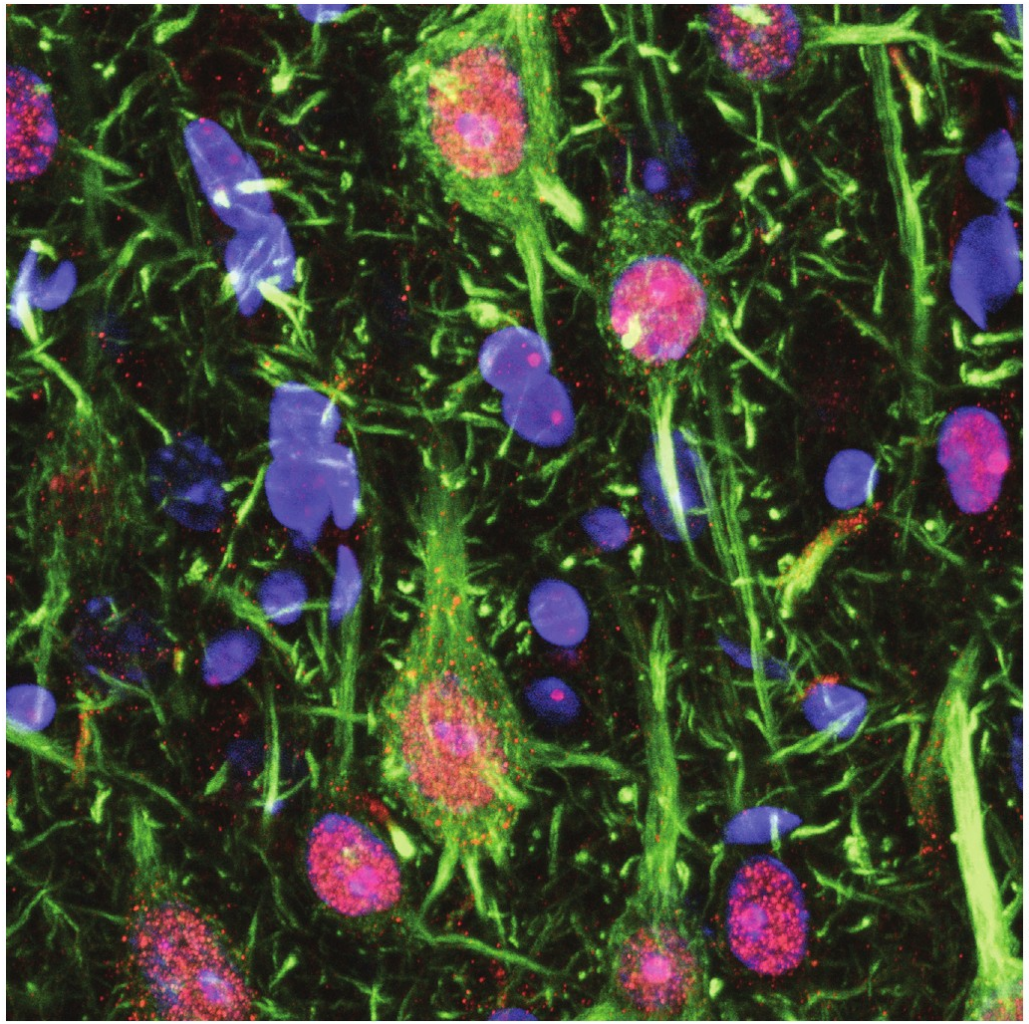


Carlson and Birkett (2022), chapter opener, p. 91. Photo: C. J. Guerin / Science Source.

BEHAVIORAL NEUROSCIENCE 392

Drugs, Receptors, and Neural Signaling

Chemical Control of Behavior: Drugs and Neural Signaling

Fall 2026

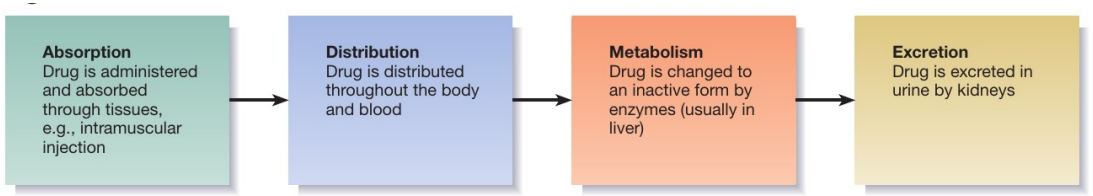
Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering

Chemicals change behavior by acting on living cells

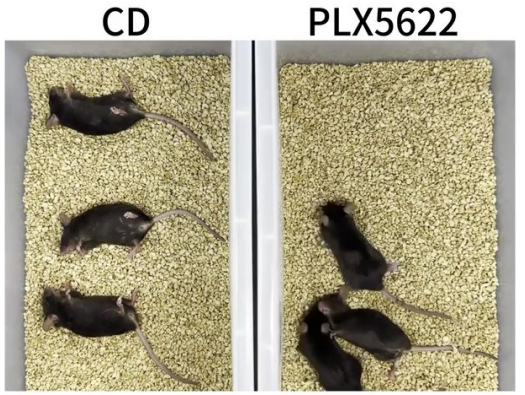
Psychopharmacology studies how drugs change nervous-system activity and behavior. A **drug** is a chemical introduced into the body that changes cellular activity at relatively low amounts. Many drugs act on proteins in the cell membrane. Changing one of these proteins can alter electrical signaling, chemical release, or the cell's response to a neurotransmitter.

A drug's **site of action** is the place where it produces a molecular change. Its **effect** is the resulting physiological or behavioral response. An anesthetic may act on a receptor in the brain while its effect is measured as loss of movement or responsiveness. The receptor and the behavioral response occupy different steps in the mechanism.

General anesthesia depends partly on **microglia**, immune cells that also regulate signaling in brain tissue. In mice, reducing microglial populations delayed the loss of the righting reflex after anesthetic administration. The animals also regained that reflex sooner. The same anesthetic therefore produced a shorter period of behavioral unresponsiveness when this brain-cell population was depleted.



Carlson and Birkett (2022), Fig. 4.2, p. 94.



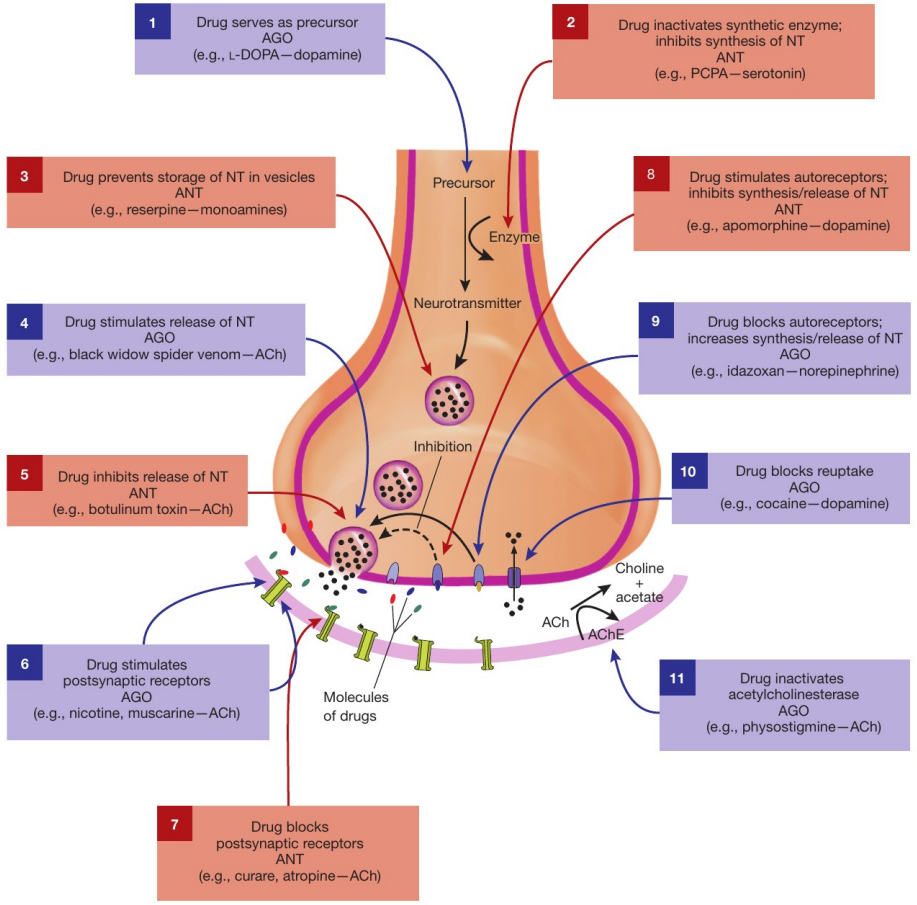
He et al. (2023), video1. Loss of the righting reflex after anesthesia.

A drug and a neurotransmitter can act on the same receptor

A **neurotransmitter** is a chemical released by a neuron that changes another cell's activity. A **receptor** is a protein that binds a chemical signal and changes its own activity in response. Some receptors contain ion channels. Others activate proteins inside the cell, which then regulate enzymes, channels, or release machinery.

Endogenous substances are produced within the body; **exogenous** substances enter from outside it. Dopamine released from an axon terminal is endogenous. A drug that activates a dopamine receptor is exogenous. Both can occupy the receptor's binding site, but they may differ in how readily they reach it and how long they remain available.

A drug can also occupy a receptor without activating it. While that drug occupies the binding site, the normal neurotransmitter has fewer available receptors to activate. A receiving neuron can consequently respond less strongly even if the releasing neuron continues to discharge action potentials and deliver the same neurotransmitter into the synaptic gap.



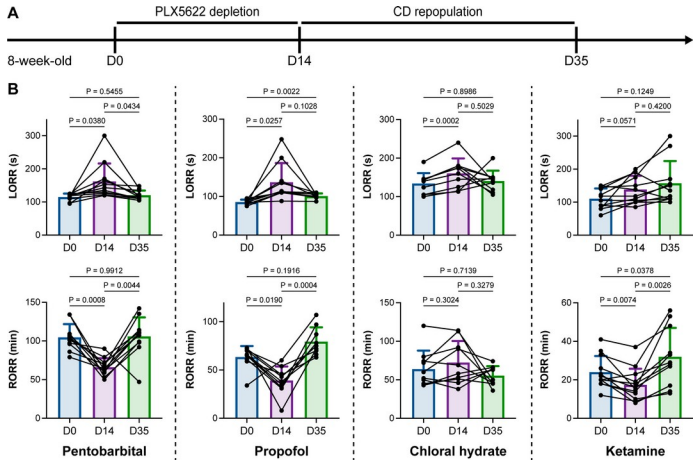
Carlson and Birkett (2022), Fig. 4.7, p. 100.

Microglial depletion changes the response to anesthesia

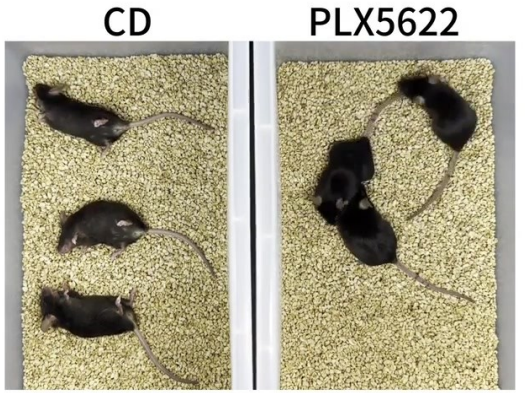
Microglia require signaling through **CSF1R**, a receptor that helps these cells survive. He and colleagues fed mice **PLX5622**, a CSF1R inhibitor, to reduce microglial populations. They then administered general anesthetics and timed loss and recovery of the **righting reflex**, the movement that returns an animal from its back to an upright position.

After microglial depletion, mice took longer to lose the righting reflex and recovered it earlier. This pattern occurred with several anesthetics, including pentobarbital, propofol, and ketamine. These compounds do not all act at the same receptor. Their behavioral effects changed after the manipulation of microglia despite their different immediate molecular targets.

The researchers subsequently stopped PLX5622 treatment and allowed microglia to repopulate the brain. Anesthesia induction and emergence returned toward the original responses. Depleting peripheral macrophages with a compound that did not enter the brain failed to reproduce the overall pattern. The altered anesthetic response followed depletion and restoration of brain microglia.



He et al. (2023), Fig. 1 (A–B). Anesthesia induction and emergence.



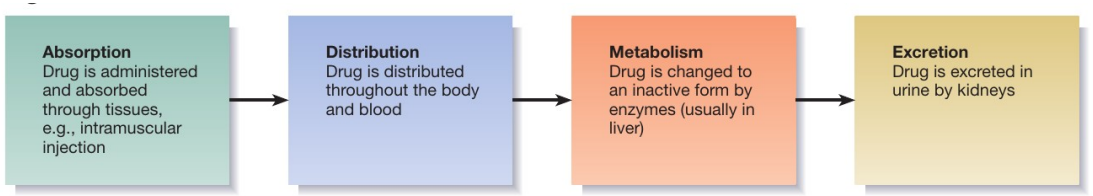
He et al. (2023), video2. Recovery of the righting reflex.

Absorption determines how a drug enters circulation

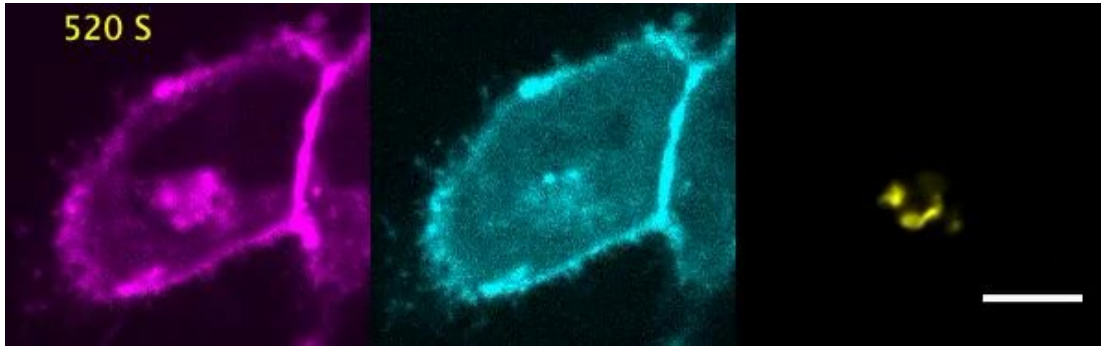
Pharmacokinetics describes the movement and removal of a drug through **absorption**, **distribution**, **metabolism**, and **excretion**. Absorption is passage from an administration site into the bloodstream. Before an orally administered drug can affect the brain, it must leave the digestive tract, enter circulation, and survive the chemical processing that occurs along that route.

An **intravenous injection** delivers a drug directly into blood, bypassing absorption across another tissue. Inhaled drugs cross the lungs' thin exchange surfaces. Oral drugs cross the digestive lining, while drugs in a skin patch pass through skin more gradually. These routes differ in the speed at which the drug's blood concentration rises.

Blood from the digestive tract passes through the liver before entering the general circulation. Liver enzymes may chemically modify an absorbed drug during this **first-pass metabolism**. **Bioavailability** is the fraction of the administered substance that reaches systemic circulation. An oral dose can therefore produce less exposure than the same amount injected directly into blood.



Carlson and Birkett (2022), Fig. 4.2, p. 94.



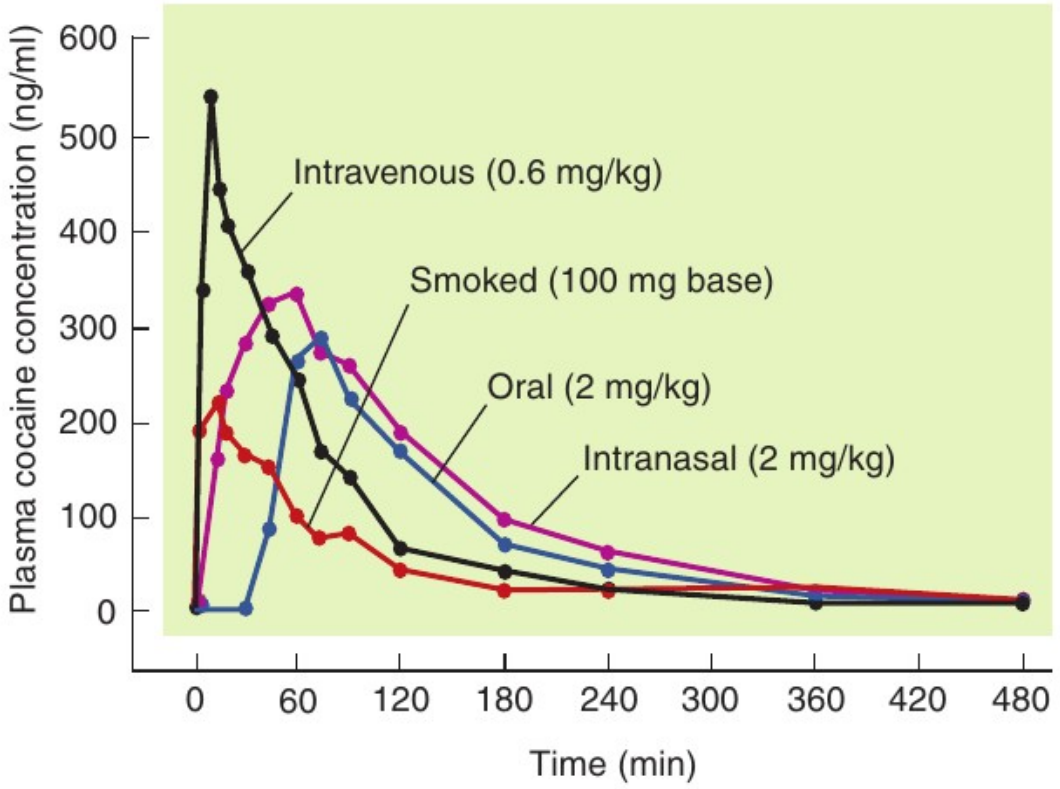
Puri et al. (2022), fig1-video1. Dopamine applied to D1 receptor-expressing HeLa cells.

Distribution determines which tissues a drug reaches

Distribution is the movement of a drug from blood into tissues. The bloodstream carries a drug past many organs, but the drug must cross local barriers to reach their cells. Drug molecules bound to blood proteins are temporarily less available to leave circulation. The unbound fraction can move into tissues and approach cellular targets.

The **blood–brain barrier** is formed by specialized brain capillaries that restrict movement from blood into neural tissue. Many lipid-soluble drugs cross cell membranes more readily than strongly charged compounds. Other substances require transport proteins. A drug can be abundant in blood yet enter the brain poorly because it cannot pass these barriers efficiently.

Metabolism changes a drug's chemical structure, and **excretion** removes the drug or its products, often through the kidneys. Together they contribute to **clearance**, removal from circulation. A **half-life** describes how long concentration takes to fall by half during a specified elimination phase. Repeated doses can accumulate when administration outpaces removal.



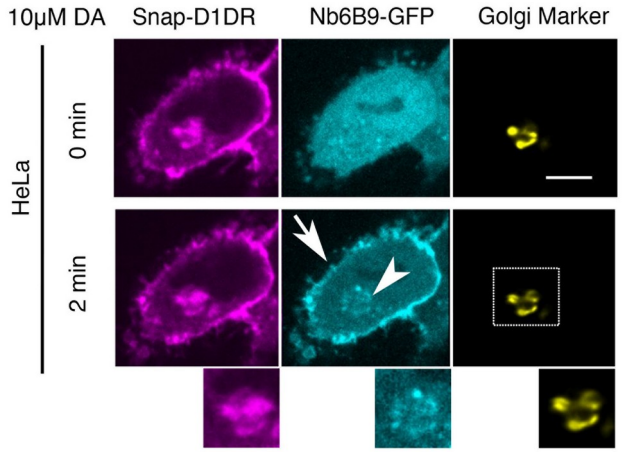
Carlson and Birkett (2022), Fig. 4.3, p. 95.

Drug access differs between surface and internal receptors

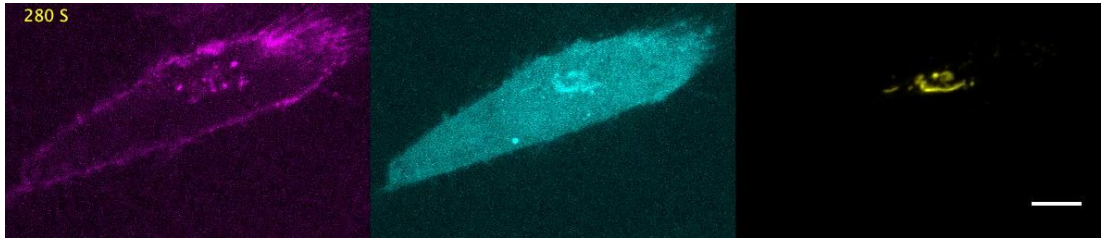
D1 receptors are dopamine receptors that activate intracellular signaling through G proteins, proteins that pass signals from receptors to other cellular components. Puri and colleagues examined D1 receptors at the **plasma membrane**, the cell's outer boundary, and at the **Golgi apparatus**, an internal membrane compartment that processes and sorts cellular proteins.

An **agonist** binds a receptor and activates it. Dopamine is a D1 agonist, but its charged structure limits passage through lipid membranes. The synthetic agonist **SKF81297** crosses membranes more readily. Applied outside a cell, these chemicals can therefore reach different receptor populations even though both activate the same D1 receptor class.

The researchers attached fluorescent labels to D1 receptors and used **Nb6B9**, a small antibody fragment that binds an activated receptor shape. Before stimulation, this detector was dispersed inside the cell. After receptor activation, it accumulated at the affected membrane. Detector movement distinguished an activated receptor population from receptors that were present but had not responded.



Puri et al. (2022), Fig. 1 (b, HeLa). D1 receptor activation at two cellular locations.



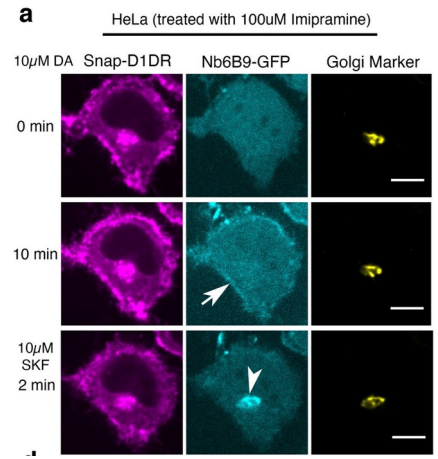
Puri et al. (2022), fig1-video3. A membrane-permeant D1 agonist in HeLa cells.

A transporter gives dopamine access to an internal target

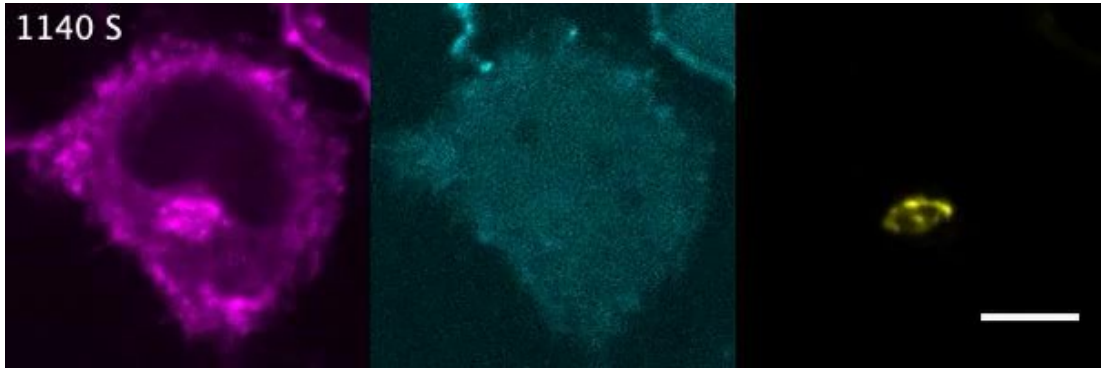
Organic cation transporter 2, or **OCT2**, carries certain positively charged molecules across membranes. Dopamine is one of its substrates. In HeLa cells, a human laboratory cell line, OCT2 occurred at both the plasma membrane and the Golgi. These transport routes allowed externally applied dopamine to reach D1 receptors that were located inside the cell.

The researchers treated cells with **imipramine** to inhibit OCT2-dependent transport. Dopamine then produced much less activation of Golgi D1 receptors. SKF81297 still activated that internal population because it could cross membranes without relying on OCT2. Receptor activation therefore persisted when a membrane-permeant agonist bypassed the inhibited transport step.

OCT2 inhibition changed delivery of the ligand, the chemical that binds the receptor. It did not remove the Golgi receptors or prevent every agonist from activating them. Dopamine needed the transport route to produce the internal response, whereas SKF81297 could reach the same receptor population through its own membrane permeability.



Puri et al. (2022), Fig. 2 (a). Dopamine response during transporter inhibition.



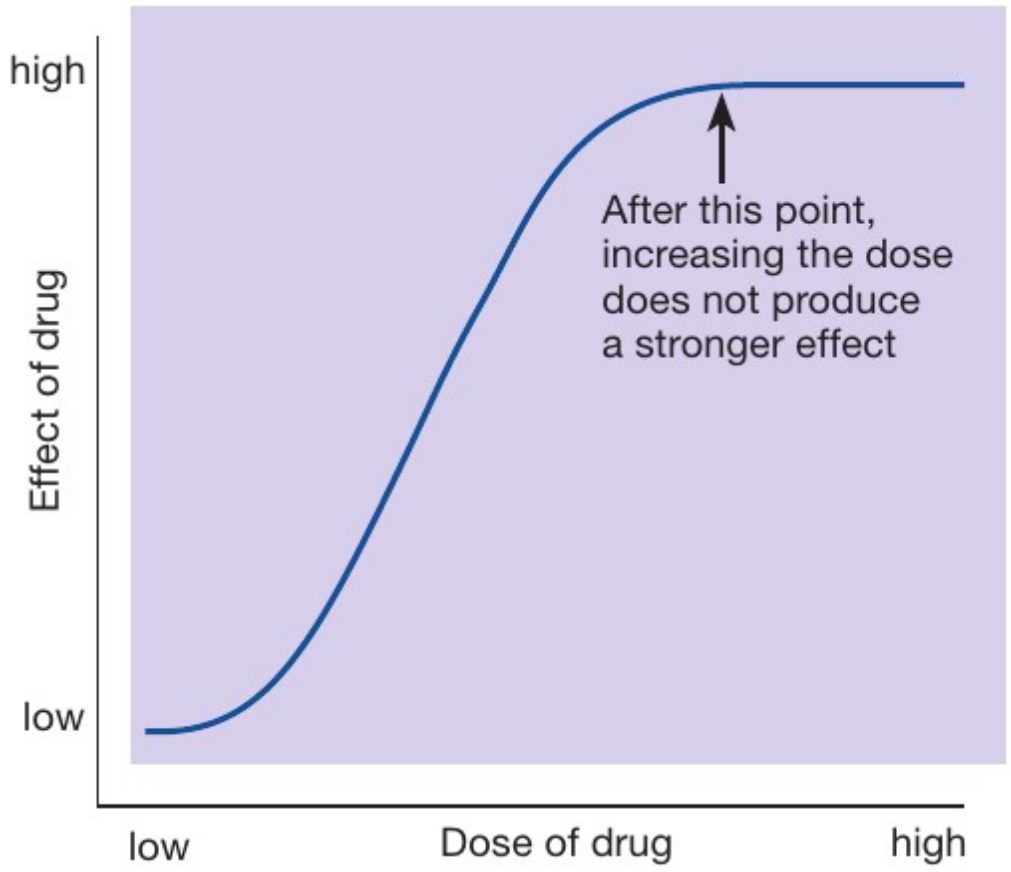
Puri et al. (2022), fig2-video1. D1 receptor response during OCT2 inhibition.

Potency and efficacy describe different properties of a drug

A **dose–response curve** relates drug exposure to a measured effect. Within the rising portion, additional drug produces a larger response. The curve eventually levels off when the measured response approaches its maximum. Increasing the dose beyond that point may increase adverse effects even though the desired response becomes little larger.

Potency describes the dose needed to produce a specified effect. A more potent drug produces that effect at a lower dose. **Efficacy** describes the maximum effect the drug can produce in the tested system. Two drugs can reach the same maximum while requiring different doses, or differ in both the required dose and maximum response.

Affinity is a ligand's tendency to bind a receptor. Binding alone does not determine the final effect: the bound receptor must also activate the relevant cellular processes. A compound may bind strongly but activate a receptor weakly. Its dose–response curve therefore reflects both receptor binding and the response produced after binding occurs.



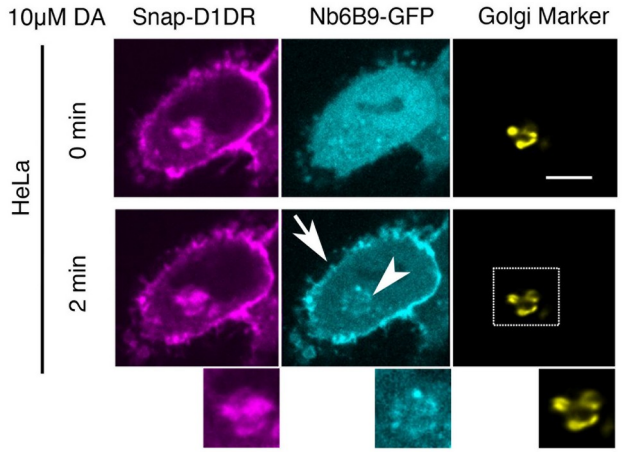
Carlson and Birkett (2022), Fig. 4.4, p. 96.

The same receptor class operates in different compartments

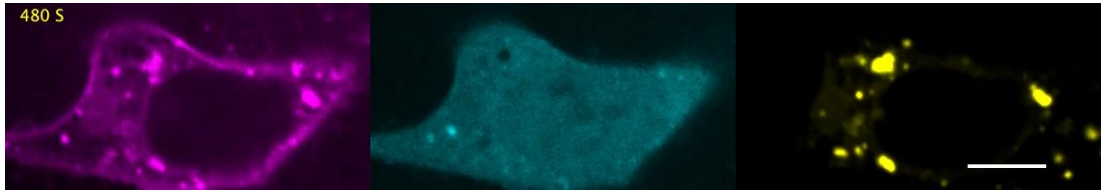
Externally applied dopamine activated surface and Golgi D1 receptors in HeLa cells. In HEK293 cells, a second laboratory cell line, it activated the surface receptors but produced little Golgi activation. The membrane-permeant agonist SKF81297 activated Golgi receptors in both cell types. The internal receptor population remained responsive even in cells where dopamine failed to reach it effectively.

HeLa cells expressed substantially more OCT2 than HEK293 cells. Introducing additional OCT2 into HEK293 cells allowed dopamine to activate their Golgi D1 receptors. Dopamine's chemical structure had not changed. The added transporter supplied a membrane-crossing route that the original cells lacked at a level sufficient to generate the internal response.

Golgi activation also persisted when the researchers blocked **endocytosis**, the process that moves surface membrane and its proteins into the cell. D1 receptors already located at the Golgi could respond after dopamine arrived. The fluorescent signal inside the cell therefore did not require activated surface receptors to travel inward during the experiment.



Puri et al. (2022), Fig. 1 (b, HeLa). D1 receptor activation at two cellular locations.



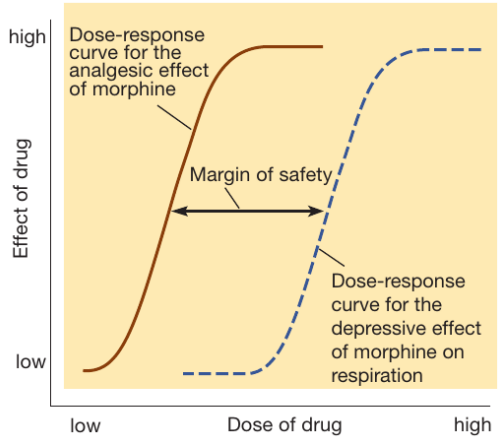
Puri et al. (2022), fig1-video2. Dopamine applied to D1 receptor-expressing HEK293 cells.

Desired and adverse effects can rise at different doses

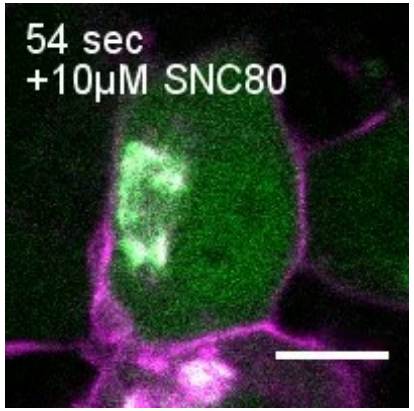
A **therapeutic effect** is a desired treatment response; an **adverse effect** is an unwanted response. Morphine can reduce pain while also depressing respiration. These effects involve opioid-sensitive neural pathways that perform different functions. The dose that produces useful pain relief and the dose that dangerously suppresses breathing need not be identical.

A **margin of safety** describes the separation between exposures producing the desired and dangerous effects. A wider separation allows more variation in exposure before severe toxicity develops. When the curves lie close together, a relatively small increase can move a patient from a therapeutic response toward serious respiratory or other physiological impairment.

The same administered amount can produce different exposure in different people. Slower elimination leaves more active drug available, while another sedating substance may add to respiratory depression. With repeated doses, residual drug can remain when the next dose arrives. The concentration can rise across doses even when the amount in each individual dose stays constant.



Carlson and Birkett (2022), Fig. 4.5, p. 97.



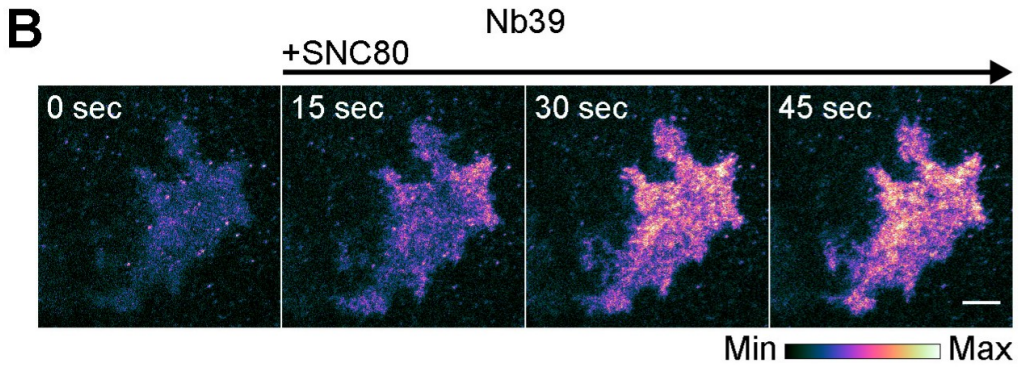
Crilly et al. (2021), video1. Nb39 recruitment to intracellular opioid receptors.

An opioid receptor's location changes its active state

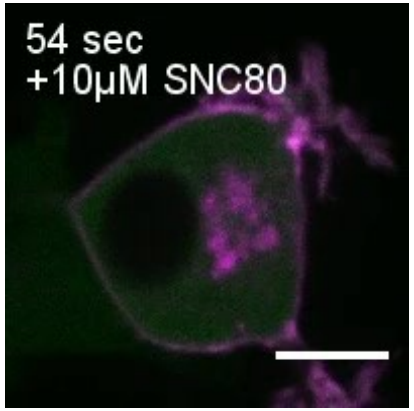
The **delta opioid receptor**, or **DOR**, is a G protein-coupled receptor found at the cell surface and in internal membranes. In cultured PC12 cells, some newly synthesized DOR remained in an intracellular compartment. The membrane-permeant agonist **SNC80** could reach both the surface population and this internal receptor population.

The fluorescent detector **Nb39** recognizes an activated opioid-receptor shape. After SNC80 was added, Nb39 rapidly accumulated at internal DOR. The antagonist **naltrindole**, which blocks DOR activation, reversed this recruitment. The detector therefore responded to an agonist-dependent receptor state rather than simply binding every DOR protein already present inside the cell.

A second detector, **miniGsi**, mimics the receptor-binding surface of an inhibitory G protein. SNC80 recruited miniGsi to surface DOR but not to internal DOR during the same interval. Internal receptors recruited Nb39 instead. The two receptor locations therefore favored different detectable protein interactions even though the agonist and receptor class were the same.



Crilly et al. (2021), Fig. 1 (B). Opioid receptor biosensor recruitment.



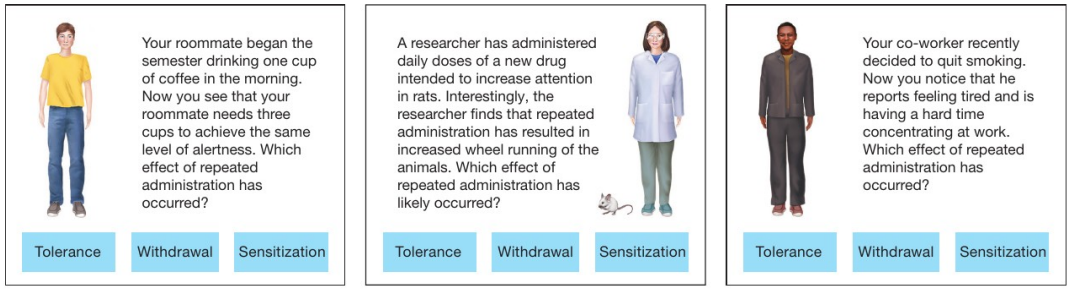
Crilly et al. (2021), video2. A second biosensor tests intracellular opioid receptors.

Repeated exposure can decrease or increase a response

Tolerance is a reduced response after repeated drug exposure. The same dose now produces less effect than it did initially. One cause is faster drug metabolism, which lowers exposure. Another is reduced cellular responsiveness, such as fewer receptors at the surface or weaker signaling after those receptors are activated by the drug.

Sensitization is an increased response after repeated exposure. Particular stimulant effects can become larger across treatments even while other effects develop tolerance. These terms describe changes in specified responses, not a single condition affecting every drug action equally. Movement, reward-related behavior, sedation, and autonomic responses can adapt at different rates within the same animal.

Repeated exposure can also produce **physical dependence**, in which cellular adjustments compensate for the drug's continuing action. Removing the drug leaves those adjustments temporarily unopposed. The resulting **withdrawal symptoms** may oppose the original drug effects. A nervous system adapted to persistent suppression can become unusually excitable when the suppressing substance is suddenly withdrawn.



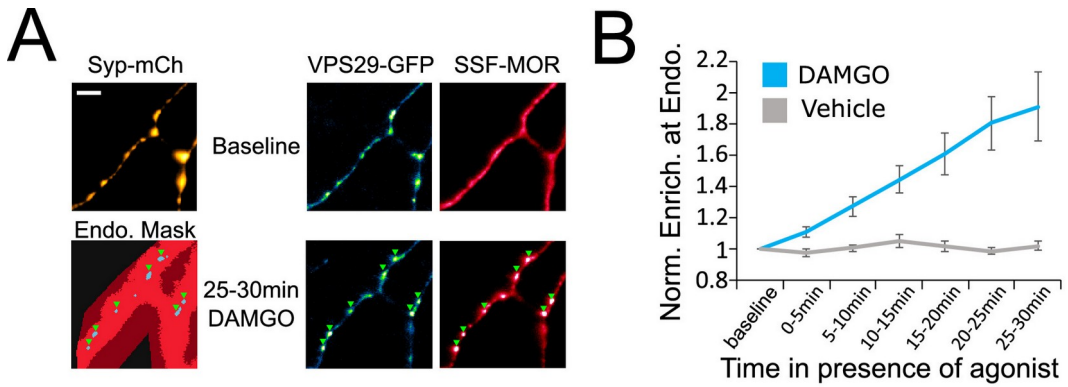
Carlson and Birkett (2022), Fig. 4.6, p. 98.

Opioid exposure removes receptors from an axon's surface

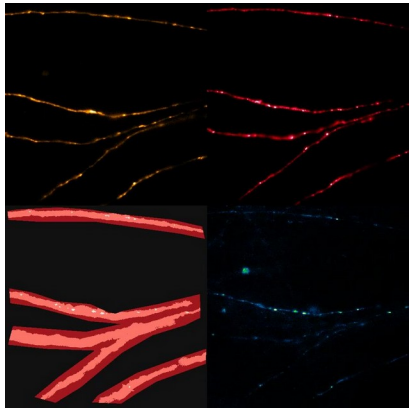
Jullié and colleagues studied **mu opioid receptors**, or **MORs**, in cultured striatal neurons. Activating these presynaptic receptors inhibited synaptic vesicle release. After prolonged exposure to the MOR agonist **DAMGO**, a fresh agonist challenge inhibited release less effectively. The neurons had developed tolerance to this particular presynaptic action of the opioid.

DAMGO also moved MORs from the axon surface into **endosomes**, membrane-bound compartments inside the cell. Surface-labeled receptors accumulated beside an endosomal marker. This movement reduced the receptor population available to extracellular agonist. A terminal with fewer accessible MORs could maintain less opioid-mediated inhibition when electrically stimulated to release transmitter.

MOR uptake required **phosphorylation**, the addition of phosphate groups to the receptor's intracellular tail. Removing selected phosphorylation sites strongly reduced uptake and prevented the usual development of tolerance. Inhibiting the enzymes GRK2 and GRK3 also reduced both processes. These enzymes normally add phosphate groups that promote removal of the receptor from the axon surface.



Jullié et al. (2022), Fig. 2 (A–B). Mu opioid receptor endosomal recruitment.



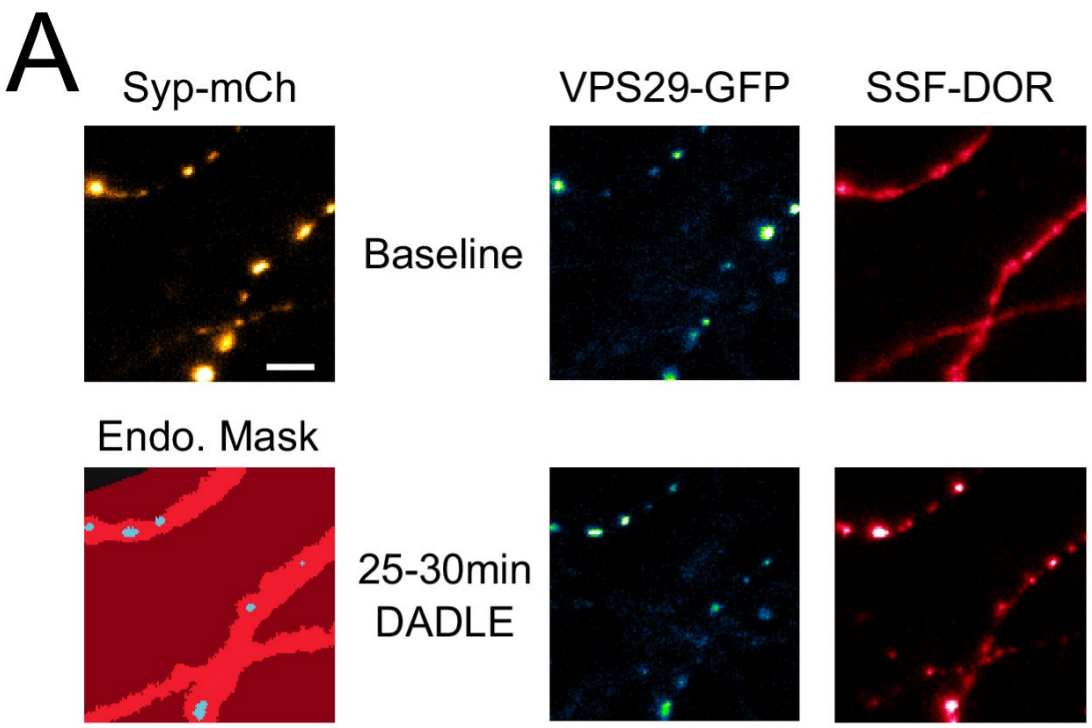
Jullié et al. (2022), fig2-video1. Mu opioid receptor internalization in axons.

Opioid receptor subtypes differ in their recycling

After entering an endosome, a receptor can return to the plasma membrane through **recycling**. Receptor-containing vesicles fuse with the membrane and restore receptors to the surface. This return competes with continuing uptake. Surface availability therefore depends on how quickly receptors leave the membrane and how quickly internal receptors are delivered back.

Both mu and delta opioid receptors underwent agonist-dependent uptake in cultured axons. Delta receptors returned to the surface less efficiently than mu receptors. During prolonged treatment, the delta receptor population consequently became more depleted at the axon surface. Delta-mediated inhibition of vesicle release also developed a larger loss of responsiveness in this preparation.

Tolerance remained selective for the activated receptor subtype. In neurons expressing both receptors, prolonged exposure to a mu-selective agonist reduced the mu response while preserving the delta response. A delta-selective agonist produced the converse pattern. The terminal retained its capacity for opioid inhibition through the receptor population that had not undergone the same prolonged activation.



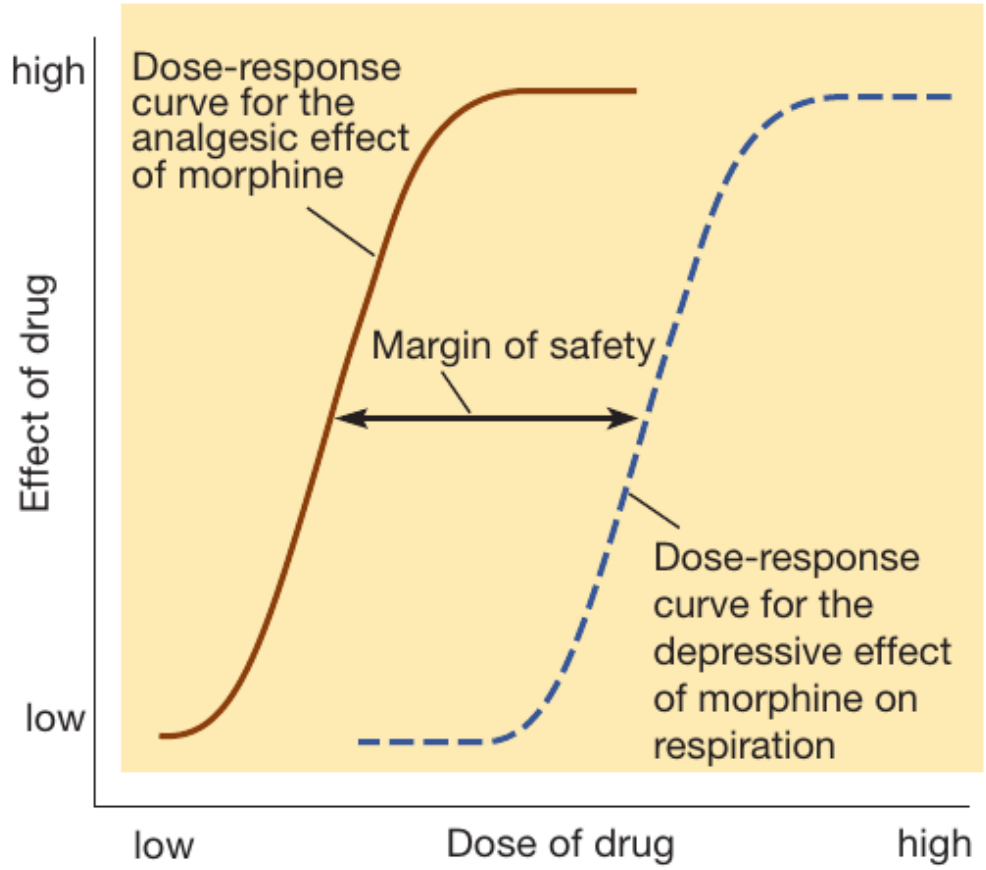
Jullié et al. (2022), Fig. 4 (A). Delta opioid receptor endosomal recruitment.

Expectations and conditioning change measured responses

A **placebo** resembles a treatment but lacks the active component being evaluated. A **placebo response** is a change that occurs in that treatment setting. Expectations and learned associations can change pain processing and reported symptoms. Such effects involve nervous-system activity even when the administered substance does not contain the tested analgesic drug.

Pain-relieving expectations can activate endogenous pain-control systems, including pathways that use opioid messengers. A previously effective treatment can also make its associated setting or cue acquire an effect. After that learning, the cue can influence later responses. The chemical action of a drug and the nervous system's response to treatment cues can occur together.

In a **blinded comparison**, participants or evaluators do not know which treatment was given. A matched placebo preserves features such as pill appearance and administration routine. Comparing the active drug with that condition reduces differences caused by treatment expectations. The remaining difference estimates the added response produced when the active chemical is present.



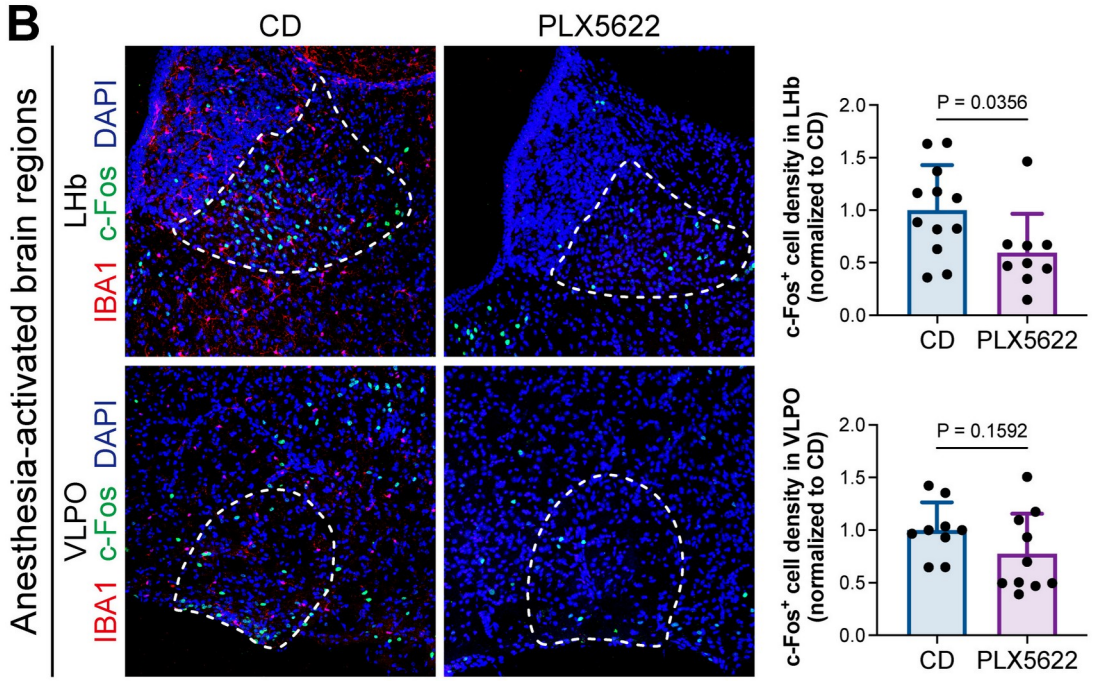
Carlson and Birkett (2022), Fig. 4.5, p. 97.

A drug response depends on the state of neural networks

Microglial depletion changed activity differently across brain regions. The researchers measured **c-Fos**, a protein produced after neuronal activation. Its labeling decreased in the **lateral habenula** and **supraoptic nucleus**, regions associated with anesthesia, but increased in the **locus coeruleus**, a brainstem region involved in arousal. The same cell manipulation favored different regional activity patterns.

Electrical recordings examined excitatory and inhibitory inputs to individual neurons. After microglial depletion, supraoptic neurons received less excitatory input, while locus coeruleus neurons received more. Inhibitory input changed little in these comparisons. The balance of synaptic input shifted toward less excitation in one region and more excitation in the other, favoring different activity patterns.

Microglia also use **P2Y12 receptors** to detect extracellular nucleotide signals and raise intracellular calcium. Blocking P2Y12 delayed anesthesia induction and accelerated emergence. Raising calcium signaling in microglia produced the opposite pattern, while disrupting their calcium signaling increased resistance to anesthesia. Drug sensitivity changed when this microglial signaling pathway was manipulated without removing the anesthetic.



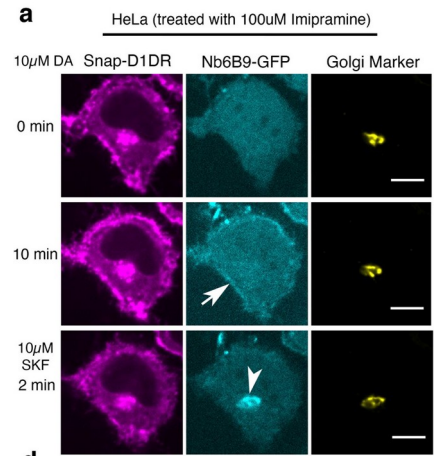
He et al. (2023), Fig. 7 (B, LHb and SON). Regional neuronal activation during anesthesia.

Transporters control access to an internal receptor

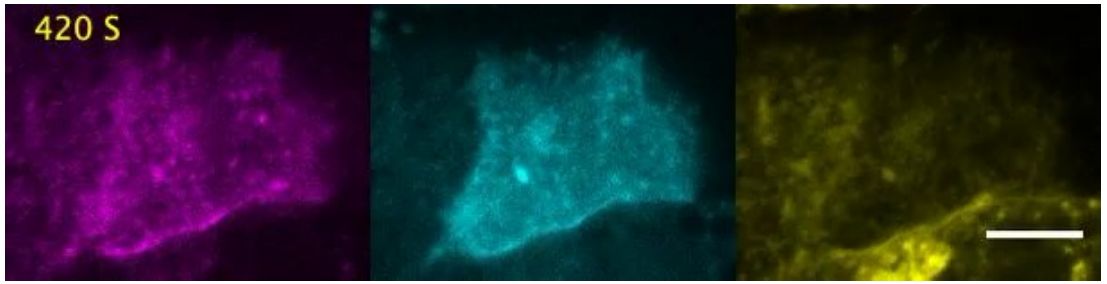
HEK293 cells expressed relatively little OCT2 and responded to extracellular dopamine mainly through surface D1 receptors. The researchers introduced additional OCT2 and then applied dopamine again. The fluorescent activation detector now accumulated at Golgi receptors as well. Increasing the transport protein allowed dopamine to activate an internal population that previously responded poorly.

The reverse manipulation reduced OCT2 production in HeLa cells. The researchers used **short hairpin RNA**, an RNA sequence that directs destruction of a selected messenger RNA, to reduce the instructions available for making OCT2. Dopamine then produced less Golgi receptor activation. The membrane-permeant agonist SKF81297 continued to activate the internal receptor population.

Transport inhibition, increased transporter expression, and reduced transporter production each changed dopamine access in the predicted direction. Surface receptors remained accessible directly from the extracellular fluid, while Golgi receptors required membrane crossing. Cells with the same receptor class can therefore respond differently because their transporter populations determine which internal receptors the applied transmitter reaches.



Puri et al. (2022), Fig. 2 (a). Dopamine response during transporter inhibition.



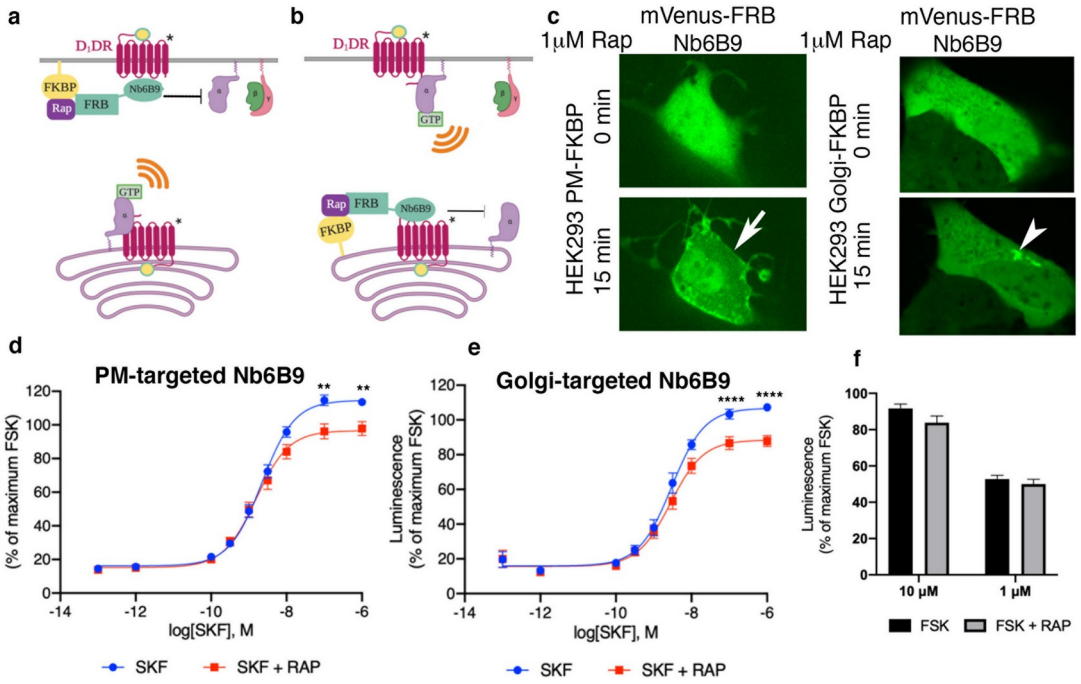
Puri et al. (2022), fig2-video2. Dopamine response in cells expressing OCT2.

Internal D1 receptors contribute to a second messenger

Activated D1 receptors stimulate **Gs**, a G protein that increases activity of the enzyme **adenylyl cyclase**. This enzyme produces **cyclic AMP**, or **cAMP**, from ATP, a molecule that supplies energy for many cellular reactions. cAMP is a **second messenger**: the receptor detects dopamine at a membrane, and cAMP carries the signal onward inside the cell to regulate other proteins.

The researchers concentrated an inhibitory antibody fragment at either the plasma membrane or the Golgi. At high local concentration, the fragment interfered with the receptor's access to G proteins. Blocking surface D1 signaling left a cAMP response generated by the internal population. Blocking Golgi D1 signaling reduced the contribution from that internal receptor population.

The cAMP signal can activate **protein kinase A**, or **PKA**, an enzyme that adds phosphate groups to target proteins. Golgi-localized D1 activation also activated nearby PKA. Surface D1 activation alone did not produce the same local PKA response in the tested cells. Where cAMP is produced can affect which nearby signaling proteins respond.



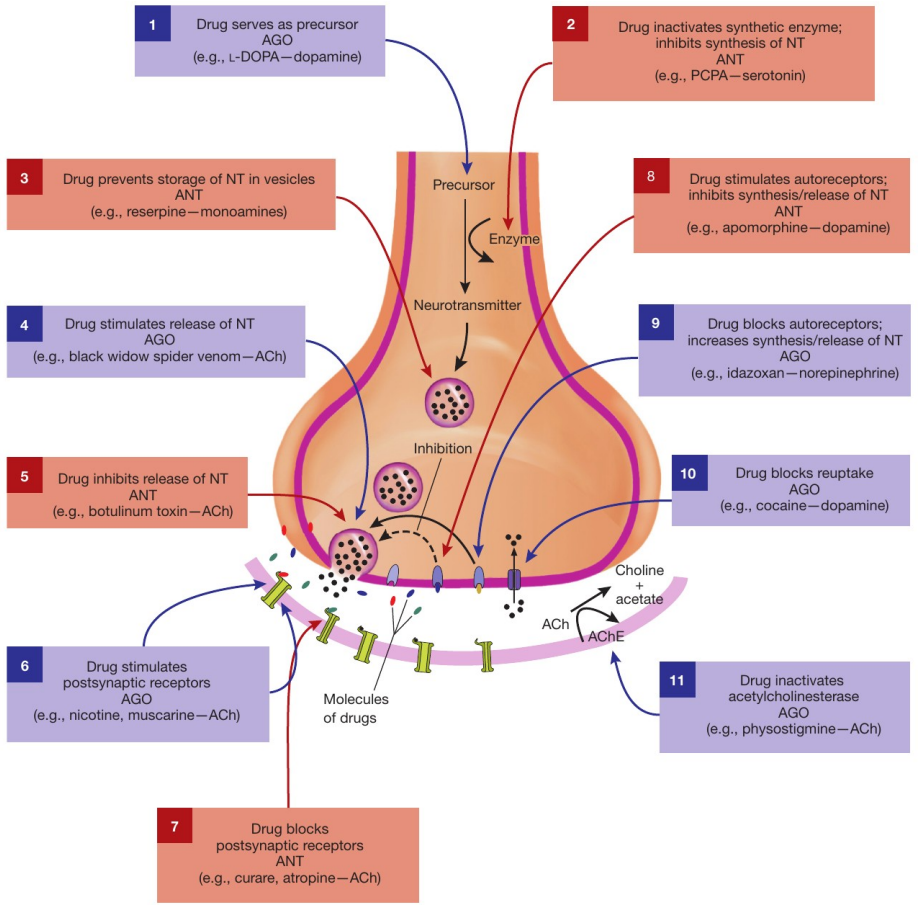
Puri et al. (2022), Fig. 4 (a–f). Compartment-specific D1 receptor inhibition.

Drugs can alter several stages of synaptic communication

At a chemical synapse, the presynaptic neuron synthesizes transmitter and loads it into vesicles. An action potential opens calcium channels in the terminal. Calcium entry triggers vesicle fusion, releasing transmitter into the synaptic gap. The transmitter then binds receptors on the receiving cell or on the releasing terminal itself.

Drugs can intervene before transmitter reaches a receptor. Blocking a synthetic enzyme reduces production of new transmitter. Blocking vesicle loading leaves less transmitter available for release. Preventing vesicle fusion reduces the amount entering the gap after an action potential. Each intervention decreases chemical delivery through a different molecular step in the releasing neuron.

After release, transporters and enzymes help terminate the signal. Blocking reuptake leaves transmitter outside cells for longer, while blocking its breakdown preserves molecules that would otherwise be destroyed. A receptor agonist instead activates the receiving protein directly. A receptor antagonist blocks activation even when the normal transmitter has been released into the gap.



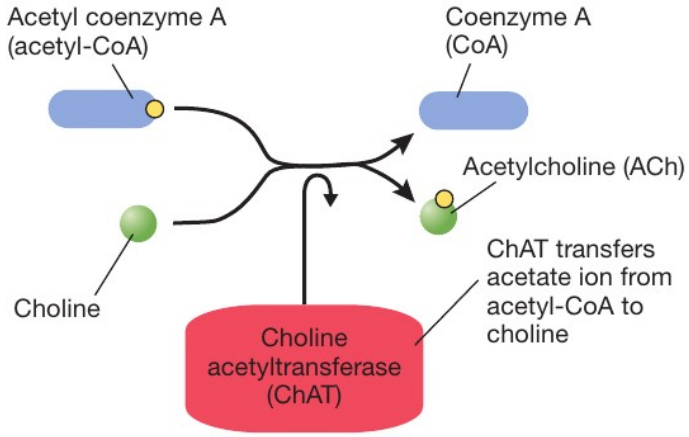
Carlson and Birkett (2022), Fig. 4.7, p. 100.

Proteins and precursors supply chemical signaling

Neurons need raw materials to make chemical signals. A **precursor** is a starting substance that an enzyme changes into another molecule. Acetylcholine, for example, is made from choline and an acetyl group supplied by acetyl coenzyme A. The enzyme **choline acetyltransferase** joins these components, producing transmitter that can then be loaded into vesicles.

Reducing the supply of a precursor can reduce transmitter production. Blocking the enzyme that performs the chemical reaction can have a similar effect, even when the raw materials remain available. Drugs acting at these steps change how much transmitter the neuron can make, rather than directly activating or blocking receptors on the receiving cell.

Receptors and enzymes are themselves proteins that neurons must produce. After a new receptor is made, it must travel to the membrane where it will function. In cultured neurons, fluorescent labeling allows newly produced glutamate-receptor cargo to be followed as it moves through dendrites. Chemical communication therefore requires both transmitter production and delivery of the proteins that detect it.



Carlson and Birkett (2022), Fig. 4.12, p. 107.



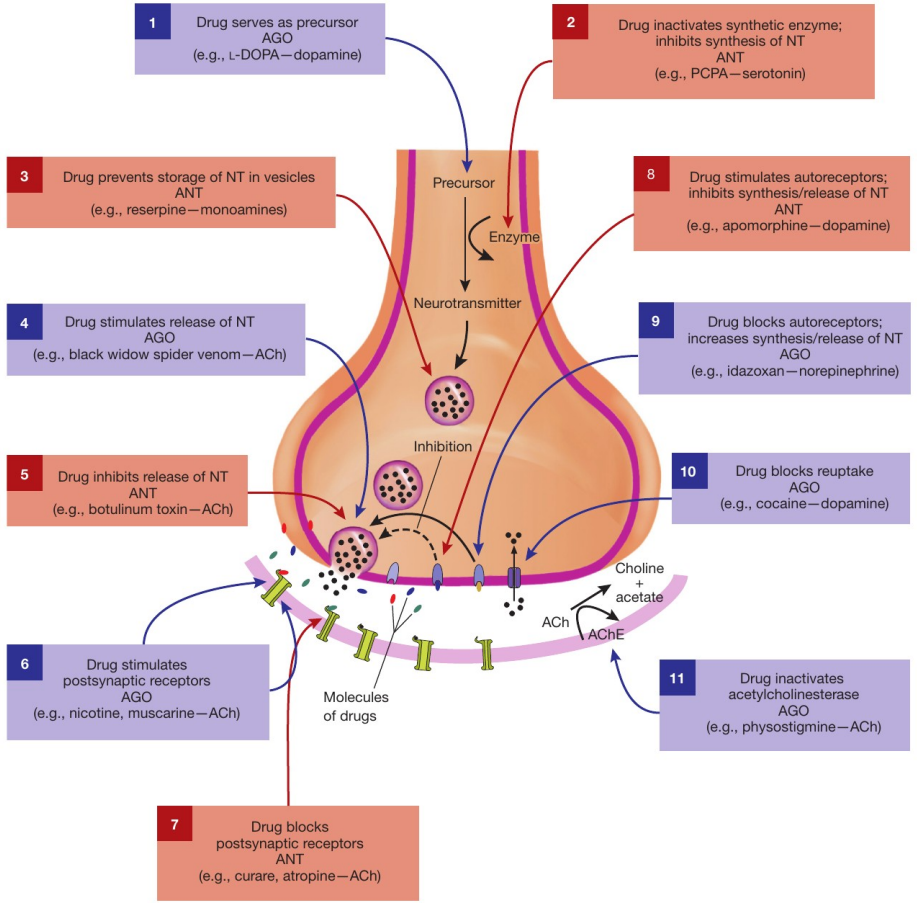
Wong et al. (2024), fig1-video5. Receptor cargo in an unstimulated dendrite.

Vesicle storage and release offer presynaptic targets

Making a transmitter does not immediately make it available for release. Many transmitters must first enter **synaptic vesicles**, small membrane-bound sacs inside an axon terminal. Transport proteins move transmitter into these sacs. A drug that blocks this loading step can leave a terminal with transmitter in its cytoplasm but little packaged for release.

An arriving action potential opens voltage-sensitive calcium channels in the terminal membrane. Calcium enters and activates proteins that help a vesicle fuse with the membrane. Fusion opens a route from the vesicle interior to the synaptic gap. The released transmitter can then reach receptors on another neuron, a muscle cell, or a gland.

Botulinum toxin interferes with proteins required for vesicle fusion. At a motor nerve terminal, this prevents acetylcholine release, so the muscle receives less chemical stimulation and contracts less. Black widow spider venom instead triggers extensive transmitter release. These substances change communication before the transmitter binds a receptor on the receiving cell.



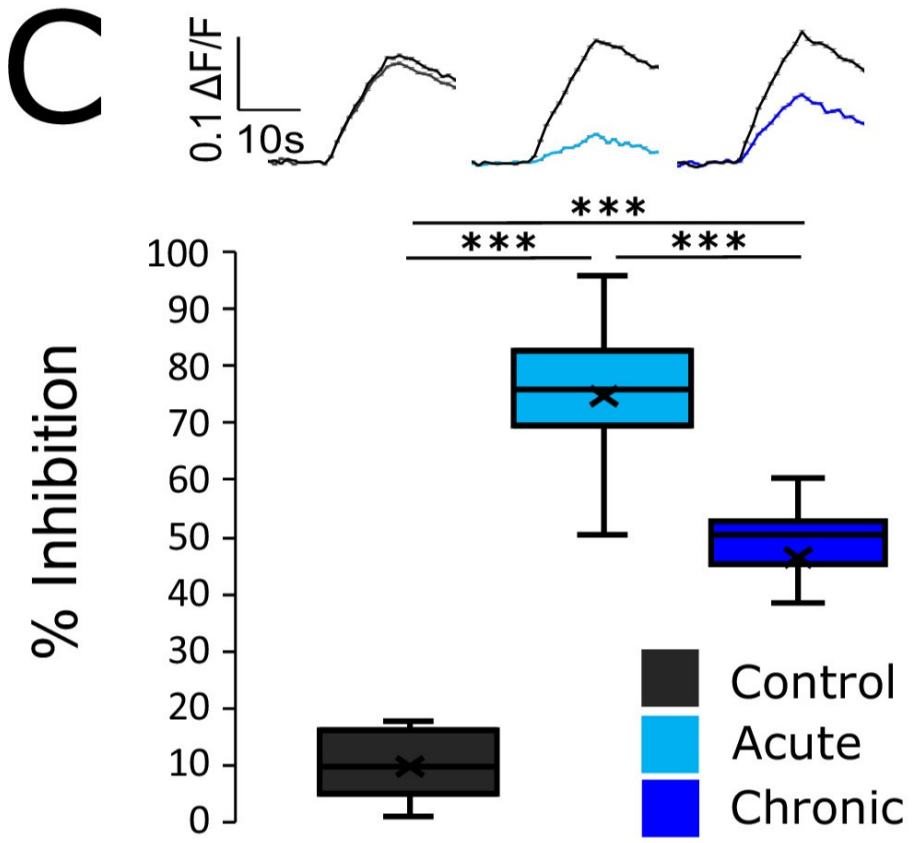
Carlson and Birkett (2022), Fig. 4.7, p. 100.

Opioids reduce release through a presynaptic receptor

Opioid receptors can reduce communication by acting on the releasing side of a synapse. In the cultured-neuron study, electrical stimulation normally caused synaptic vesicles to fuse with the terminal membrane. Activating mu opioid receptors with DAMGO reduced that fusion response, so less vesicle release occurred in response to the same stimulation.

The researchers measured release using a fluorescent protein attached to a vesicle protein. Its fluorescence was weak inside an acidic vesicle and became brighter when fusion exposed it to the fluid outside the cell. A smaller stimulation-evoked increase therefore meant that fewer vesicle proteins had been exposed through fusion at the terminal surface.

After prolonged DAMGO treatment, the same opioid challenge produced less suppression of release. The dose–response curve also shifted, so more agonist was needed to obtain a comparable effect. Both the amount required and the greatest inhibition changed. This was tolerance to the opioid's presynaptic effect, measured directly as a change in vesicle release.



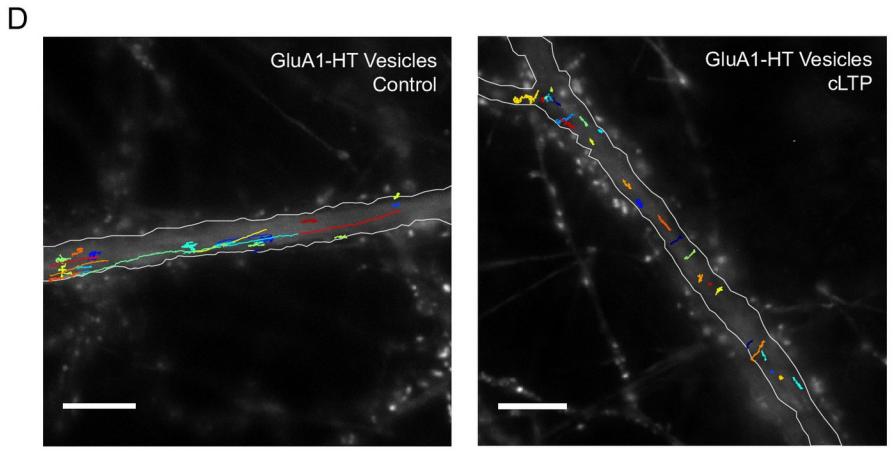
Jullié et al. (2022), Fig. 1 (C). Presynaptic opioid responses after chronic treatment.

Receptor-containing cargo moves inside dendrites

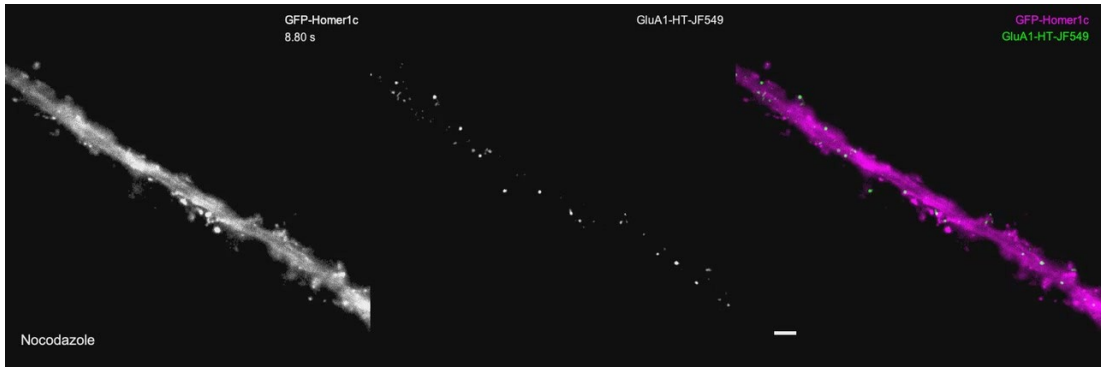
Receptors must reach the cell surface before they can respond to transmitter outside the neuron. An **AMPA receptor** is a glutamate-sensitive ion channel that contributes to excitatory signaling. Researchers labeled **GluA1**, one of its protein components, in cultured hippocampal neurons. This allowed receptor-containing vesicles to be followed inside dendrites, the branches that receive many synaptic inputs.

Some vesicles traveled along the dendrite in directed runs, while others moved more slowly within a local area. Long-distance transport uses **microtubules**, protein structures that serve as internal tracks for motor proteins. A receptor-containing vesicle can therefore move from where its cargo was produced toward a region where receptors will be inserted into the membrane.

The researchers labeled newly produced receptors separately from older receptor populations, making individual moving structures easier to distinguish. They also used a label that could not cross the membrane to identify receptors already on the surface. Internal transport and surface movement could then be examined separately, rather than treating every fluorescent spot as the same stage of receptor delivery.



Wong et al. (2024), Fig. 1 (D). AMPA receptor cargo in neuronal dendrites.



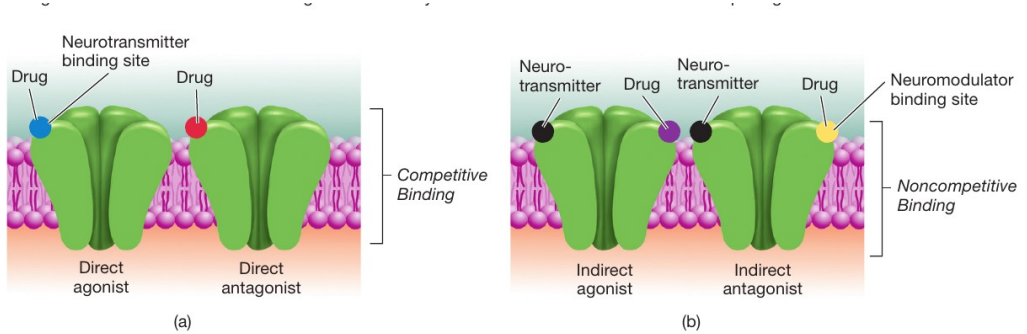
Wong et al. (2024), fig1-video2. Receptor cargo after microtubule disruption.

Agonists and antagonists alter receptor activation

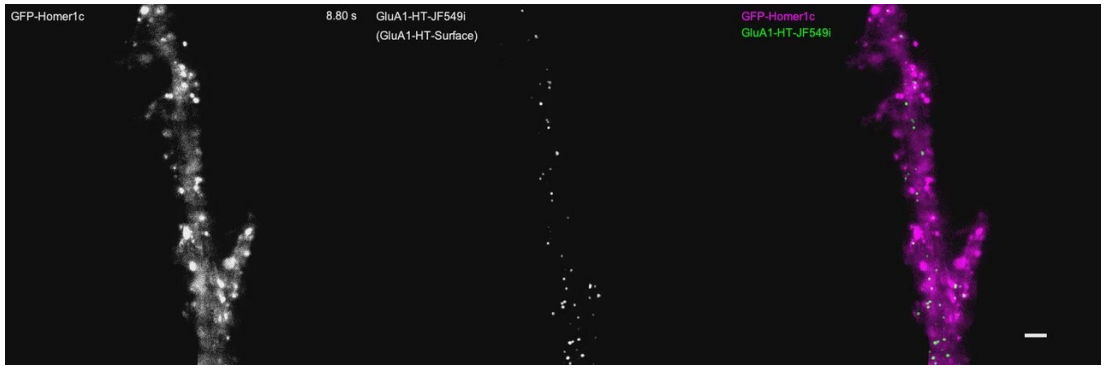
An agonist binds a receptor and promotes the change that activates it. An **antagonist** binds without producing that activation and reduces the response to an agonist. At a transmitter's binding site, these molecules compete for access. When more receptors are occupied by antagonist, fewer are available for the transmitter to activate.

An **allosteric site** is a different binding location on the same receptor. A molecule acting there can change how readily the receptor responds without occupying the transmitter's own binding site. Its effect may increase or decrease receptor activity. The receptor's response therefore depends on both the transmitter and substances acting at these additional sites.

Receptor activation and receptor number are also separate sources of change. Blocking a receptor reduces signaling while the receptor remains present. Removing receptors from the surface reduces the number available to respond. Inserting additional receptors can increase the receiving cell's sensitivity even when the amount of transmitter released into the synaptic gap stays unchanged.



Carlson and Birkett (2022), Fig. 4.8, p. 102.



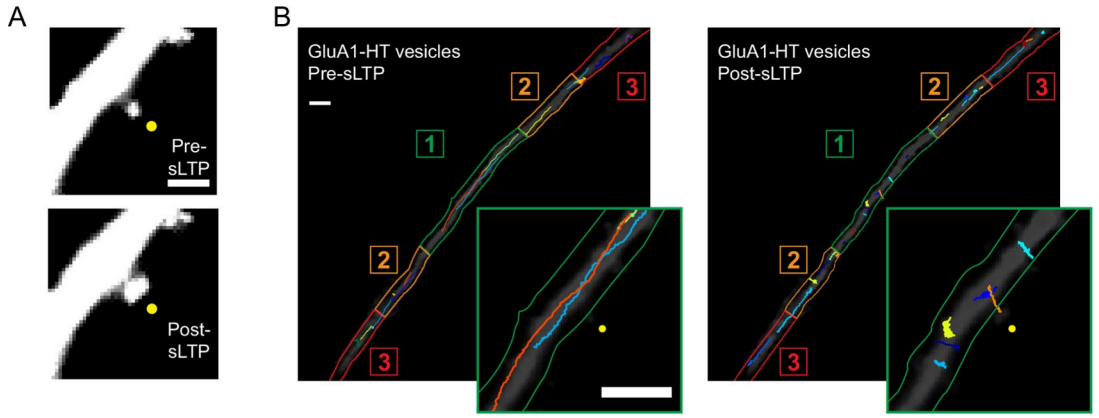
Wong et al. (2024), fig1-video3. Newly labeled receptors at the dendritic surface.

Synaptic stimulation changes local receptor transport

A **dendritic spine** is a small projection that receives an excitatory synaptic contact. Its receptors help determine how strongly it responds to released glutamate. The receptor-transport study stimulated a selected spine by using light to release glutamate from an inactive chemical form. This method activated a small region rather than stimulating the entire neuron equally.

After stimulation, the targeted spine enlarged and contained more labeled GluA1. Nearby receptor-containing vesicles also changed their movement within the dendrite. They made fewer long directed runs and moved less freely through the surrounding region. Receptor cargo that had previously traveled along the branch became more restricted near the stimulated synapse.

This restriction occurred mainly near the stimulated spine, while more distant dendritic regions retained their earlier movement patterns. Local activity can therefore change receptor delivery within one part of a neuron. The cell does not have to alter every synapse in the same way when a particular input undergoes a sustained increase in strength.



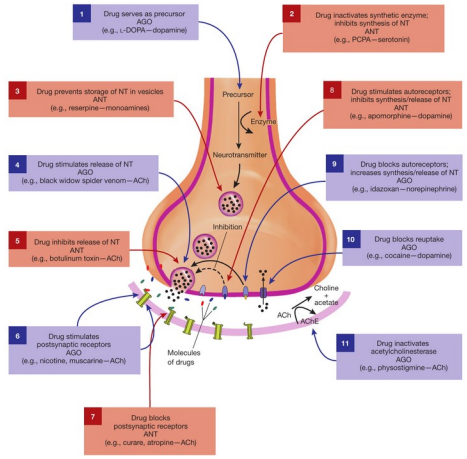
Wong et al. (2024), Fig. 2 (A–B). Receptor cargo near a stimulated dendritic spine.

Receptor delivery changes a cell's responsiveness

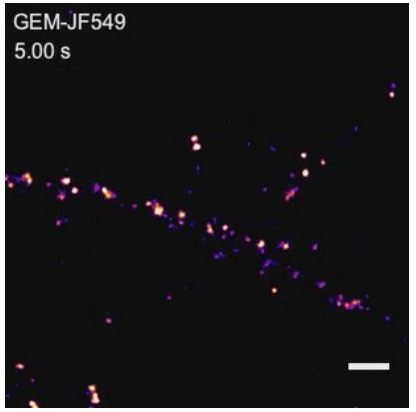
A receptor inside a transport vesicle cannot yet detect transmitter in the synaptic gap. For that receptor to become available, the vesicle must reach the surface and fuse with the cell membrane. The receptor is then part of the membrane, with its transmitter-binding region exposed outside the cell and its signaling regions facing the appropriate cellular compartments.

The number of available receptors can change as insertion and removal continue. More insertion can increase the surface population; more removal can reduce it. These changes alter how much response a given amount of transmitter produces. A receiving neuron can therefore become more or less sensitive without requiring a matching change in transmitter release.

Cargo also moves through the cell's crowded interior, where protein structures can restrict its path. Recordings of small particles inside dendrites make this movement visible. Some particles move over longer distances, while others remain within a smaller area. Such physical restrictions help explain why delivery depends on the organization of the cell interior as well as receptor production.



Carlson and Birkett (2022), Fig. 4.7, p. 100.



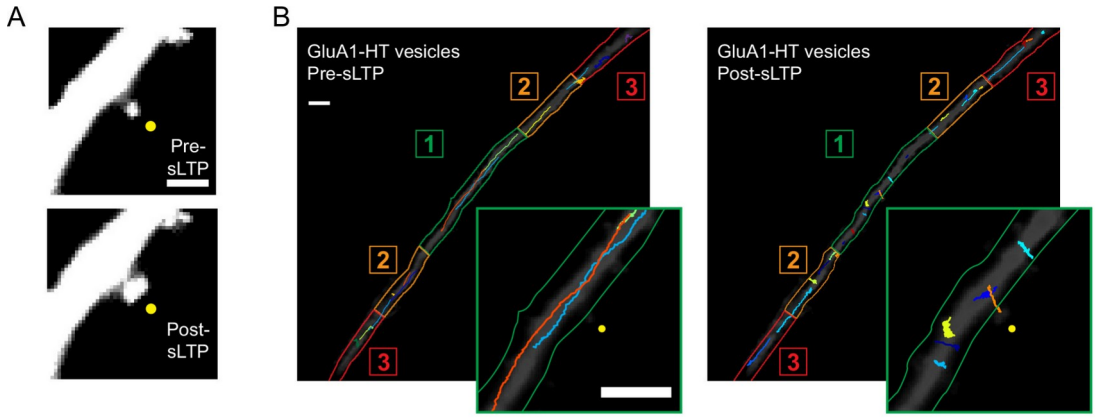
Wong et al. (2024), fig4-video1. Inert particles moving inside a dendrite.

Local stimulation concentrates receptor cargo nearby

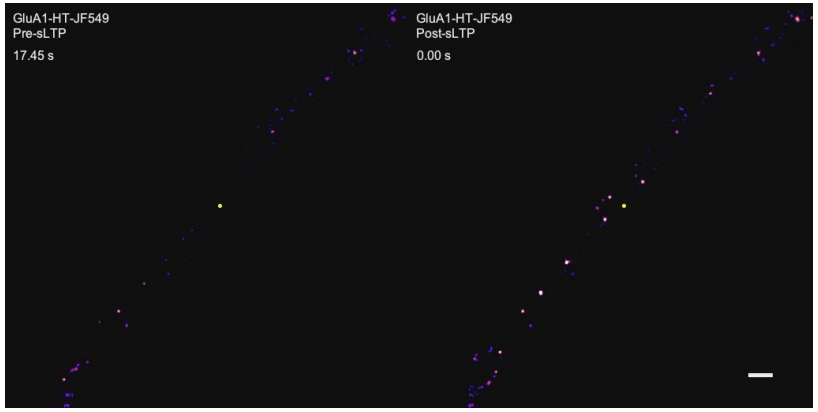
After a spine was stimulated, receptor-containing vesicles near that spine traveled less freely along the dendrite. The researchers compared nearby and more distant regions of the same branch. The change was strongest close to the activated synapse, where vesicles made fewer directed journeys and their less-directed movement also slowed.

Keeping cargo nearby increases the opportunity for local delivery because a vesicle does not immediately pass through to a distant part of the dendrite. The receptor proteins remain available within the region where activity has increased. Some of the movement changes persisted after the stimulation ended, so the local restriction outlasted the brief glutamate-release procedure.

The restriction depended on actual glutamate release. Applying the light procedure without the compound that supplied glutamate did not produce the same change in vesicle movement. Stimulation also increased actin assembly near the spine. This local change in the cell's internal framework provided a physical explanation for how activity could restrict moving receptor cargo.



Wong et al. (2024), Fig. 2 (A–B). Receptor cargo near a stimulated dendritic spine.



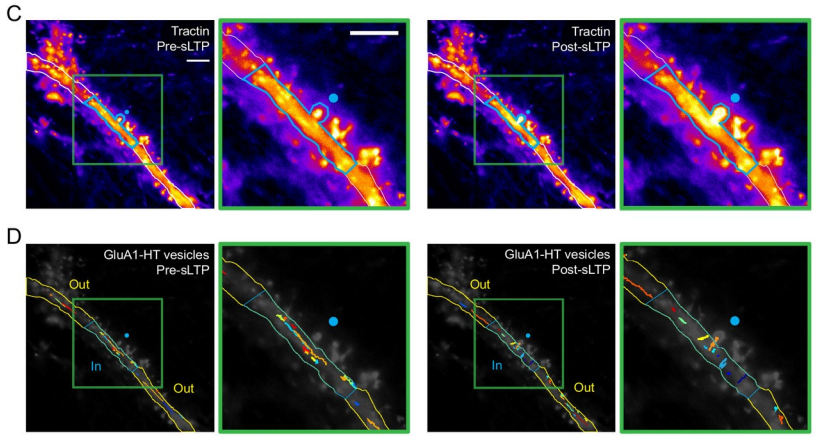
Wong et al. (2024), fig2-video1. Receptor cargo before and after local stimulation.

Actin and myosin help retain and deliver receptor cargo

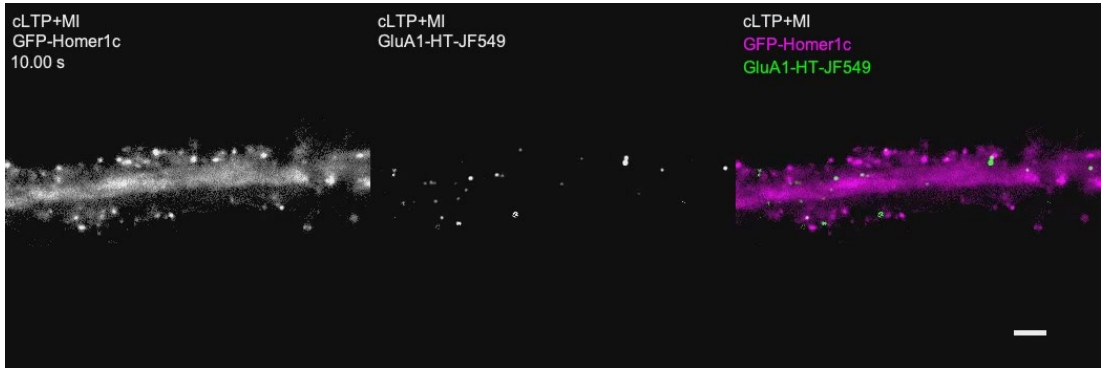
Actin is a protein that assembles into filaments, forming part of the **cytoskeleton**, the cell's internal framework. In the receptor-transport study, synaptic stimulation increased actin filaments in the nearby dendrite. As this network developed, receptor-containing vesicles moved less freely through that region and remained closer to the stimulated spine.

The researchers disrupted actin assembly with **latrunculin A**. This reduced the normal restriction of receptor-vesicle movement after stimulation. Actin filaments can crowd the space through which cargo moves, so reorganizing this network changes the routes available inside a dendrite. The effect concerns physical movement of vesicles, not simply the opening of receptors already at the surface.

Myosin is a motor protein that uses energy to generate force against actin. Blocking myosin did not prevent vesicle confinement in the same way as disrupting actin assembly. It did, however, reduce the stimulation-related increase in receptor insertion. Restricting cargo nearby and completing its delivery to the surface therefore required partly different cellular processes.



Wong et al. (2024), Fig. 3 (C–D). Actin and receptor cargo near stimulated spines.



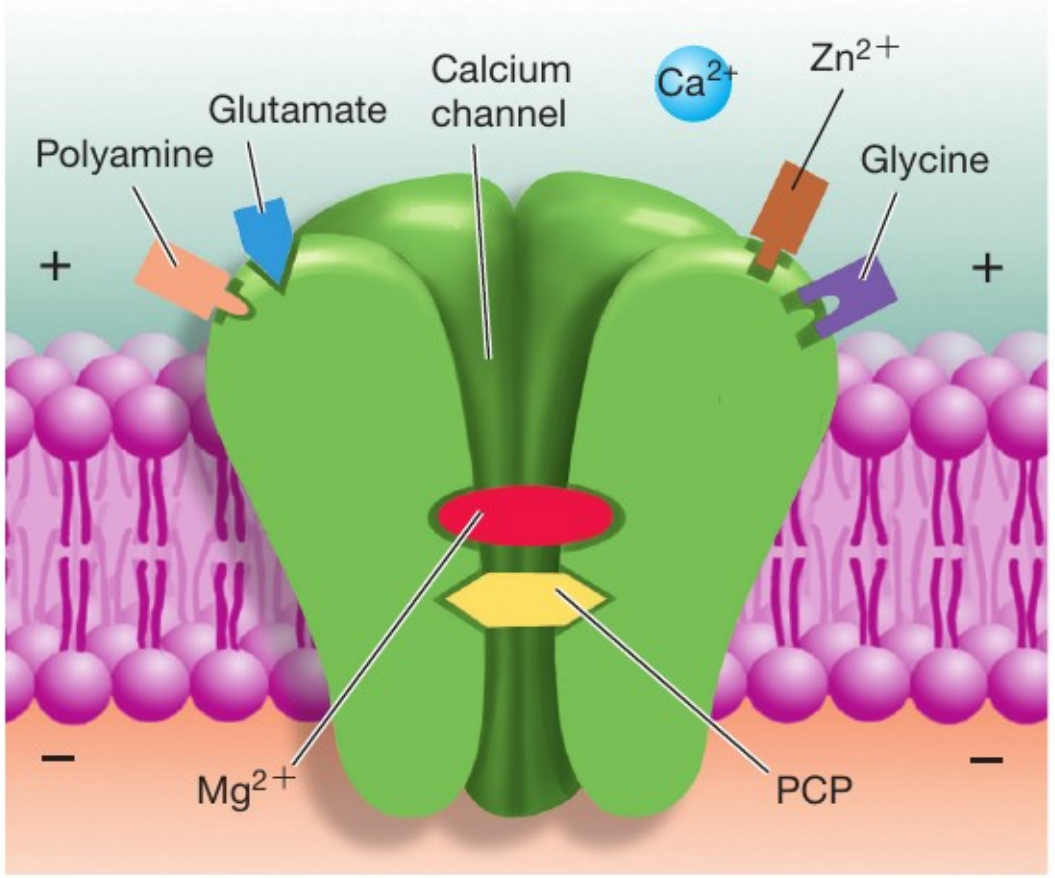
Wong et al. (2024), fig3-video3. Receptor cargo during myosin inhibition.

Glutamate activates several receptor classes

Glutamate is the main excitatory neurotransmitter in the mammalian brain. At many synapses, it opens **AMPA receptors**, allowing positive ions to enter the receiving neuron. This makes the membrane less negative and increases the chance that other excitatory inputs will help trigger an action potential. Changing the number of AMPA receptors changes the strength of this response.

NMDA receptors also respond to glutamate, but their channels are blocked by magnesium when the membrane is sufficiently negative. Depolarization helps relieve that block. Glutamate and the appropriate electrical condition must therefore occur together for substantial current to pass. The open channel admits calcium as well as other ions, allowing electrical activity to initiate changes inside the cell.

Calcium can activate enzymes that alter synaptic strength and receptor delivery. Glutamate also acts on **metabotropic receptors**, which use G proteins rather than containing the ion channel themselves. Drugs can act differently at these receptor families. Ketamine, for example, blocks NMDA receptor channels, changing one part of glutamate signaling rather than eliminating every response to the transmitter.



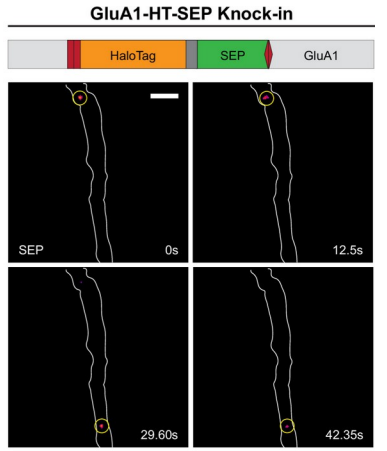
Carlson and Birkett (2022), Fig. 4.9, p. 104.

Receptor vesicles fuse near a stimulated synapse

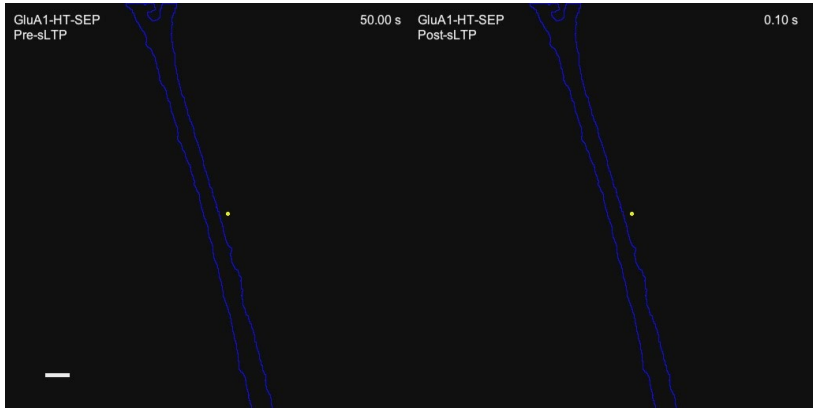
Receptor delivery is completed when a cargo vesicle fuses with the neuronal membrane, a process called **exocytosis**. The AMPA-receptor researchers used a fluorescent tag that was dim inside an acidic vesicle and bright after exposure to the fluid outside the cell. Each brief brightening identified a receptor-insertion event at the surface.

Synaptic stimulation increased these insertion events, especially near the stimulated spine. Nearby vesicles had already become more restricted in their movement through the dendrite. The neuron could therefore draw on a local supply of receptor cargo while adding receptor proteins to its surface. Delivery changed most strongly in the region where the synapse had been activated.

Disrupting actin assembly or inhibiting myosin reduced the increase in receptor insertion. Actin helped retain cargo near the active region, while myosin contributed to the later delivery process. A stronger synaptic response can consequently involve movement and membrane insertion of receptor proteins, in addition to changes in how receptors already at the synapse function.



Wong et al. (2024), Fig. 5 (A, image sequence). AMPA receptor insertion at the cell surface.



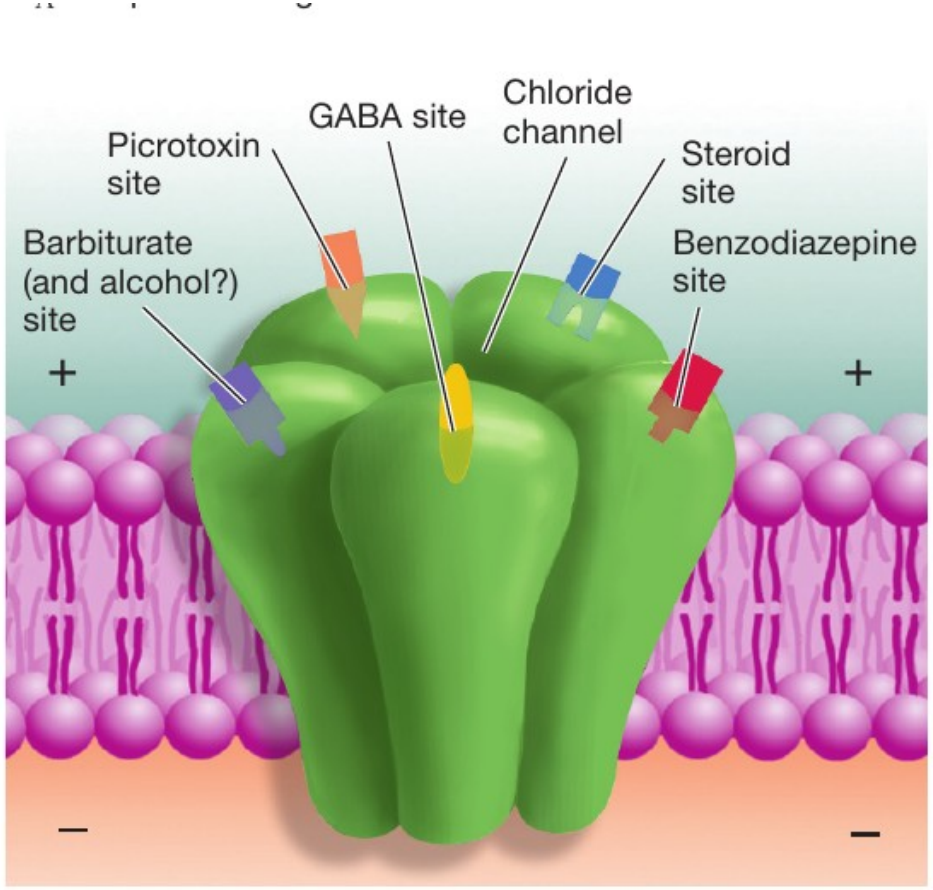
Wong et al. (2024), fig5-video3. Receptor insertion before and after local stimulation.

GABA controls excitability through channels and G proteins

GABA is a major inhibitory neurotransmitter in the mature brain. It is produced from glutamate by an enzyme inside GABA-releasing neurons. After release, GABA can activate **GABA-A receptors**, which contain channels permeable mainly to chloride. Opening these channels changes the membrane's electrical behavior and usually reduces the ability of excitatory inputs to trigger firing.

The effect depends on the chloride gradient, the difference in chloride concentration across the membrane. Opening the channel moves voltage toward the value favored by that gradient. In mature neurons, the resulting current commonly opposes excitation or makes it harder for other inputs to change voltage enough to generate an action potential.

GABA-B receptors work differently because they activate G proteins instead of directly containing the chloride channel. Their signaling can open potassium channels or reduce calcium entry. At a releasing terminal, less calcium entry can reduce transmitter release. GABA can therefore suppress communication through both an immediate channel response and slower changes produced through intracellular signaling.



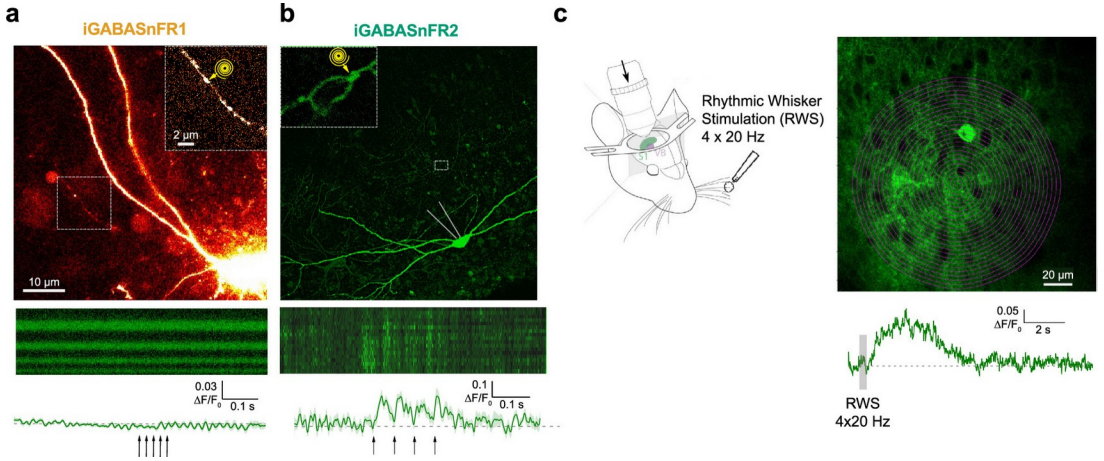
Carlson and Birkett (2022), Fig. 4.10, p. 105.

A fluorescent sensor detects extracellular GABA release

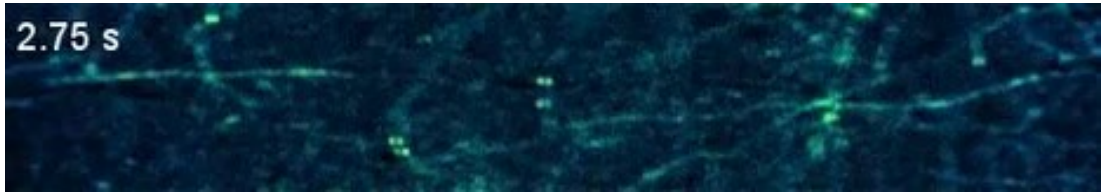
GABA release can be measured using a sensor that becomes brighter when GABA binds to it. **iGABASnFR2** is an engineered fluorescent sensor placed on the outside of cell membranes. Unlike a receptor that changes electrical activity, its main experimental function is to report nearby extracellular GABA through a change in fluorescence.

In brain slices, researchers triggered bursts of action potentials in a GABA-releasing neuron while recording fluorescence near its axon terminals. The newer sensor detected signals at individual release sites more effectively than the earlier version. The fluorescence increase followed transmitter release near the terminal, rather than simply reporting calcium changes inside the releasing neuron.

The researchers also stimulated mouse whiskers while imaging the **barrel cortex**, a cortical region that processes whisker information. Whisker stimulation produced a brief extracellular GABA signal. Released GABA can diffuse away from individual terminals, so the recording measured a chemical response across nearby tissue as sensory information entered the cortical network.



Kolb et al. (2026), Fig. 6 (a–c). Fluorescent detection of extracellular GABA.



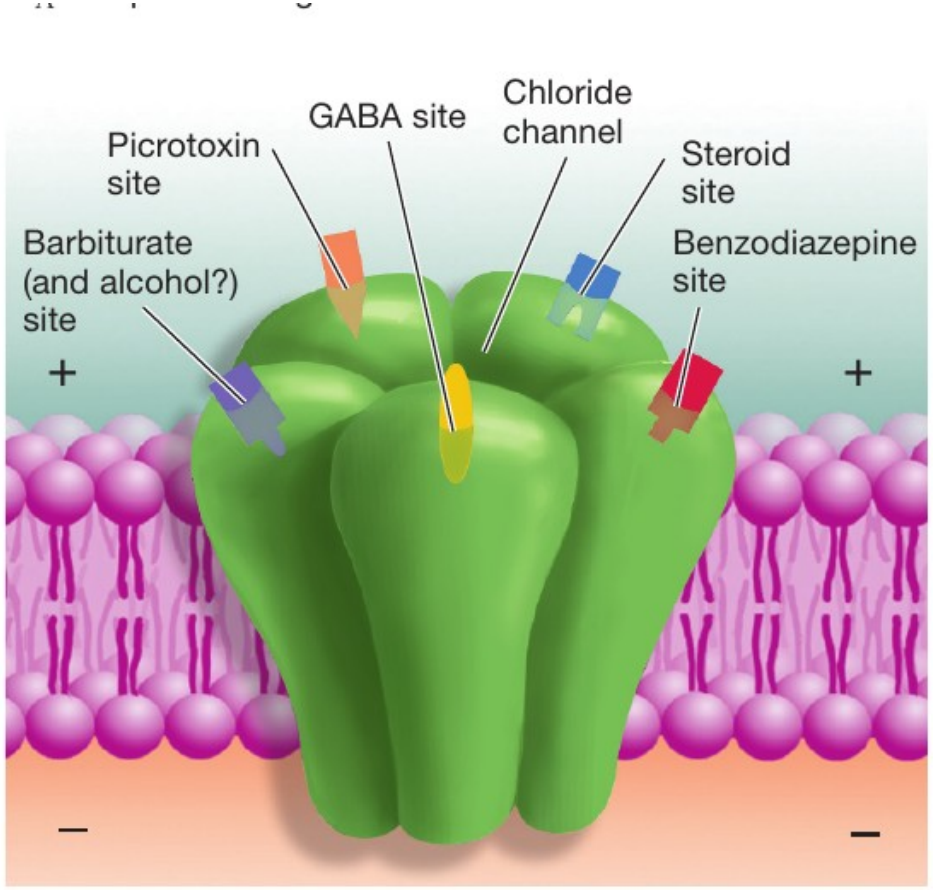
Kolb et al. (2026), fig6-video1. Whisker-evoked extracellular GABA fluorescence.

Allosteric drugs modify how a receptor responds

An allosteric drug changes a receptor's response by binding outside the transmitter's own binding site. The GABA-A receptor has several such sites. **Benzodiazepines** enhance the receptor's response to GABA, increasing the inhibitory influence of GABAergic signaling. They do not simply replace GABA at its usual binding location on the receptor.

Barbiturates act at a different site and can also increase GABA-A channel activity. Because the receptor contains several interacting binding regions, different drugs can modify the same channel through different molecular contacts. The resulting inhibition depends on which substance is present, its concentration, and how it changes the receptor's opening behavior.

The receptor's chloride channel can also be blocked. **Picrotoxin** interferes with channel function, reducing the inhibitory current even when GABA is available. Enhancing a receptor's response and blocking its channel therefore push signaling in opposite directions. These actions change communication by modifying a protein already present in the receiving membrane, rather than changing GABA synthesis or release.



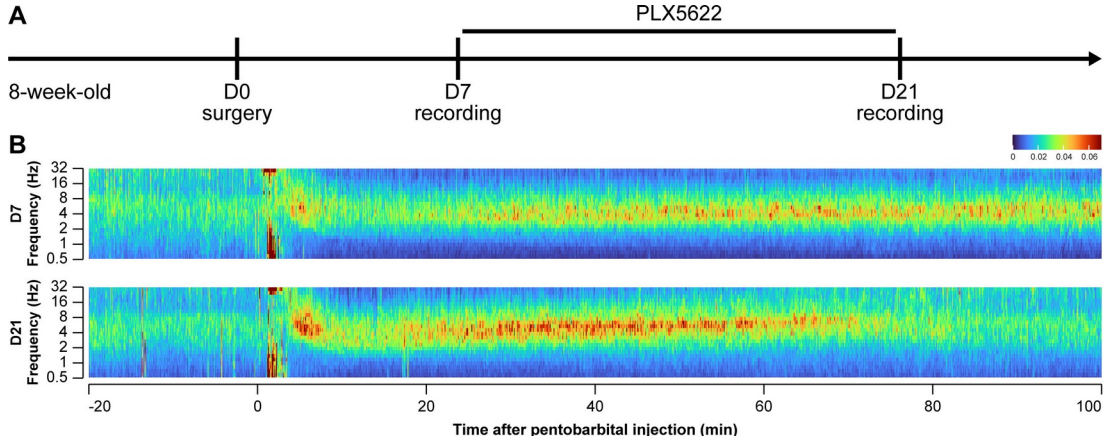
Carlson and Birkett (2022), Fig. 4.10, p. 105.

Anesthesia alters brain activity as well as movement

An animal that stops moving has not necessarily entered the same brain state as an anesthetized animal. The microglia researchers therefore recorded **electroencephalography**, or **EEG**, which measures electrical activity from the brain, alongside **electromyography**, or **EMG**, which measures muscle activity. These recordings followed changes in brain rhythms and movement during anesthetic induction and emergence.

After microglial depletion, the changes associated with anesthesia began later and ended earlier. Muscle activity also disappeared later and returned sooner. Before anesthetic administration, the measured awake EEG and muscle activity were not strongly altered in the same way. The shortened anesthetic period therefore appeared during the response to drug exposure rather than as continuous behavioral inactivity.

EEG recordings contain activity at different frequencies, allowing the balance of brain rhythms to be followed over time. That frequency pattern changed during induction and recovery. Comparing it with muscle activity and the righting reflex gave several measurements of the same transition, including a brain signal that did not depend solely on whether the animal moved.



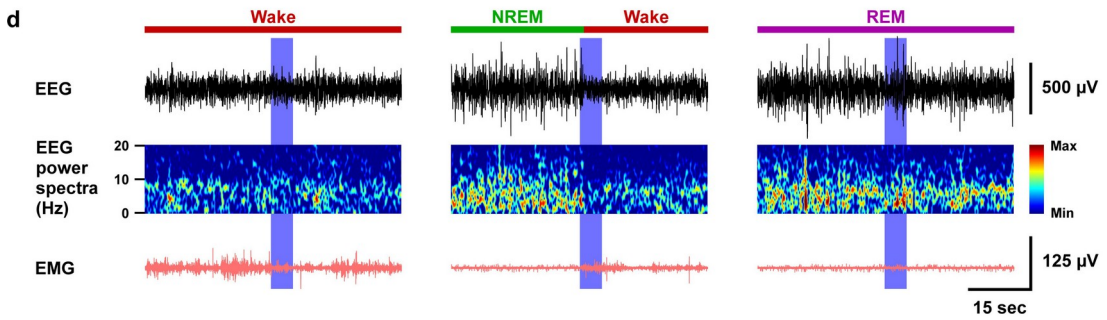
He et al. (2023), Fig. 4 (A–B). Brain electrical activity during anesthesia.

GABAergic neurons contribute to the control of wakefulness

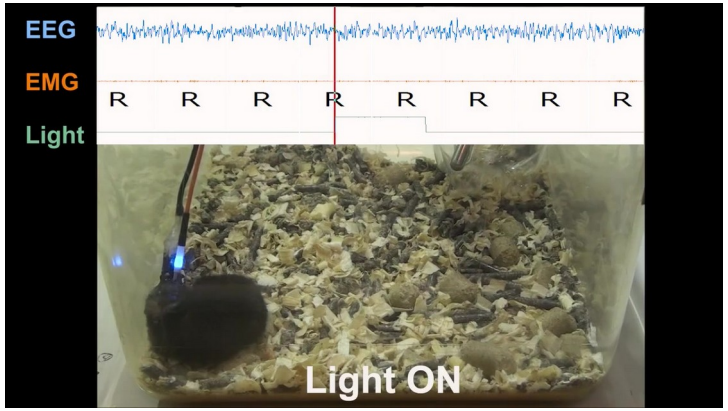
Some GABA-releasing neurons help maintain sleep by inhibiting cells that promote wakefulness. Researchers examined such neurons in the **ventral tegmental area**, or **VTA**, a midbrain region containing several neuronal types. They identified a GABAergic population using **GAD67**, an enzyme involved in GABA production, and manipulated those cells while monitoring sleep and wake states.

The researchers expressed a light-sensitive inhibitory channel in these neurons. During **non-REM sleep**, a sleep state containing prominent slow brain rhythms, briefly opening the channel suppressed the targeted neurons and rapidly produced wakefulness. The same manipulation during REM sleep did not produce the same immediate transition. Its effect depended on the animal's starting sleep state.

These VTA neurons also sent inhibitory input to **orexin neurons** in the lateral hypothalamus, which help promote wakefulness. Activating their terminals reduced orexin-neuron firing. Blocking GABA-A receptors prevented the measured inhibitory synaptic response. GABA release along this pathway can therefore suppress a wake-promoting population, helping explain how an inhibitory neuron can promote sleep.



Chowdhury et al. (2019), Fig. 3 (d). State-dependent EEG and muscle responses.



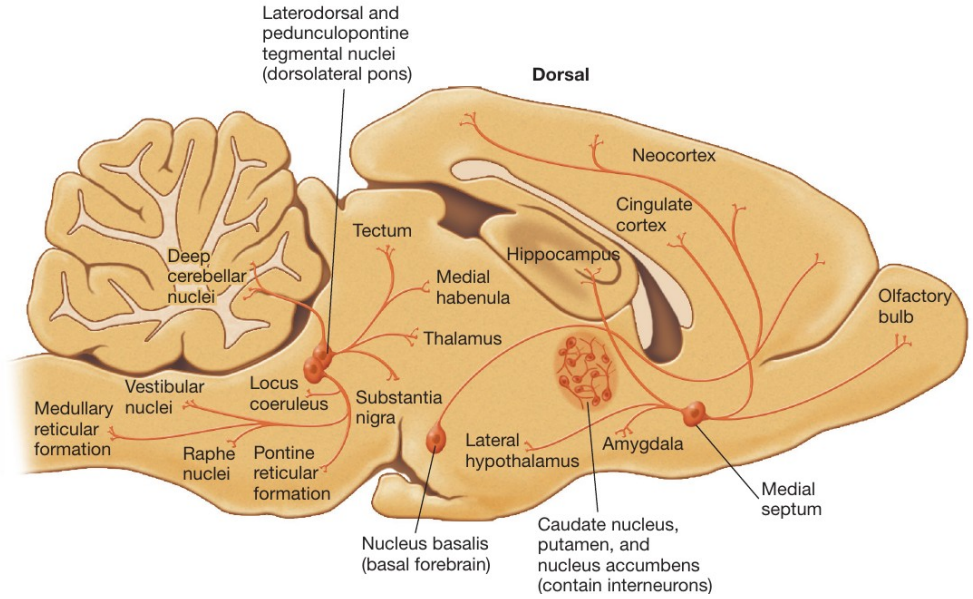
Chowdhury et al. (2019), video1. Inhibiting GABA neurons across sleep and wake states.

Acetylcholine acts through two major receptor families

Acetylcholine, or **ACh**, is released onto skeletal muscles and within several brain pathways. Neurons in the basal forebrain send cholinergic axons to the cortex, while other groups supply additional regions. **Cholinergic** means that a cell releases acetylcholine. The response depends on the receptor family expressed by the receiving cell.

Nicotinic receptors contain ion channels that open when acetylcholine binds. At the skeletal neuromuscular junction, positive ions enter through these channels and help initiate muscle activation. **Muscarinic receptors** instead activate G proteins, producing slower changes in channels and enzymes. Acetylcholine can therefore change cellular activity through either a direct channel opening or an intracellular signaling pathway.

The enzyme **acetylcholinesterase** rapidly breaks acetylcholine into choline and acetate after release. Choline can be taken back into the terminal and reused to make new transmitter. Drugs that inhibit this enzyme prolong acetylcholine's availability. They change the duration of signaling without having to activate either receptor family directly or increase the neuron's initial transmitter release.



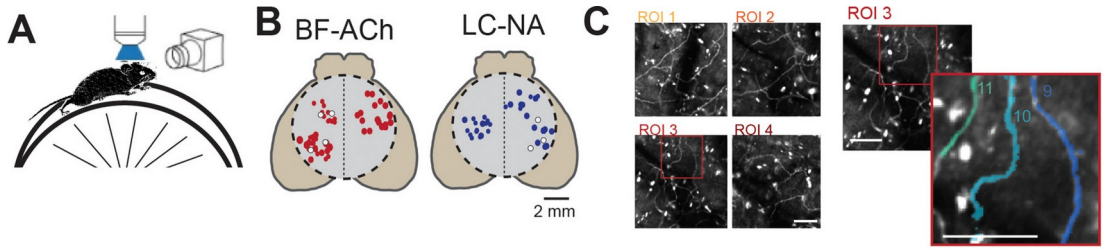
Carlson and Birkett (2022), Fig. 4.11, p. 107.

Cholinergic axons respond during behavioral engagement

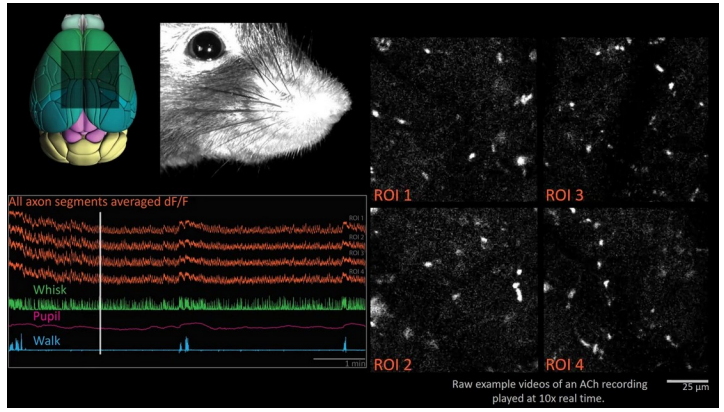
Acetylcholine release in the cortex is supplied partly by neurons whose cell bodies lie in the basal forebrain. Researchers recorded from their cortical axons in awake mice using a fluorescent calcium indicator. Calcium commonly rises when an axon is active, so changing fluorescence allowed activity in these thin neuronal processes to be followed during behavior.

The mice could remain still, move their whiskers, or walk on a wheel while their heads stayed in position for imaging. The researchers measured these movements along with pupil size. Cholinergic axon activity increased during periods containing whisker and facial movement, including periods when the animal was not walking around on the wheel.

Several cortical regions could be recorded during the same session. Many cholinergic axons became active together as the mouse moved and became more behaviorally engaged. The measurement reported calcium inside axons, not acetylcholine concentration outside them. It therefore described when the transmitter-supplying fibers were active while the animal changed its behavior.



Collins et al. (2023), Fig. 1 (A–C). Recording labeled cholinergic and noradrenergic axons.



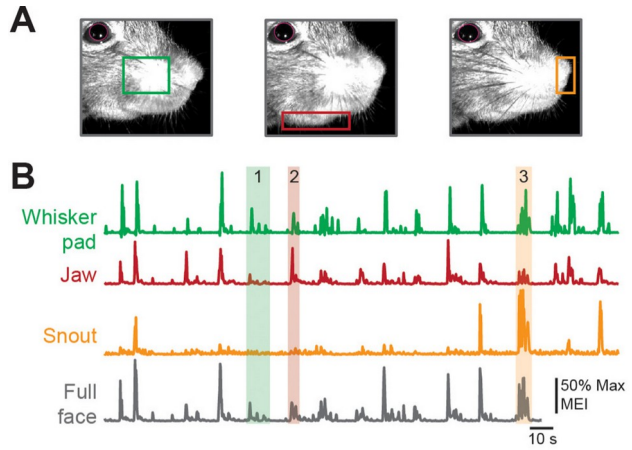
Collins et al. (2023), video1. Axon imaging alongside behavior measurements.

Two transmitter systems track facial movement

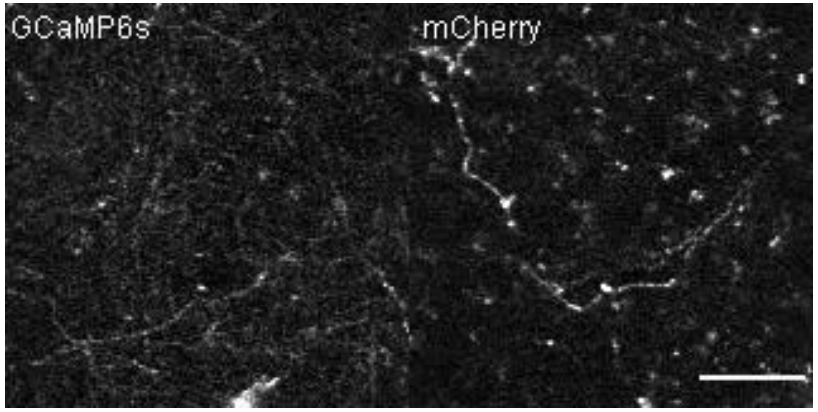
The same experiment examined cortical axons that supply **norepinephrine**, also called noradrenaline. Their cell bodies lie in the **locus coeruleus**, a small brainstem region. Both these axons and the cholinergic axons became more active during behavioral engagement. The researchers followed the systems separately while measuring the mouse's facial movements, pupils, and walking.

Whisker-pad movement tracked axonal activity especially closely. Activity could increase during brief facial movements even when the animal did not begin walking. Measuring only locomotion would therefore miss part of the behavioral change occurring alongside these signals. The face supplied a more continuous record of small movements across quiet and more active periods.

A calcium-sensitive fluorescent indicator changed brightness during axonal activity, while a comparison fluorescent protein without calcium sensitivity did not change in the same way. This comparison helped distinguish activity-related fluorescence from simple movement of the preparation. The signal being measured was a change inside the labeled axons occurring alongside the animal's changing behavioral state.



Collins et al. (2023), Fig. 3 (A–B). Face tracking and behavioral measurements.



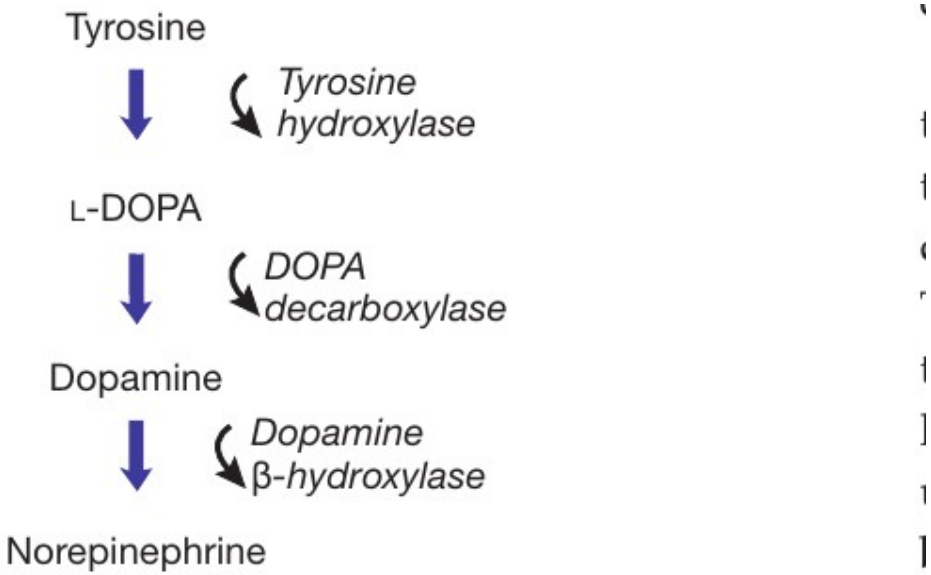
Collins et al. (2023), video2. Activity-sensitive and reference fluorophore recordings.

Catecholamine synthesis creates a sequence of drug targets

Catecholamines are a group of related chemical messengers that includes dopamine, norepinephrine, and epinephrine. Neurons build them through a sequence of enzyme-controlled reactions. The amino acid **tyrosine** is first changed into **L-DOPA** by tyrosine hydroxylase. Another enzyme then converts L-DOPA into dopamine, which can be packaged into vesicles and released.

In norepinephrine-producing cells, an additional enzyme converts dopamine into norepinephrine. The final transmitter therefore depends on which enzymes the cell makes. Blocking an early reaction can reduce the supply of several later products, whereas blocking a later reaction affects a narrower part of the sequence. A synthetic pathway consequently contains several distinct places where a drug can act.

L-DOPA is used to increase dopamine production in Parkinson's disease because it can enter the brain more readily than dopamine itself. Once inside, it can be converted into dopamine by the remaining enzymatic step. Giving a precursor increases the material available for transmitter synthesis, rather than activating dopamine receptors directly in the way a receptor agonist does.



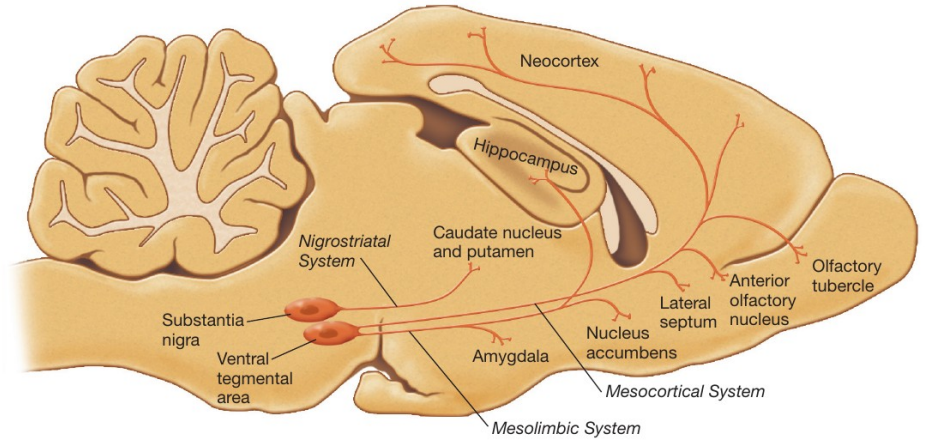
Carlson and Birkett (2022), Fig. 4.15, p. 110.

Dopamine pathways influence different functions

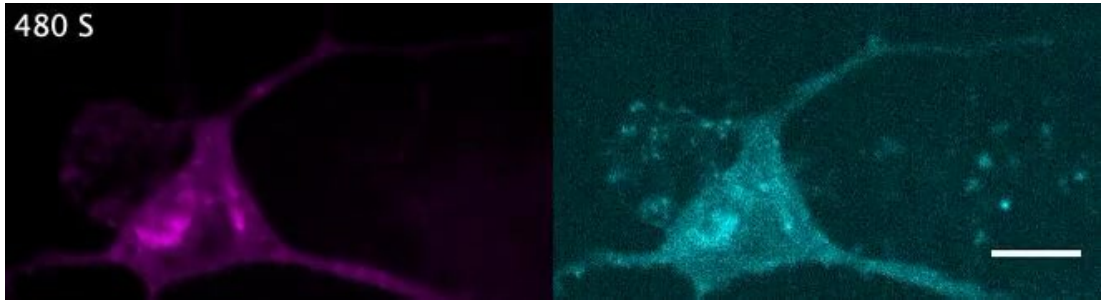
Dopamine reaches different brain regions through distinct groups of projecting neurons. The **nigrostriatal pathway** runs from the substantia nigra to the dorsal striatum and helps regulate movement. Loss of many dopamine-producing neurons in this pathway contributes to the difficulty initiating and controlling movement in Parkinson's disease. Reduced transmitter supply changes how the receiving striatal circuits operate.

Other dopamine neurons begin in the ventral tegmental area. **Mesolimbic** projections reach regions involved in motivation and reinforcement, including the nucleus accumbens. **Mesocortical** projections reach the cerebral cortex, including prefrontal regions involved in planning and cognitive control. Dopamine can therefore influence different functions depending on which cells release it and which circuits receive their axons.

Dopamine receptors also differ in how they change intracellular signaling. D1-like receptors generally increase cAMP production through G proteins, while D2-like receptors generally reduce it. These receptors occur in different cell populations and locations. A drug that reaches dopamine-sensitive tissue can consequently change several pathways, with effects shaped by the receptors and cells it activates.



Carlson and Birkett (2022), Fig. 4.14, p. 109.



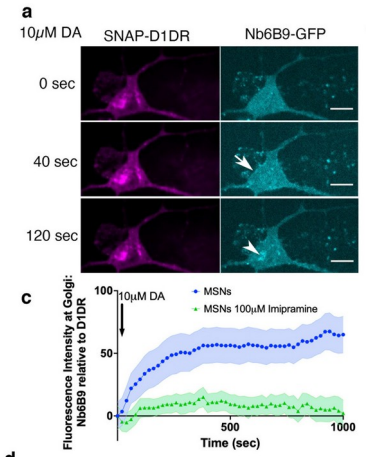
Puri et al. (2022), fig3-video1. Dopamine response in striatal neurons.

Dopamine activates Golgi receptors in striatal neurons

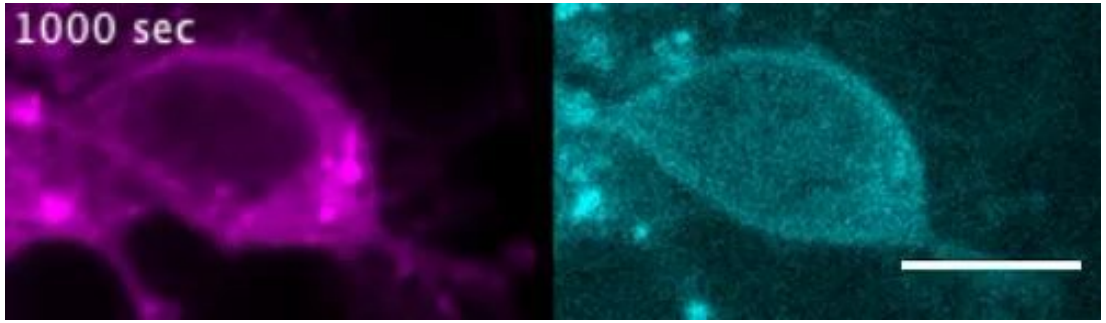
Dopamine receptors inside a neuron can respond as well as those on its outer membrane. Puri and colleagues examined **medium spiny neurons**, a major neuronal population in the striatum. These cells had D1 receptors at the surface and near the Golgi apparatus, an internal structure involved in processing and sorting proteins.

When dopamine was added outside the cultured neurons, the fluorescent activation detector moved toward receptors at both locations. The neurons also contained OCT2 transporters at the cell surface and Golgi. This transport protein supplied a route by which dopamine could reach an internal receptor population rather than acting only on receptors exposed directly to the surrounding fluid.

Inhibiting OCT2 reduced dopamine's activation of Golgi receptors, while the membrane-permeant agonist SKF81297 still activated them. A separate detector also detected receptor interaction with G proteins at the Golgi. The internal receptors could therefore initiate signaling inside these neurons, and their response depended partly on whether the applied chemical could reach that compartment.



Puri et al. (2022), Fig. 3 (a, c). D1 receptor activity in striatal neurons.



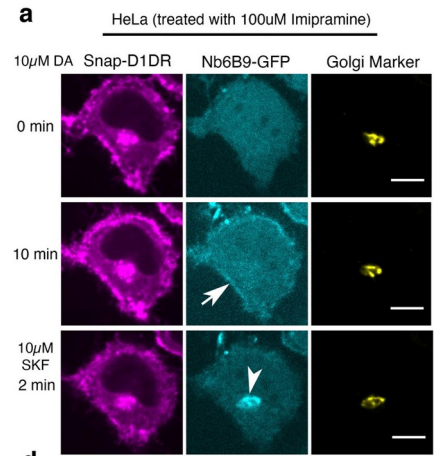
Puri et al. (2022), fig3-video2. Striatal D1 response during transporter inhibition.

Transporter reduction alters dopamine signaling

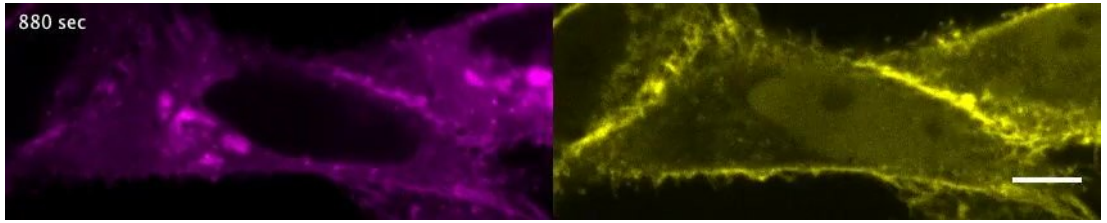
The amount of OCT2 a cell produces affects dopamine access to internal receptors. The gene **SLC22A2** contains the instructions for making OCT2. Researchers reduced its expression using targeted RNA molecules, leaving cells with less transporter protein. They then tested whether dopamine could still activate the D1 receptor population located at the Golgi.

After transporter reduction, dopamine produced less internal receptor activation. The membrane-permeant agonist SKF81297 retained its effect, so the receptors could still respond when an activating compound reached them. The change concerned delivery across membranes rather than a complete loss of receptor function. Surface receptors remained accessible from the fluid outside the cell.

The researchers checked transporter production as well as receptor responses. Two targeted RNA sequences reduced OCT2 expression, while a comparison sequence did not produce the same pattern. Reducing the protein's production and blocking its transport activity both weakened dopamine's access to Golgi receptors. The cell's transporter supply therefore changed where the same transmitter could initiate signaling.



Puri et al. (2022), Fig. 2 (a). Dopamine response during transporter inhibition.



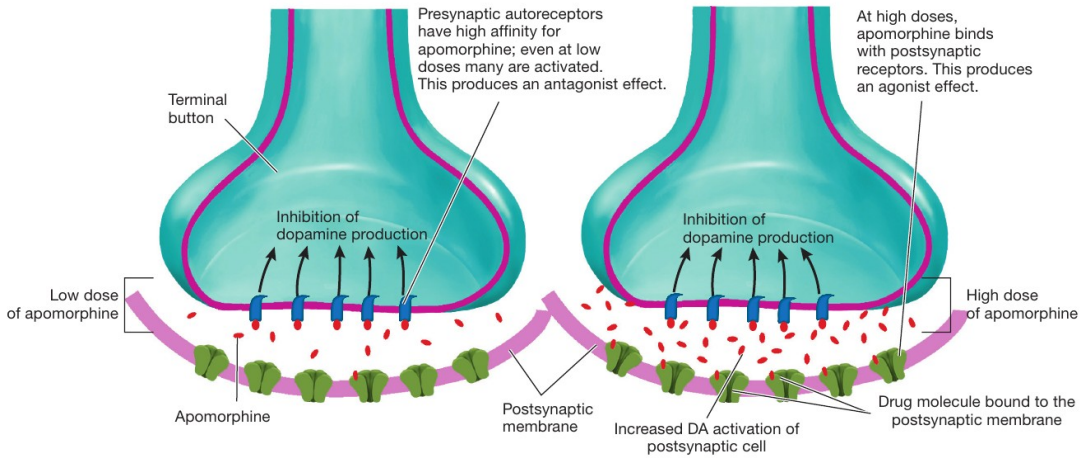
Puri et al. (2022), fig2-video3. D1 signaling after OCT2 reduction.

Autoreceptors let a neuron regulate its own output

An **autoreceptor** is a transmitter receptor located on the same kind of neuron that releases that transmitter. Dopamine can activate D2 autoreceptors on dopamine-producing neurons and their terminals. This feedback reduces further dopamine production or release, so an increase in available transmitter can limit the neuron's subsequent chemical output.

The drug **apomorphine** activates dopamine receptors, but its behavioral effect changes with dose. At low doses, activation of sensitive autoreceptors can reduce dopamine output. Less transmitter then reaches other dopamine-sensitive cells. A receptor agonist can therefore decrease a downstream response by activating inhibitory feedback on the releasing neuron rather than blocking the receiving cell's receptors.

At higher doses, apomorphine also strongly activates receptors on other dopamine-responsive cells. These direct effects can produce a different behavioral response from the low-dose reduction in release. The drug has not changed from an agonist into a different chemical. Increasing exposure has changed which receptor populations are recruited strongly enough to influence the observed response.



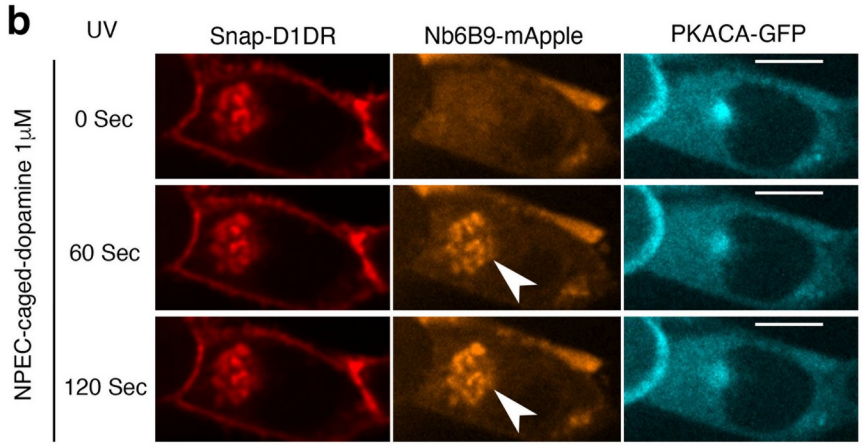
Carlson and Birkett (2022), Fig. 4.16, p. 111.

Local dopamine release activates a compartmental signal

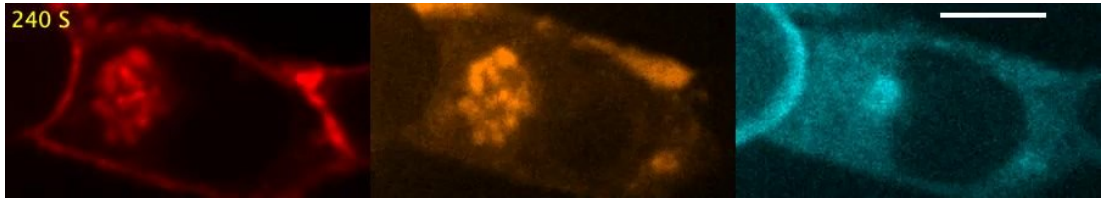
Researchers can control where an active chemical becomes available by first applying an inactive, light-sensitive form. In the D1 study, **caged dopamine** crossed cell membranes but did not activate the receptors before illumination. A brief light exposure removed its chemical cage, releasing active dopamine after the compound had entered the cells.

Following illumination, a fluorescent detector accumulated at activated Golgi D1 receptors. This was followed by a change in nearby **protein kinase A**, or **PKA**, an enzyme activated through cAMP signaling. Its active components moved away from their anchored location. The recording therefore followed receptor activation and a subsequent intracellular response after dopamine was released locally.

The order of these changes depended on the signaling pathway. Dopamine activated D1 receptors, the receptors stimulated G proteins and cAMP production, and cAMP activated PKA. The chemical signal was converted into enzyme activity inside the cell. Releasing dopamine after membrane entry bypassed the transport problem that otherwise limited its access to the internal receptor population.



Puri et al. (2022), Fig. 6 (b). Intracellular dopamine release and receptor activity.



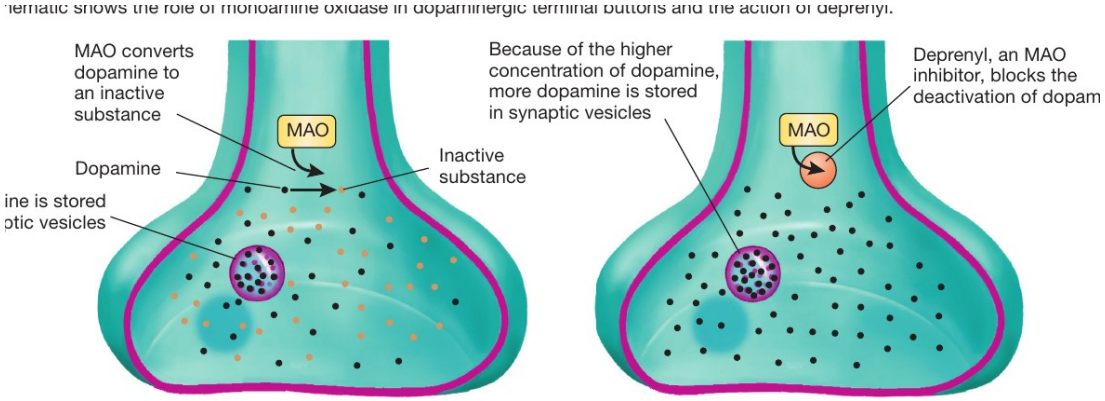
Puri et al. (2022), fig6-video1. D1 receptor and PKA responses to uncaged dopamine.

Reuptake and enzymes terminate monoamine signals

After a monoamine transmitter is released, **reuptake transporters** move much of it back into cells. Dopamine transporters clear dopamine, while other transporters handle norepinephrine or serotonin. Removing transmitter from the extracellular fluid reduces the amount available to activate receptors. The duration of a chemical message therefore depends partly on how quickly these transporters clear it.

A reuptake inhibitor blocks this removal step. Cocaine, for example, interferes with monoamine transporters and prolongs transmitter availability outside cells. The receiving receptors can continue encountering transmitter that would otherwise have been taken up. This differs from a receptor agonist, which activates a receptor directly rather than preserving the neuron's own released messenger.

Inside cells, **monoamine oxidase**, or **MAO**, helps break down monoamines. Inhibiting MAO reduces this chemical destruction and can increase the transmitter available for later signaling. Reuptake and breakdown thus act at different stages: one moves transmitter across a membrane, while the other changes its chemical structure. Drugs affecting either stage can prolong or increase monoamine signaling.



Carlson and Birkett (2022), Fig. 4.17, p. 112.

Dopamine helps coordinate several motor programs

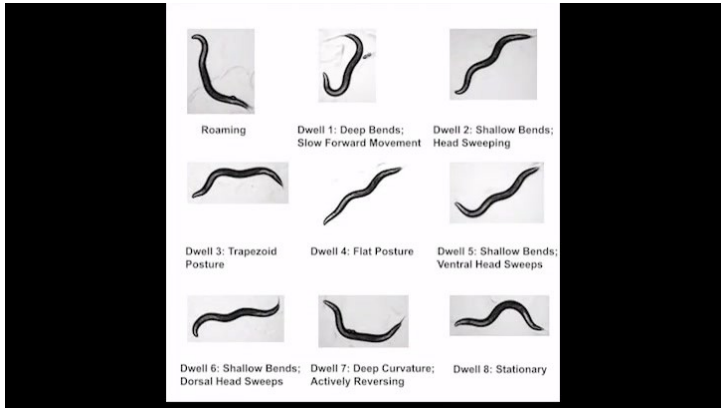
An animal often needs to coordinate several actions at once, such as moving, feeding, and laying eggs. Cermak and colleagues recorded these behaviors in **Caenorhabditis elegans**, a small nematode worm. Their tracking system followed individual animals as they moved across food, allowing several motor activities to be measured together over extended periods.

The worms alternated between **roaming**, with more extensive forward travel, and **dwelling**, with slower movement and frequent local changes in posture. Feeding, body movements, and egg-laying did not occur independently of these states. Egg-laying was especially frequent during roaming, so reproductive movements were coordinated with the way the animal explored its food source.

Dopamine contributed to this coordination. Worms with impaired dopamine production or selected dopamine receptors lost much of the normal increase in egg-laying during roaming. They could still perform egg-laying and locomotion, but the timing between them changed. Dopamine therefore helped regulate when one motor program occurred in relation to another during the animal's ongoing behavior.



Cermak et al. (2020), Fig. 1 (A–B). Worm behavior recording and animal.



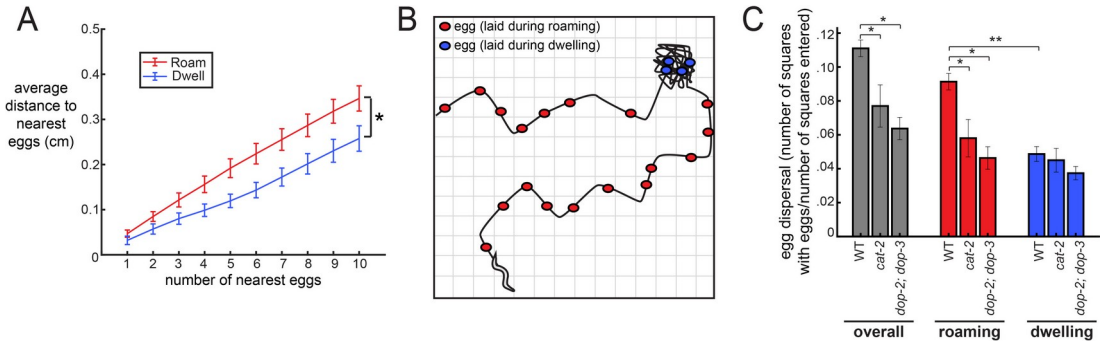
Cermak et al. (2020), video1. Worm locomotor and egg-laying states.

Dopamine couples egg laying to a locomotor state

Egg-laying during roaming spreads eggs across a larger area because the worm continues moving between laying events. During dwelling, repeated events occur closer together. The researchers mapped where eggs were deposited along each animal's path and found greater dispersal during roaming. Dopamine-deficient worms showed reduced egg dispersal alongside the altered timing of egg-laying.

Roaming increased egg-laying beyond what could be explained by movement speed alone. At comparable speeds, worms still laid eggs more frequently during roaming than during dwelling. The difference concerned the animal's behavioral state as well as how fast it traveled. Dopamine signaling helped maintain this state-dependent increase in the probability of laying an egg.

Briefly activating dopamine neurons increased egg-laying, while silencing them reduced egg-laying especially during roaming. These changes occurred during the manipulation rather than requiring a lifetime without dopamine. The neurons were therefore regulating the ongoing behavior. They changed the frequency of reproductive events while the worm continued to coordinate movement and other activities around the food source.



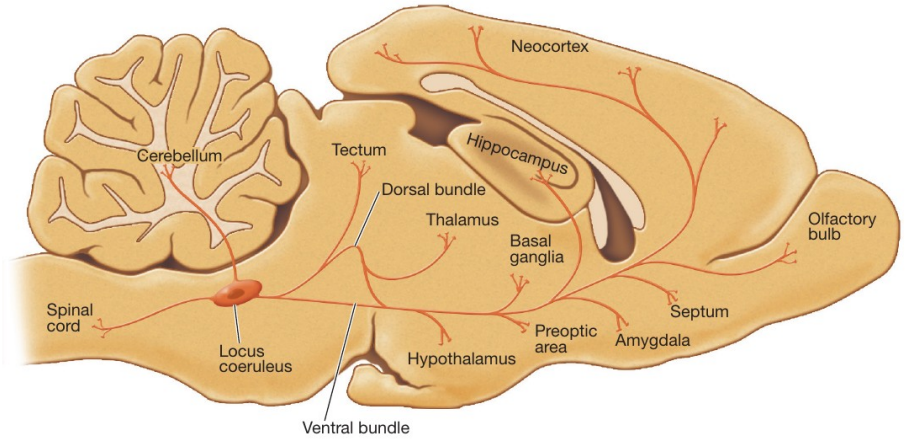
Cermak et al. (2020), Fig. 4 (A–C). Egg-laying and locomotor-state measurements.

Norepinephrine acts through several receptor types

Norepinephrine, also called **noradrenaline**, is released by neurons in both the brain and peripheral nervous system. In the brain, many norepinephrine-containing axons originate in the locus coeruleus and reach widespread targets. Changing activity in these neurons can alter signaling across several regions involved in arousal, attention, and responses to the environment.

Norepinephrine acts through **alpha** and **beta adrenergic receptors**, all of which use G proteins. These receptor families produce different cellular responses. Alpha2 receptors can inhibit transmitter release when they act as autoreceptors, while other receptor types change the activity of receiving cells. The effect depends on the receptor expressed at the location reached by the transmitter.

The destination of an axon also changes the function of its signal. A norepinephrine pathway entering the spinal cord can alter pain processing, while projections to the cortex influence other aspects of behavior. Drugs that increase norepinephrine availability may affect several of these pathways together. Selectively manipulating one projection allows its contribution to be examined more directly.



Carlson and Birkett (2022), Fig. 4.18, p. 112.



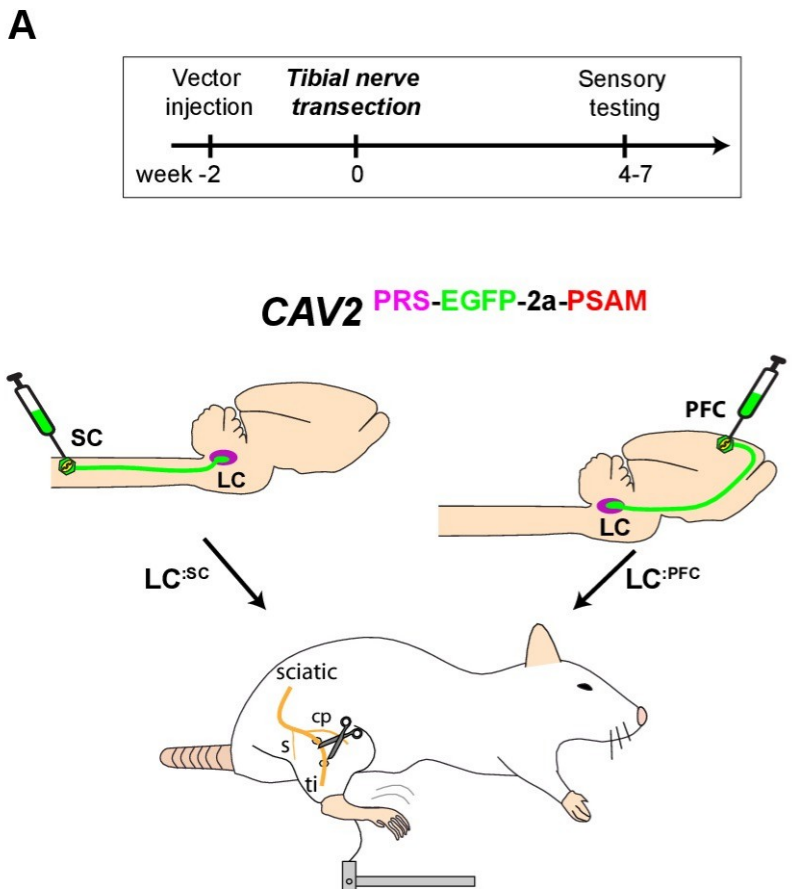
Hirschberg et al. (2017), video1. Pain behavior during projection-targeted activation.

A noradrenergic projection suppresses pain behavior

Hirschberg and colleagues targeted locus coeruleus neurons whose axons reached the spinal cord. The researchers gave these neurons an engineered ion channel that opened in response to a particular compound. Opening the channel increased neuronal firing, allowing the spinal projection to be activated without stimulating every norepinephrine-producing neuron in the brain at once.

In rats with peripheral nerve injury, activating this pathway reduced sensitivity to mechanical and cold stimulation. It also reduced spontaneous lifting of the affected foot and improved weight-bearing on the injured limb. The rats retained their performance on a movement-coordination task. The behavioral changes therefore included reduced pain-related responses while general motor performance remained available.

Norepinephrine acted through spinal **alpha2 receptors** to produce this effect. Blocking those receptors with yohimbine prevented the reduction in hypersensitivity. The reuptake inhibitor **reboxetine** instead strengthened and prolonged the response by keeping released norepinephrine available for longer. Activation increased transmitter delivery, while reuptake inhibition extended the period during which that transmitter could act.



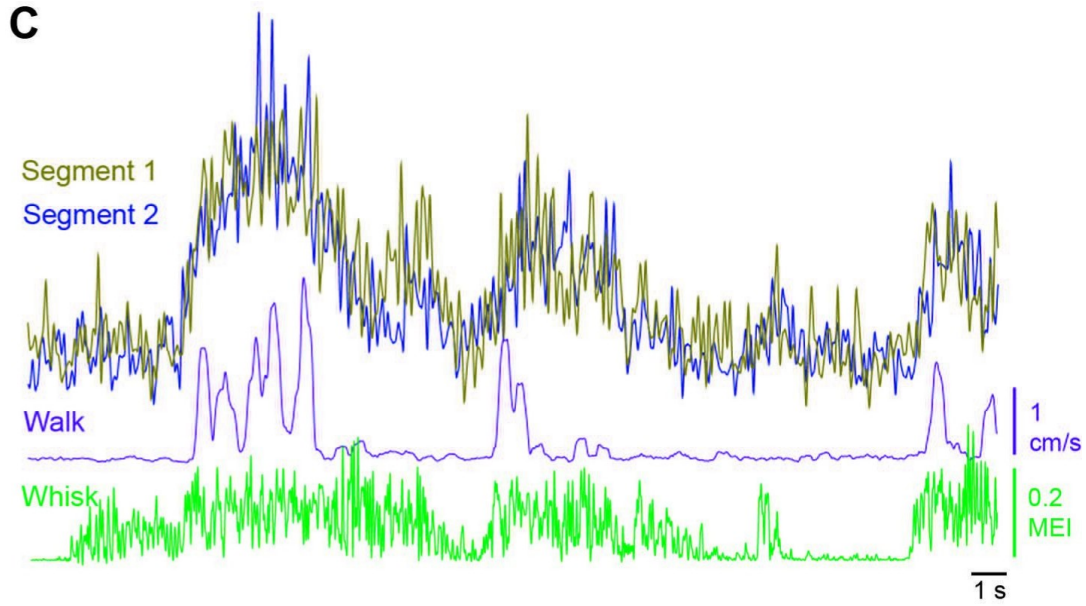
Hirschberg et al. (2017), Fig. 5 (A). Noradrenergic projection-targeting experiment.

Neuromodulatory axons carry a broad cortical state signal

Individual cholinergic and noradrenergic axons can become active together across different cortical regions. In the awake-mouse imaging study, many axon segments shared slow changes in their calcium signals during facial movement and other behavioral activity. This common pattern was present across a broad part of the cortex, rather than being restricted to one small recording site.

The researchers also compared different segments of the same axon. Signals within one axon provided a reference for how closely measurements could match when they came from a single neuronal process. Separate axons shared substantial activity, but their signals were not identical. Some differences remained between fibers even after the activity common to the population was considered.

The shared signal tracked changes in behavior, including whisker movement, while individual axons could also carry additional activity. A widespread transmitter system can therefore combine a broad change in behavioral state with differences between local fibers. Measuring several regions at once helps distinguish activity distributed across the cortex from changes confined to a particular axon or location.



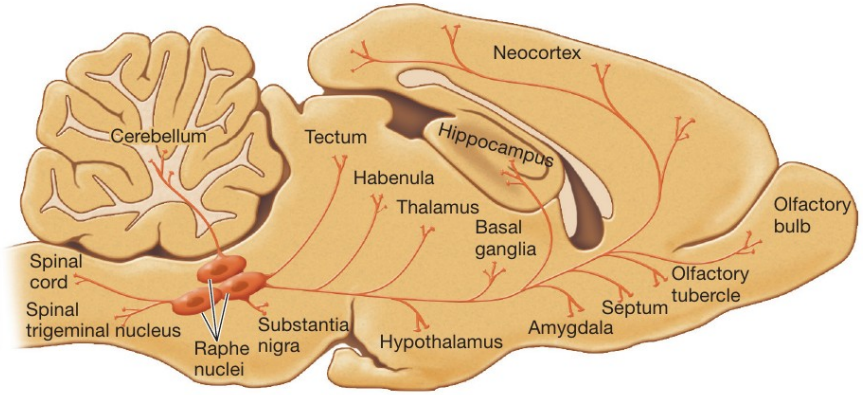
Collins et al. (2023), Fig. 4 (C). Activity along individual labeled axons.

Serotonin acts through several widespread pathways

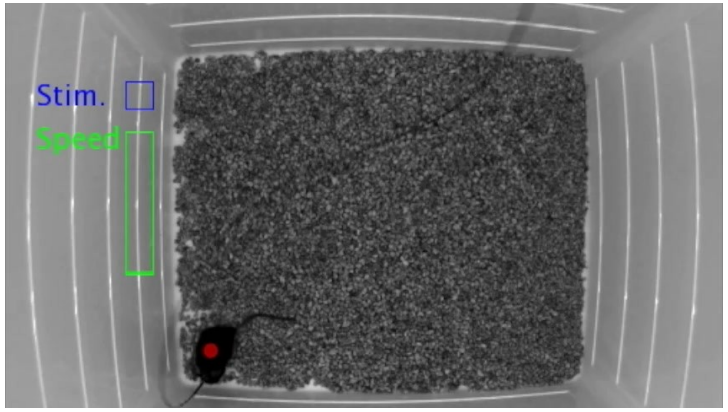
Serotonin, also called **5-hydroxytryptamine** or **5-HT**, is made from the amino acid tryptophan. Many serotonin-producing neurons lie in the **raphe nuclei**, groups of cells in the brainstem. Their axons reach different brain and spinal targets, allowing serotonin to influence functions that include movement, feeding, sleep, and pain processing through distinct receiving circuits.

Most serotonin receptors act through G proteins, while the **5-HT3 receptor** contains an ion channel. Different receiving cells therefore respond through different molecular processes. Activating one serotonin pathway can change a behavior differently from activating another. The transmitter's effect depends on the receptor and circuit involved, rather than producing the same response throughout the nervous system.

The **serotonin transporter**, or **SERT**, removes serotonin from the extracellular fluid after release. **Selective serotonin reuptake inhibitors**, or **SSRIs**, block this transporter and prolong the availability of released serotonin. Their immediate molecular action is reduced clearance. Serotonin synthesis, neuronal firing, and receptor activation remain separate processes that can also change the amount and effect of serotonergic signaling.



Carlson and Birkett (2022), Fig. 4.19, p. 113.



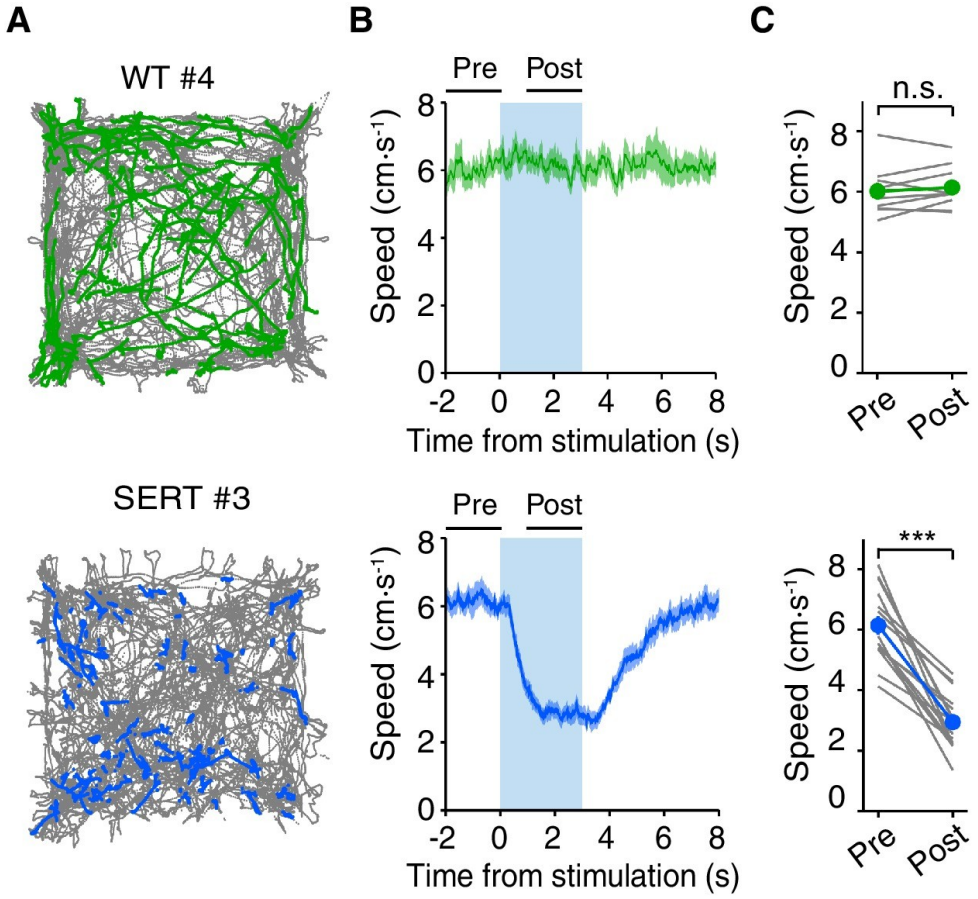
Correia et al. (2017), media1. Mouse movement during serotonin-neuron stimulation.

Serotonin-neuron activation changes movement across time

Researchers activated serotonin neurons in the **dorsal raphe nucleus**, a brainstem group that sends axons to other regions. The neurons expressed a light-sensitive protein, so brief illumination increased their activity. When mice explored an open arena, this stimulation rapidly reduced their walking speed. Most of the decrease developed within about a second after stimulation began.

The mice could still perform movements when the task demanded them. Stimulation did not produce the same slowing when the animals crossed a narrow track toward a destination, and performance on a rotating-rod coordination task remained intact. The reduced spontaneous walking in the open arena therefore differed from a general inability to coordinate or generate movement.

Repeated daily stimulation produced a different change over several weeks: the animals' usual movement speed gradually increased. This increase was present even during intervals without stimulation. The immediate slowing and the longer-term increase occurred over different timescales. Repeated activation had changed the background level of locomotion while brief stimulation still produced its rapid effect during individual trials.



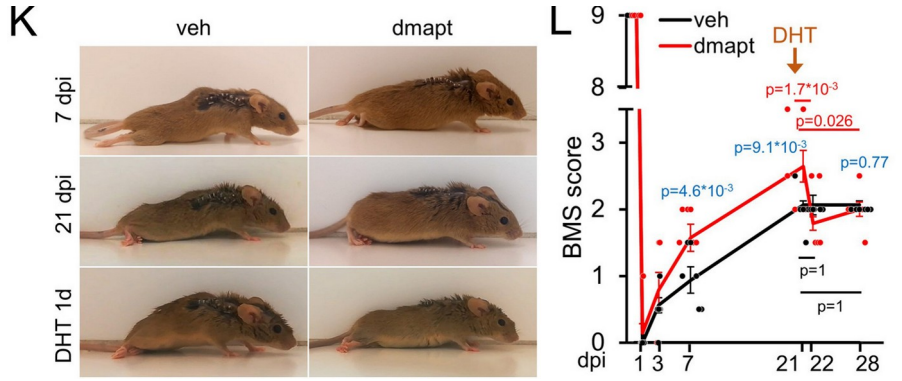
Correia et al. (2017), Fig. 2 (A–C). Locomotion during dorsal raphe stimulation.

Regenerated serotonin pathways aid recovered movement

Some serotonin neurons send **raphespinal axons** from the brainstem raphe nuclei into the spinal cord, where they influence motor circuits. A severe spinal crush interrupts these descending pathways. Leibinger and colleagues examined whether treatment with **DMAPT**, a compound that promotes axon growth, improved recovery of these fibers and hindlimb movement after injury in mice.

DMAPT-treated mice recovered more useful hindpaw placement than the control-treated animals. The researchers also found greater growth of serotonin-containing axons beyond the injury. Their recordings followed changes in the animals' ability to place the bottom of the paw against the surface, rather than simply measuring whether any ankle movement could be produced.

After recovery had developed, the researchers administered **5,7-dihydroxytryptamine**, a toxin that depleted serotonin neurons and their axons. The additional movement improvement produced by DMAPT disappeared. The recovered behavior therefore required the serotonergic pathway to remain present. Removing that pathway after treatment reversed the benefit, even though the earlier growth-promoting treatment had already been given.



Leibinger et al. (2023), Fig. 7 (K–L). Movement recovery after serotonin-pathway depletion.



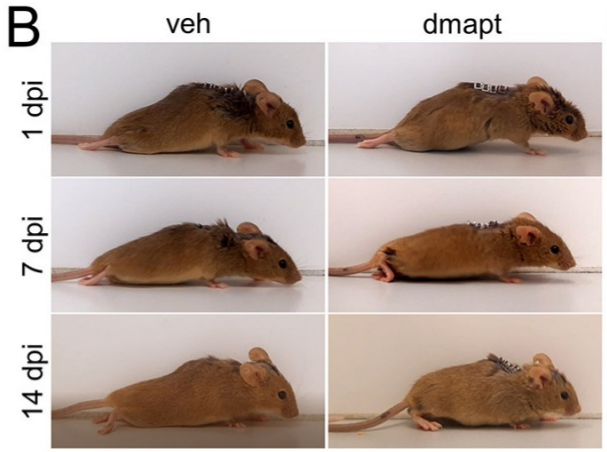
Leibinger et al. (2023), video4. Movement recovery before and after serotonin depletion.

A prodrug changes cellular machinery after injury

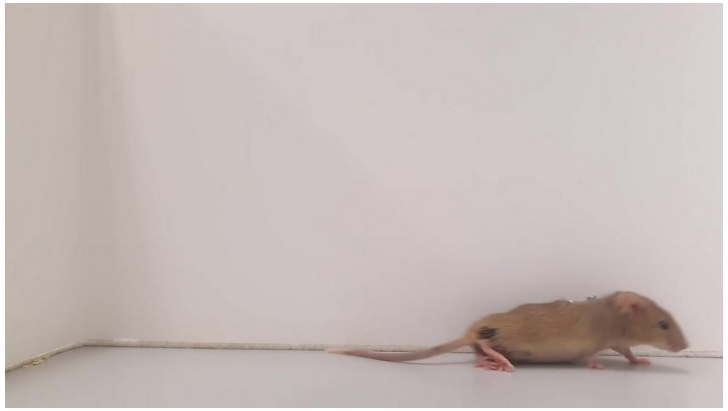
A **prodrug** is a compound that is converted into an active drug after entering the body. DMAPT is a more water-soluble derivative of **parthenolide**, the active compound used in this regeneration research. Improving solubility makes a substance easier to administer, while conversion supplies the compound that acts on the cellular process being targeted.

Parthenolide reduces **detyrosination**, removal of a tyrosine amino acid from **tubulin**, the protein that builds microtubules. These structures form internal supports and transport tracks in an axon. Changing their chemical modification can alter growth behavior. The treatment promoted axon extension by changing the cytoskeleton rather than directly activating serotonin receptors on spinal neurons.

After spinal injury, treated mice developed longer serotonin-containing axons beyond the damaged region and improved hindpaw placement. These were measured as separate outcomes: anatomical growth in the tissue and movement in the animal. A drug can therefore influence a transmitter pathway by changing the growth of its axons, increasing the tissue through which that pathway can later communicate.



Leibinger et al. (2023), Fig. 5 (B). Hindlimb posture after spinal injury.



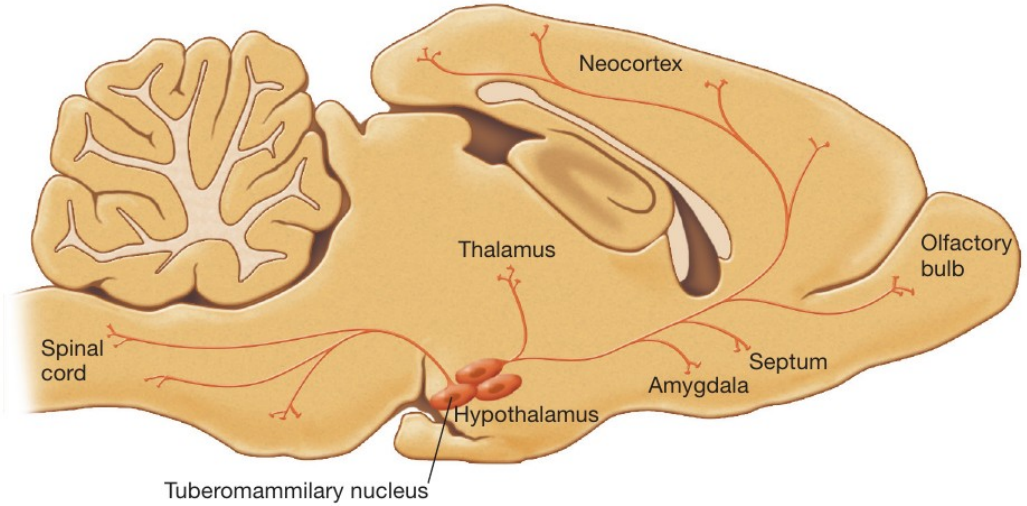
Leibinger et al. (2023), video2. Mouse movement at intervals after spinal injury.

Histamine receptor location changes behavioral effects

Histamine is produced by neurons in the **tuberomammillary nucleus**, a group of cells in the hypothalamus. Their axons reach many brain regions and help maintain wakefulness. Histamine also acts outside the brain, including in allergic responses. The same chemical messenger can therefore participate in different functions depending on where it is released and which receptors receive it.

Drugs that block **H1 receptors** can reduce histamine's effects. Older antihistamines often enter the brain and cause drowsiness because they interfere with central wake-promoting signaling. Other antihistamines enter the brain less readily and are less sedating. Their effects differ partly because of drug distribution, even when both are used to reduce an allergic response elsewhere in the body.

H3 receptors can act as autoreceptors that restrain histamine release. Activating this feedback reduces the amount of transmitter supplied to receiving cells, whereas blocking it can increase availability. Receptor location changes the result of an intervention: blocking a receiving receptor reduces the response there, while blocking an inhibitory autoreceptor can increase the transmitter released onto other receptors.



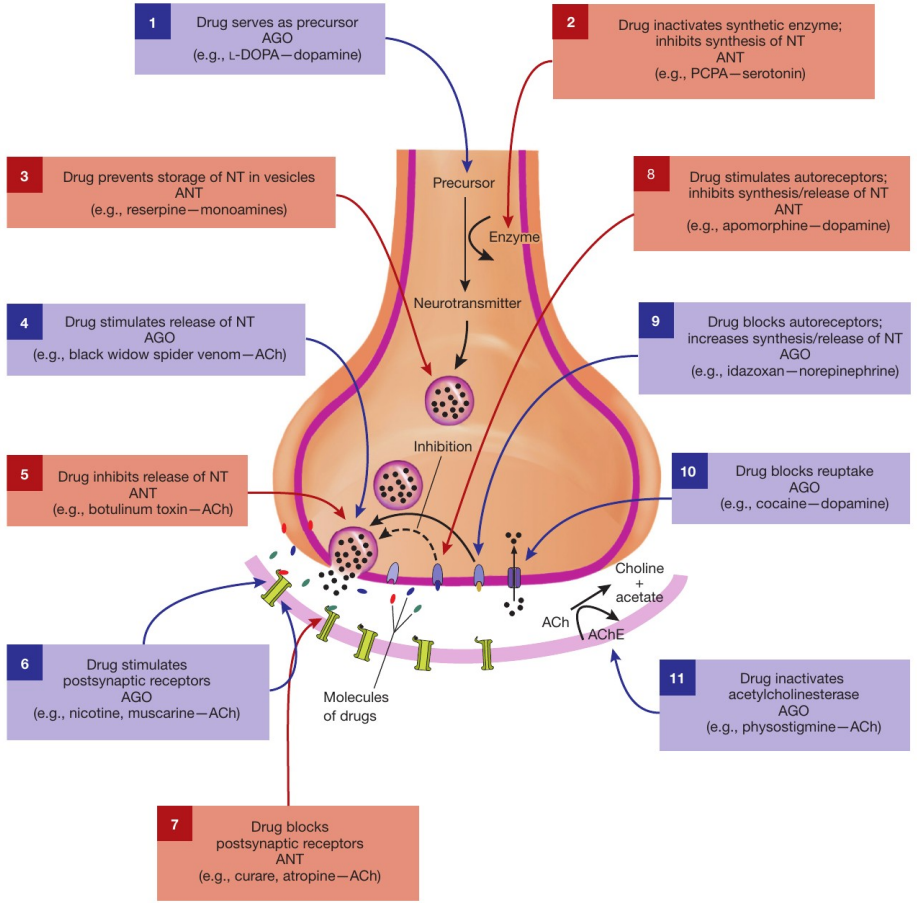
Carlson and Birkett (2022), Fig. 4.21, p. 114.

Peptide transmitters require different packaging

Peptide transmitters are short chains of amino acids that neurons use as chemical messengers. Unlike many small-molecule transmitters, they are made as parts of larger proteins in the cell body. Enzymes cut these larger proteins into active peptides, which are packaged into vesicles before being transported toward the locations where the neuron will release them.

The **endoplasmic reticulum** and **Golgi apparatus** help produce, process, and package these proteins. Peptide-containing vesicles then travel along the axon using the cell's transport machinery. A terminal cannot replenish its peptide supply simply by recycling the released molecules from the synaptic gap. New peptide cargo must be produced and delivered from the cell body.

After release, many peptides act on G protein-coupled receptors and produce changes that last longer than a brief ion-channel opening. Examples include endogenous opioids, which regulate pain-related signaling, and orexin, which helps maintain wakefulness. The response still depends on the receiving cell: a peptide changes the activity of a particular pathway through the receptors expressed there.



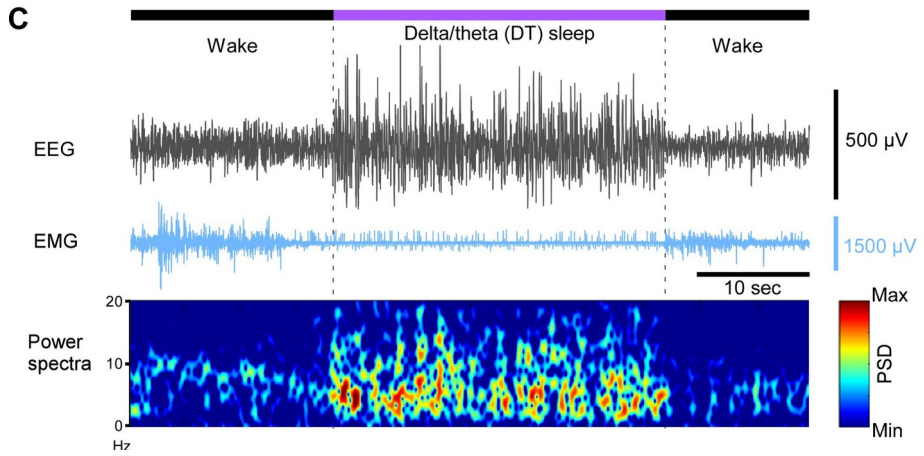
Carlson and Birkett (2022), Fig. 4.7, p. 100.

Loss of two peptide-neuron populations alters sleep states

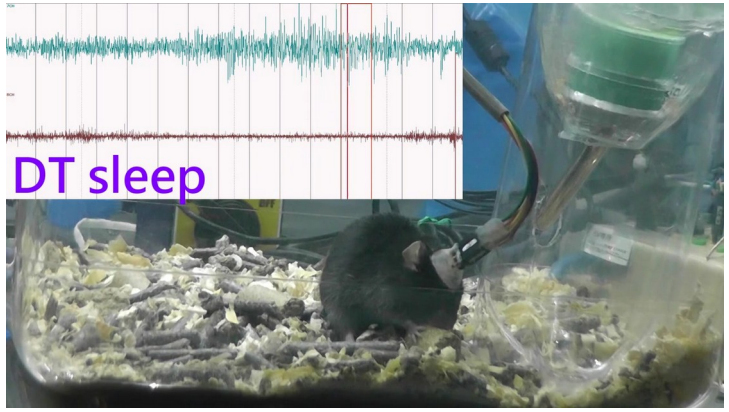
Orexin and **melanin-concentrating hormone**, or **MCH**, are peptide messengers produced by different hypothalamic neurons. Researchers removed both cell populations in adult mice and recorded sleep, muscle activity, and behavior. Removing the neurons eliminated their ongoing output to receiving circuits, allowing changes in sleep-state control to be followed as the cell populations were lost.

The mice developed sudden episodes in which movement stopped and muscle activity fell. Their brain recordings contained unusually strong **delta** and **theta** activity, two different ranges of brain-wave frequency. The researchers called this state **delta–theta sleep**. It began during an ongoing period of wakefulness and was followed by a return to wakefulness.

These episodes differed from ordinary non-REM sleep, REM sleep, and cataplexy in their electrical pattern. They also occurred outside the nest more often than ordinary sleep did. Loss of the two peptide-producing populations therefore changed how stable waking behavior was maintained, allowing a distinct sleep-like state to interrupt activity while the mice were in their home cages.



Hung et al. (2020), Fig. 3 (C). Delta–theta sleep with low muscle activity.



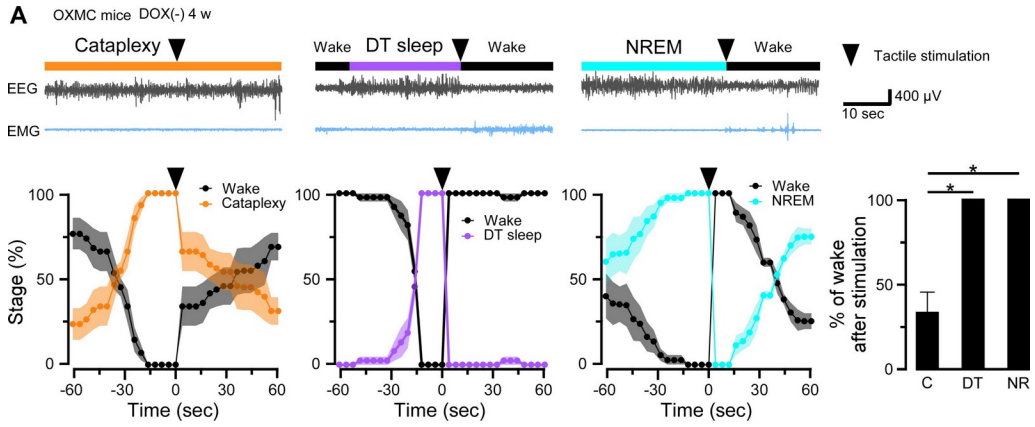
Hung et al. (2020), video1. Delta–theta sleep after two peptide-cell populations are removed.

Cataplexy differs from an ordinary sleep attack

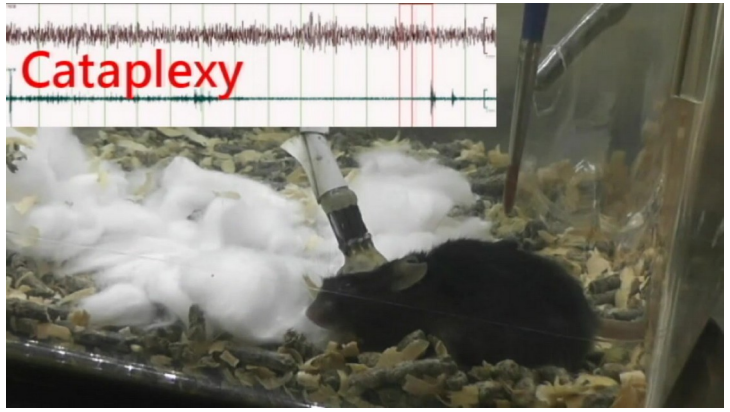
Cataplexy is a sudden loss of muscle tone during wakefulness. The dual-cell-ablation mice displayed cataplexy as well as delta–theta sleep episodes, although both could look like abrupt pauses in movement. Researchers recorded brain rhythms and muscle activity together because stillness alone could not distinguish these different physiological states in the behaving animal.

The researchers gently touched mice during each state to test their responses. During non-REM sleep and delta–theta sleep, the animals responded by waking and moving away from the stimulation. During cataplexy, they rarely made the same escape response. Similar-looking behavioral arrests therefore differed in how readily the animal regained active movement after a tactile stimulus.

The brain recordings also differed across these episodes. Cataplexy and delta–theta sleep were not interchangeable labels for an animal lying still. One state combined its own electrical pattern with weak responsiveness to touch, while the other permitted rapid escape. These measurements described how the nervous system controlled muscle tone and responsiveness during each kind of behavioral interruption.



Hung et al. (2020), Fig. 6 (A). Electrical and movement responses to tactile stimulation.



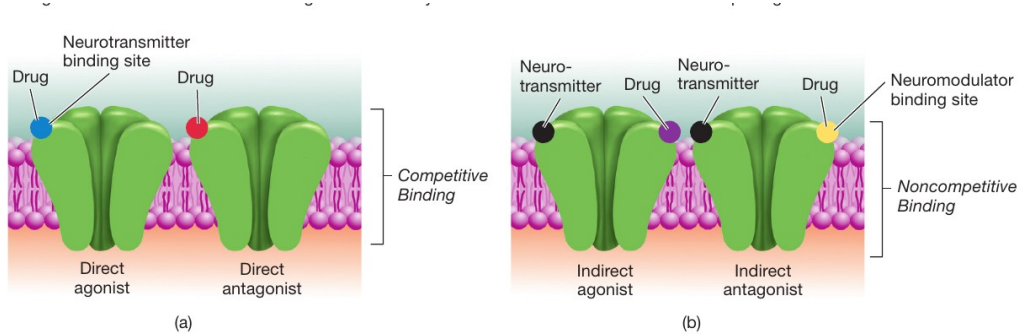
Hung et al. (2020), video2. Tactile stimulation during cataplexy.

Opioid receptors regulate pain and other functions

Opioid receptors respond to endogenous peptide messengers, including enkephalins and endorphins, as well as drugs such as morphine. The major receptor families are **mu**, **delta**, and **kappa**. They activate G proteins that can reduce calcium entry or increase potassium conductance, making neurons less excitable or reducing the amount of transmitter released from their terminals.

Opioids can reduce pain by suppressing transmission through pain-related pathways in the spinal cord and brain. Their effects are broader because opioid receptors also occur in circuits controlling breathing, reward, and digestion. A drug can therefore produce analgesia alongside respiratory depression or constipation, with each response arising from the cells and pathways affected at that location.

Naloxone blocks opioid receptors and can reverse opioid-induced respiratory depression by preventing continuing receptor activation. The antagonist competes with opioid agonists for receptor access. In a physically dependent person, this abrupt loss of opioid signaling can also precipitate withdrawal. Reversing the immediate drug action leaves the nervous system's adaptations to repeated exposure temporarily unopposed.



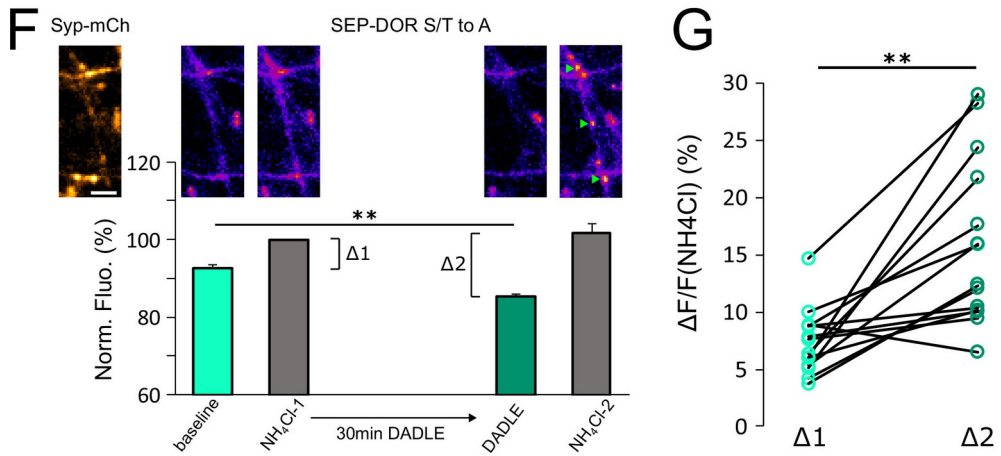
Carlson and Birkett (2022), Fig. 4.8, p. 102.

Receptor subtypes use different routes into tolerance

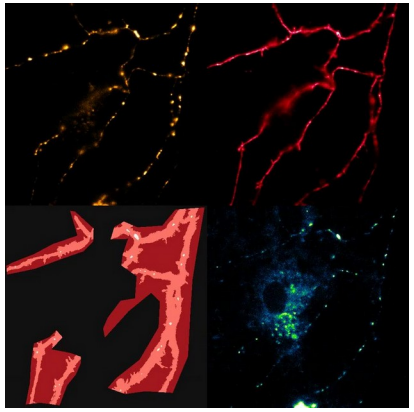
Mu and delta opioid receptors both became less effective at inhibiting vesicle release after prolonged agonist treatment in cultured neurons. Both also became less available at the axon surface. Their removal, however, depended on different intracellular requirements. Blocking the receptor-phosphorylating enzymes GRK2 and GRK3 prevented the usual mu-receptor changes but did not prevent the corresponding delta-receptor responses.

The researchers also changed amino acids in the delta receptor's intracellular tail so that the usual phosphate additions could not occur there. This altered receptor still entered endosomes after agonist exposure. It also developed reduced surface availability and tolerance. Delta-receptor uptake therefore continued even when the tail sites required for the usual mu-receptor process were unavailable.

Fluorescent measurements detected the altered delta receptors inside acidic cellular compartments after treatment. Fewer receptors remained exposed at the axon surface, leaving less receptor available to regulate release during the next agonist challenge. Similar reductions in presynaptic inhibition can therefore arise even when the two receptor subtypes differ in the biochemical steps that initiate their removal.



Jullié et al. (2022), Fig. 6 (F–G). Mutant delta receptor recycling and repeated responses.



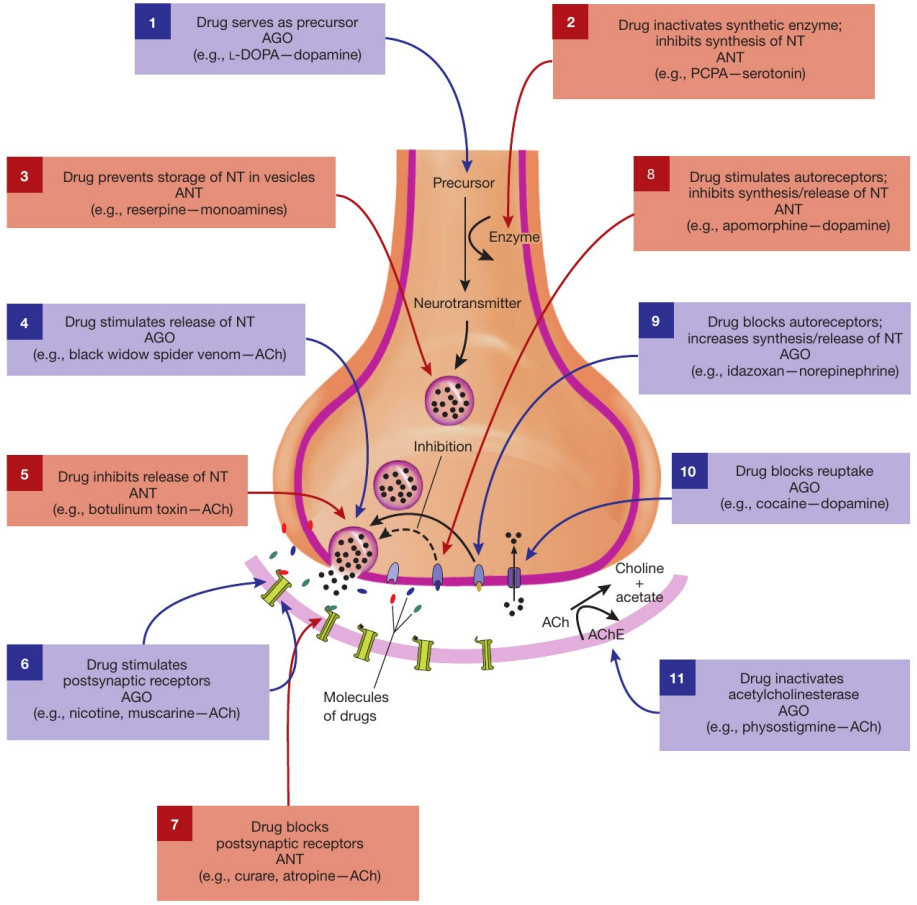
Jullié et al. (2022), fig6-video1. Mutant delta opioid receptor internalization in axons.

Endocannabinoids are synthesized on demand

Endocannabinoids are lipid-derived chemical messengers made within the body. Two major examples are **anandamide** and **2-arachidonoylglycerol**, or **2-AG**. Neurons commonly produce them when activity or receptor signaling stimulates the relevant enzymes. They are not generally stored in conventional transmitter vesicles, so release depends strongly on when and where the cell makes them.

These messengers activate **cannabinoid receptors**, including **CB1**, which is widely expressed in the nervous system. CB1 is a G protein-coupled receptor that can regulate transmitter release and other cellular processes. An external cannabinoid drug can activate part of this same signaling system, but its distribution and duration can differ from the brief local production of an endogenous messenger.

Enzymes help endocannabinoid signals end after production. **FAAH** contributes to anandamide breakdown, while **monoacylglycerol lipase**, or **MAGL**, is a major enzyme that breaks down 2-AG. Blocking degradation leaves an endogenous messenger available for longer. This increases signaling by preserving molecules the body has produced, rather than supplying an external agonist directly to the receptor.



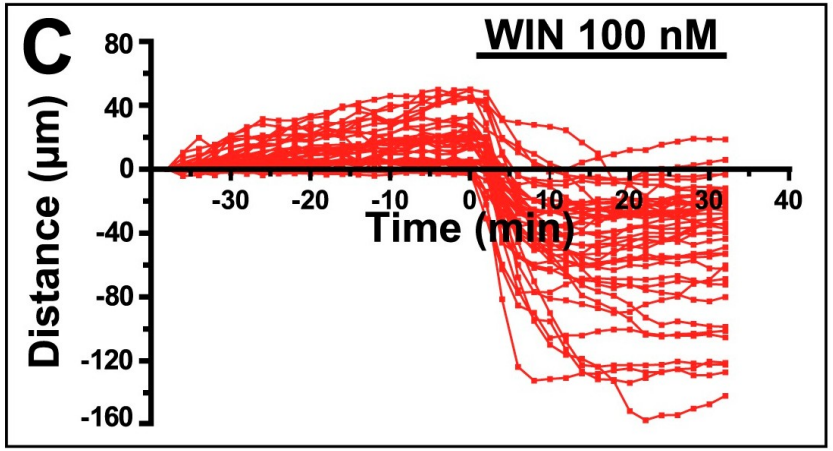
Carlson and Birkett (2022), Fig. 4.7, p. 100.

CB1 activation rapidly changes a neuronal growth cone

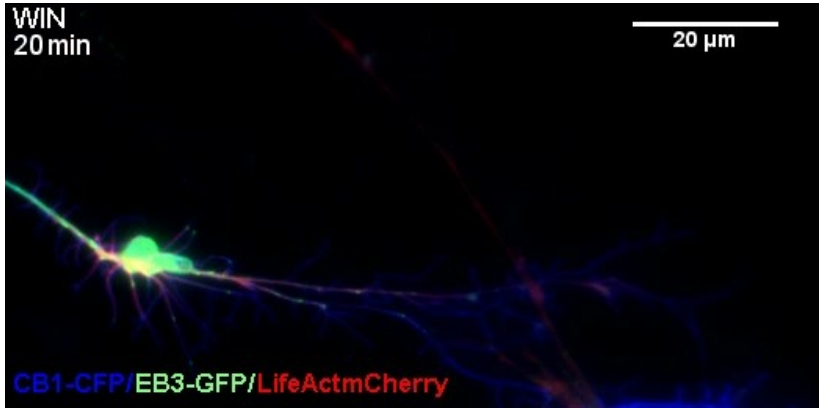
A **growth cone** is the moving structure at the tip of a growing axon. Its internal actin network helps the tip extend, change shape, and respond to local signals. Roland and colleagues followed growth cones in cultured hippocampal neurons while applying **WIN 55,212-2**, a synthetic agonist that activates cannabinoid receptors.

After WIN exposure, the growth cones retracted toward the cell body instead of continuing their earlier advance. Retraction began within a few minutes and typically approached a plateau over the following minutes. The fluorescent labels allowed movement of the actin-rich tip to be followed in the same neuron before and after the chemical was added.

A CB1-blocking drug prevented the retraction, while removing WIN allowed growth-cone movement to recover in the tested condition. The change depended on continuing receptor-sensitive signaling and could be reversible. CB1 activation was therefore altering the forces that controlled the growing tip, producing a physical movement of neuronal structure rather than only changing transmitter release at a synapse.



Roland et al. (2014), Fig. 1 (C). Growth-cone movement during CB1 activation.



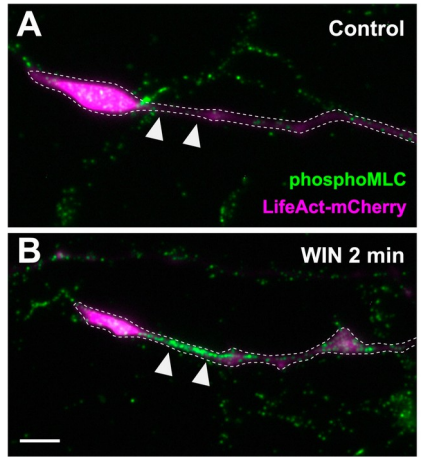
Roland et al. (2014), Video 1. Growth-cone retraction during CB1 activation.

Myosin contributes to cannabinoid-induced retraction

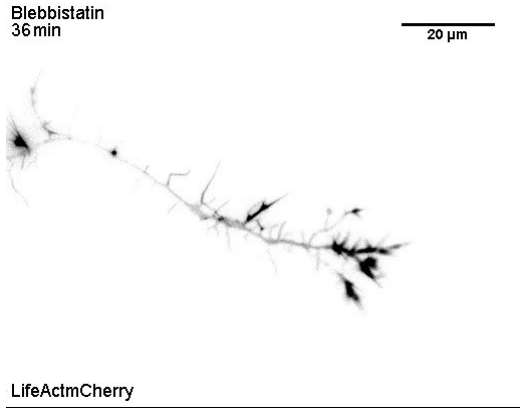
Actin filaments help form the growth cone's internal framework, while **non-muscle myosin II** generates force against those filaments. The motor uses energy from ATP to pull within the actin network. When this contraction becomes strong enough, the growing tip can retract toward the axon instead of continuing to extend away from the cell body.

In the cannabinoid study, CB1 activation increased phosphorylation of a protein that regulates myosin activity. **ROCK**, an enzyme that promotes actin–myosin contraction, was part of the signaling pathway required for retraction. The receptor's chemical response was therefore converted into greater activity of machinery that could generate physical force inside the neuronal process.

The researchers applied **blebbistatin**, which inhibits myosin II, before activating CB1 receptors. Growth-cone retraction was strongly reduced. The agonist was still available, but the motor could no longer produce its usual response. Blocking ROCK also reduced retraction, interrupting an earlier signaling step that would normally promote the contraction of the actin–myosin network.



Roland et al. (2014), Fig. 3 (A–B). Actin at a growth cone before and after CB1 activation.



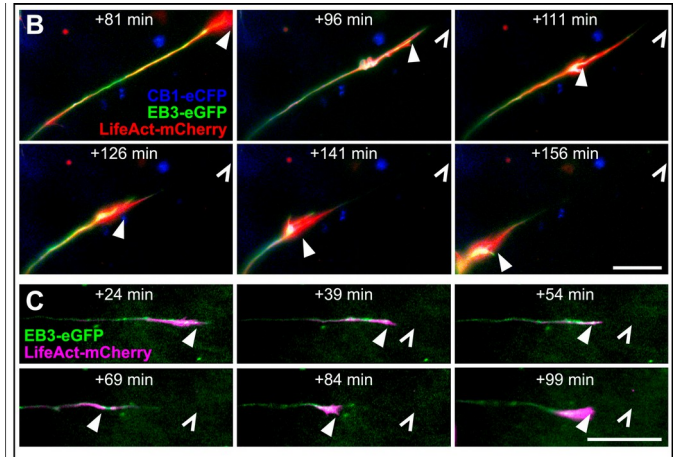
Roland et al. (2014), Video 4. Myosin inhibition before CB1 activation.

Cannabinoid activation also changes growth cones in slices

An **organotypic slice** is a tissue preparation kept alive outside the animal while preserving part of its original cellular arrangement. The cannabinoid researchers examined axons growing from cortical tissue in these slices. This allowed growth-cone movement to be followed while the axons remained among neighboring cells and tissue structures rather than growing as isolated neurons on a dish.

Applying WIN also caused actin-rich growth cones in the slices to retract. The response was observed in a preparation containing more of the developing brain's local structure. Fluorescent labeling distinguished the growing processes, allowing their positions to be followed through time as the receptor agonist reached the tissue and the tips changed their direction of movement.

Growth cones with additional CB1 expression showed a stronger response, while axons relying on their naturally expressed receptors also responded. Retraction developed more slowly in slices than in isolated cultures. The authors attributed part of that delay to the time required for the highly lipid-soluble compound to diffuse through tissue before reaching its receptor targets.



Roland et al. (2014), Fig. 4 (B–C). Growth-cone movement in brain slices.



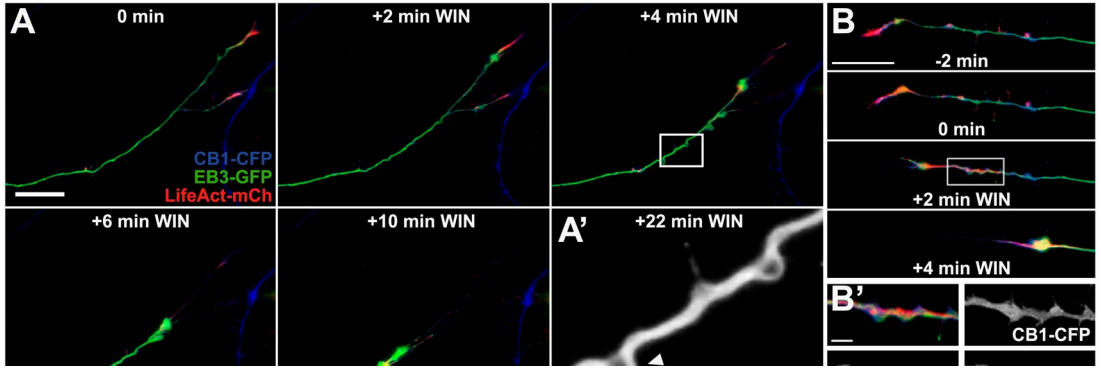
Roland et al. (2014), Video 5. Growth-cone retraction in brain slices.

Cytoskeletal inhibitors separate contraction from transport

Actin and microtubules contribute differently to a growing axon's shape and movement. Actin forms a flexible network near the growth cone, while microtubules provide internal support and transport tracks. During cannabinoid-induced retraction, actin became organized into straighter bundles and nearby microtubules bent. These changes were consistent with force acting on the internal framework rather than simple withdrawal of the membrane alone.

The researchers disrupted microtubules with **nocodazole**. This stopped normal growth-cone advance, but WIN could still cause the tip to retract. Microtubules were therefore needed for normal forward growth without being essential for the drug-induced pullback. The same structure can contribute to extension while a different part of the cytoskeleton supplies the force for contraction.

Disrupting actin with **cytochalasin D** reduced both forward growth and cannabinoid-induced retraction. Inhibiting myosin also reduced retraction. Together, these interventions identified an actin–myosin force-generating process: the receptor activated intracellular signaling, the motor pulled against actin, and the neuronal process shortened. Removing the actin network or disabling the motor interrupted that physical response.



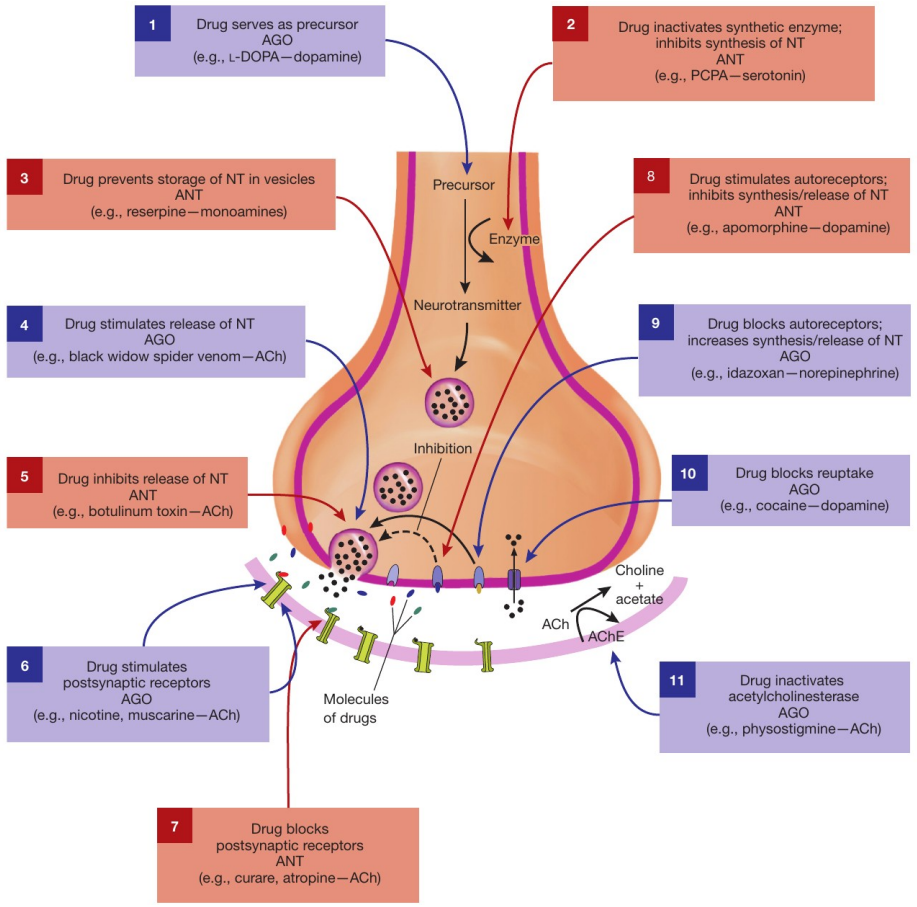
Roland et al. (2014), Fig. 2 (A–B). Cytoskeletal changes during CB1 activation.

Lipid messengers can signal back toward a nerve terminal

At some synapses, the receiving neuron can send a chemical signal back to the releasing terminal. Activity in the postsynaptic cell stimulates endocannabinoid production, and the lipid messenger reaches presynaptic CB1 receptors. This is called **retrograde signaling** because the chemical influence travels in the opposite direction from the usual transmitter message across that synapse.

Activating presynaptic CB1 receptors can reduce calcium entry and weaken the processes that release transmitter. The next action potential then produces less chemical output from the affected terminal. The receiving neuron can consequently reduce some of the input reaching it by making a messenger that acts back on the axon supplying that input.

The effect on the wider circuit depends on the terminal being suppressed. Reducing glutamate release can decrease excitatory input, while reducing GABA release can remove inhibition and make a receiving neuron more responsive. In both cases, the direct action is reduced transmitter release. The different circuit effects arise because the suppressed terminals normally perform different functions.



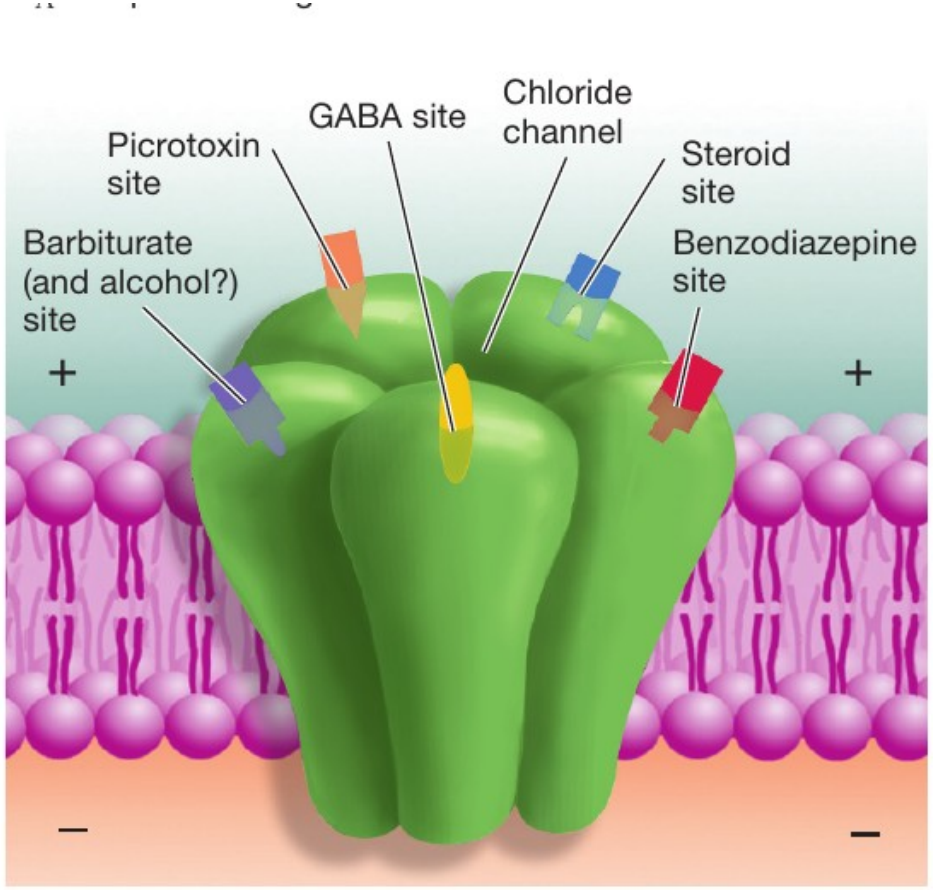
Carlson and Birkett (2022), Fig. 4.7, p. 100.

Chloride gradients determine an anion channel's effect

Opening a chloride channel does not produce the same electrical effect in every neuron. The response depends on the **chloride gradient**, the difference in concentration across the membrane. Transport proteins establish that gradient by moving ions into or out of the cell. Opening the channel then allows current that moves voltage toward the level favored by this distribution.

When chloride-channel opening makes the inside more negative, the response is called **hyperpolarization**. This often reduces firing during the immediate input. The same voltage change can also affect other channels. Some sodium channels become unavailable during sustained depolarization and recover from that state when the membrane becomes more negative, increasing the number available for a later response.

A neuron can therefore become more negative at rest while responding more strongly to a subsequent stimulus. The immediate voltage shift and later action-potential output are different measurements. In retinal ganglion cells, cannabinoid-sensitive changes in chloride handling produced this combination, making the effect of the channel depend on both the ion gradient and the neuron's recent membrane voltage.



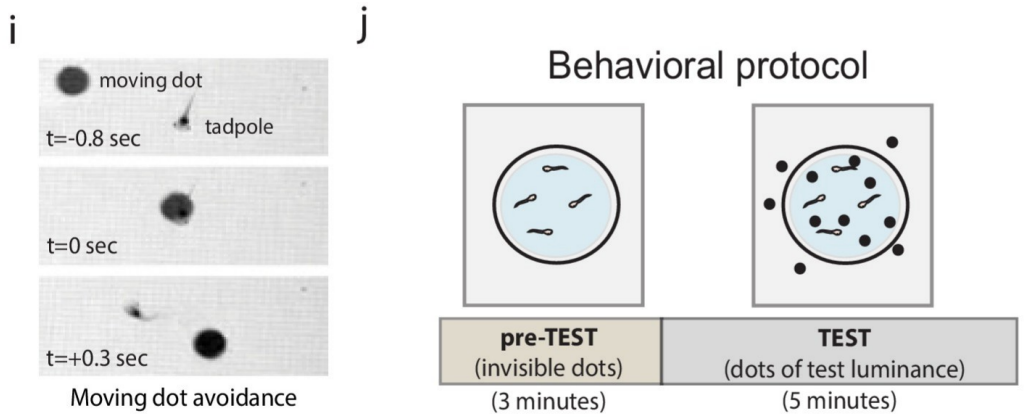
Carlson and Birkett (2022), Fig. 4.10, p. 105.

A cannabinoid study measures visual behavior in tadpoles

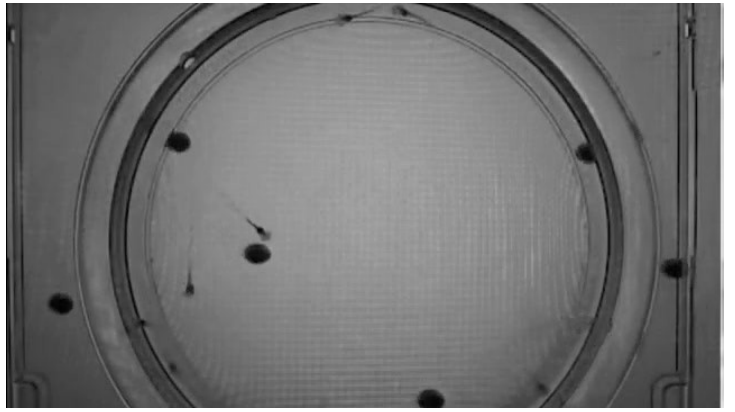
Retinal ganglion cells carry the retina's output toward the brain. Miraucourt and colleagues examined how cannabinoid signaling changed these neurons and visual behavior in **Xenopus** tadpoles. The animals swam freely above a screen displaying moving dark dots. Researchers tracked their movements to measure whether they avoided a dot as it approached them.

Under brighter **photopic** conditions, control tadpoles responded to visible dark dots with rapid avoidance movements. The recording follows the animal and the moving target in the same view. Measuring the encounter and the change in swimming made it possible to distinguish an avoidance response from movement that occurred while no target was approaching.

The researchers varied dot visibility by changing illumination and contrast, the difference in brightness between target and background. They also measured responses to a uniform background before presenting visible targets. These conditions established how often a rapid movement occurred without a detectable dot, providing a comparison for the later avoidance behavior under different visual conditions.



Miraucourt et al. (2016), Fig. 8 (i–j). Moving-dot avoidance and behavioral protocol.



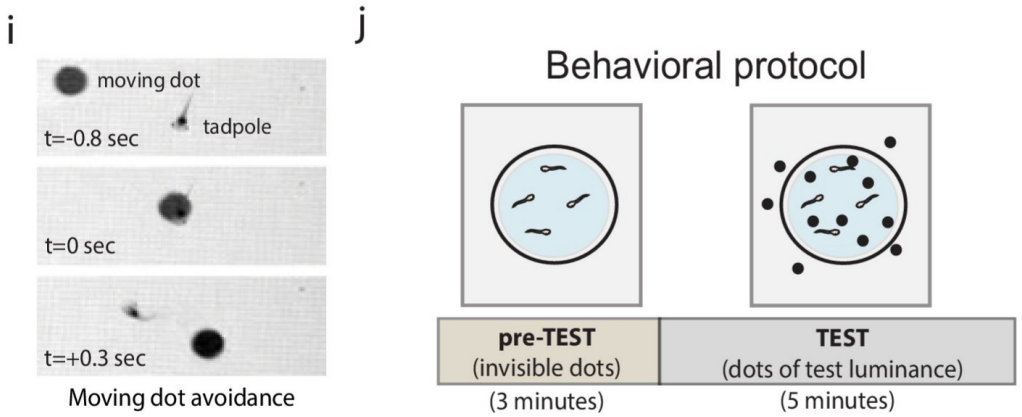
Miraucourt et al. (2016), media1. Control tadpole avoiding moving dots in brighter light.

CB1 activation improves vision under dim illumination

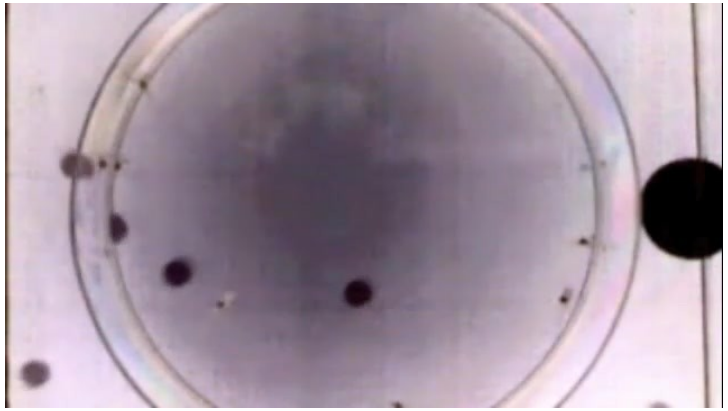
Under **mesopic** conditions, an intermediate range of dim illumination, untreated tadpoles often failed to avoid the moving dots reliably. Tadpoles treated with WIN responded more successfully under the same low-light conditions. The drug therefore changed performance when the visual target was difficult to detect, rather than requiring a different kind of stimulus or movement response.

The researchers also recorded stronger light-evoked firing in retinal ganglion cells after cannabinoid receptor activation. These neurons supply visual information to the brain, so changing their response can alter the sensory signal available before an avoidance movement is selected. The behavioral improvement occurred alongside a measured change in the output neurons of the visual system.

Ordinary swimming speed between dot encounters was similar in control and treated animals. Treated tadpoles, however, produced faster movements during successful escapes. Increasing endogenous cannabinoid availability also improved dim-light avoidance, and blocking CB1 prevented that benefit. Both receptor activation and preservation of the body's own cannabinoid messenger could therefore change performance in this particular visual task.



Miraucourt et al. (2016), Fig. 8 (i–j). Moving-dot avoidance and behavioral protocol.



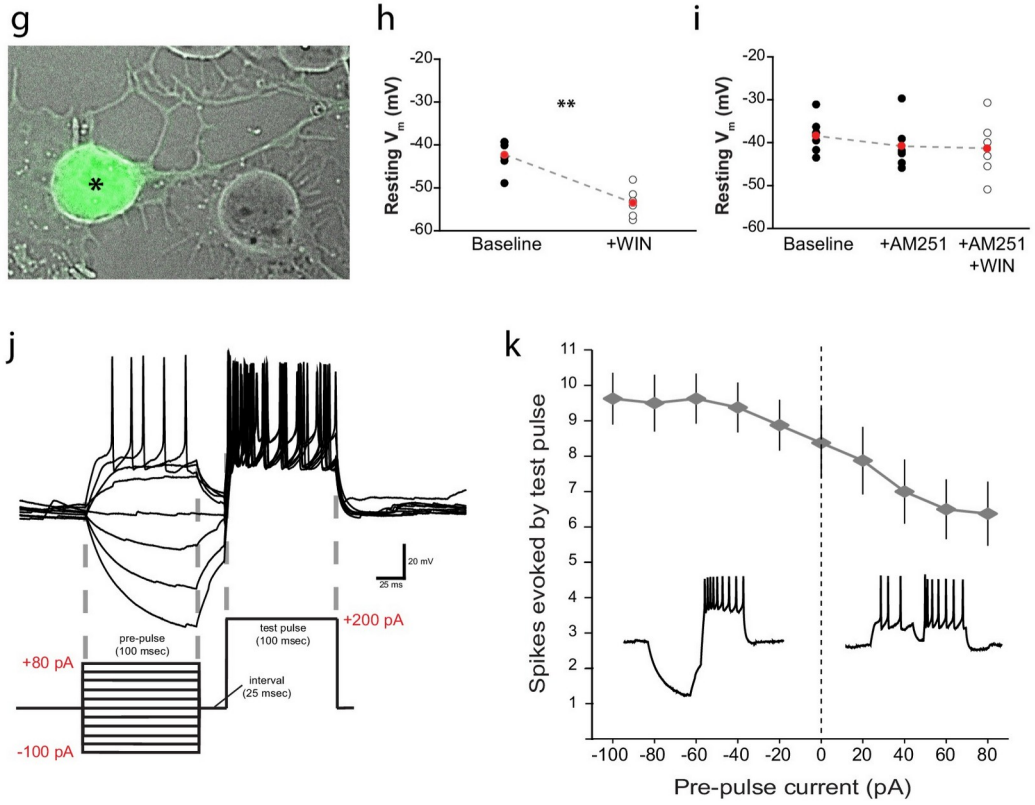
Miraucourt et al. (2016), media3. WIN-treated tadpole avoiding dots in dimmer light.

Cannabinoids change chloride-dependent excitability

The retinal study identified a change in chloride handling inside ganglion cells. CB1 activation reduced intracellular chloride through a pathway involving **NKCC1**, a transporter that normally brings chloride into the cell. Less chloride accumulated inside, changing the electrical effect of chloride-permeable channels. Fluorescent chloride measurements allowed this internal ion change to be followed during drug exposure.

The shift made the resting membrane potential more negative through chloride currents controlled by **glycine**, another neurotransmitter. This hyperpolarization increased the availability of sodium channels for later activation. When the cell subsequently received a depolarizing stimulus, it could produce more action potentials. A more negative resting voltage therefore occurred together with a stronger response to a later input in these retinal neurons.

The researchers observed hyperpolarization in isolated ganglion cells as well as altered responses in the retina. Blocking CB1 prevented the voltage change, and blocking glycine receptors prevented the usual increase in visual responsiveness. The chemical effect involved the receiving cell's ion transport and channels, which changed how visual input was converted into firing before the signal traveled onward to the brain.



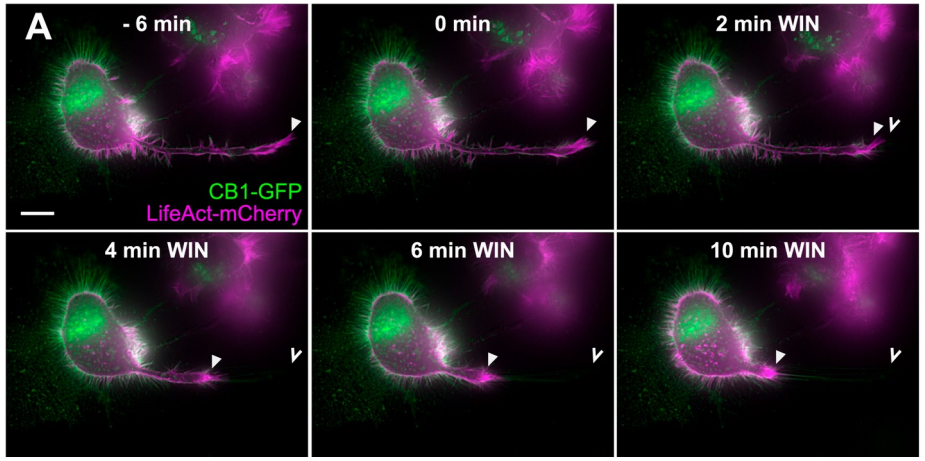
Miraucourt et al. (2016), Fig. 6 (g–k). Chloride measurements and retinal excitability.

Receptor activation changes the mechanical state of a cell

Cannabinoid signaling can also change the shape of the cell body. Roland and colleagues examined **Neuro2A cells**, a cultured neural cell line that grows identifiable processes. After CB1 activation, these processes retracted and the cell body became rounder. Actin accumulated beneath the membrane and near the ends of the retracting processes as the cell changed shape.

The researchers measured **stiffness**, the cell's resistance to being deformed. A small probe pressed against the cell surface, allowing that resistance to be compared before and after treatment. CB1 activation produced a temporary increase in stiffness alongside the structural changes. The response therefore included a measurable change in the cell's physical properties, not only a change in its outline.

Inhibiting myosin with blebbistatin prevented the usual cell-rounding response. Myosin normally generates force against the actin network beneath the membrane, so reducing its activity weakens the contraction. The agonist could still reach the receptor, but the physical response was interrupted downstream. Receptor signaling had changed cell shape by engaging an internal network capable of producing mechanical force.



Roland et al. (2014), Fig. 6 (A). Cell rounding during CB1 activation.



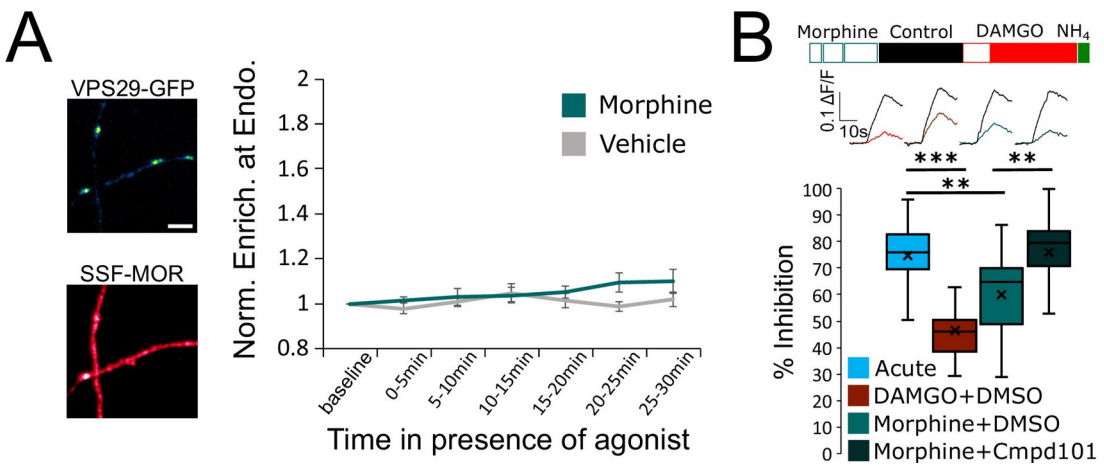
Roland et al. (2014), Video 6. Cell rounding during CB1 activation.

Agonists differ in receptor uptake and cellular tolerance

Agonists that activate the same opioid receptor can differ in how strongly they promote its removal from the surface. DAMGO produced substantial mu-receptor uptake in cultured axons. Morphine produced little detectable rapid uptake, yet prolonged morphine exposure still reduced receptor availability and weakened presynaptic inhibition. Its slower effects differed from those measured immediately after the drug was applied.

Two other agonists, **PZM21** and **TRV130**, produced little detectable receptor uptake and did not produce the same tolerance after the tested prolonged treatment. These compounds favored some receptor-signaling responses more than others. An agonist's ability to activate a receptor initially therefore need not match its ability to promote the cellular processes that later reduce responsiveness.

Blocking GRK2 and GRK3 also prevented the tolerance produced by prolonged morphine treatment in this preparation. These enzymes add phosphate groups that regulate the receptor's movement and availability. The later loss of opioid responsiveness depended partly on that regulatory process, even though morphine had produced little measurable rapid uptake during the shorter initial observation period.



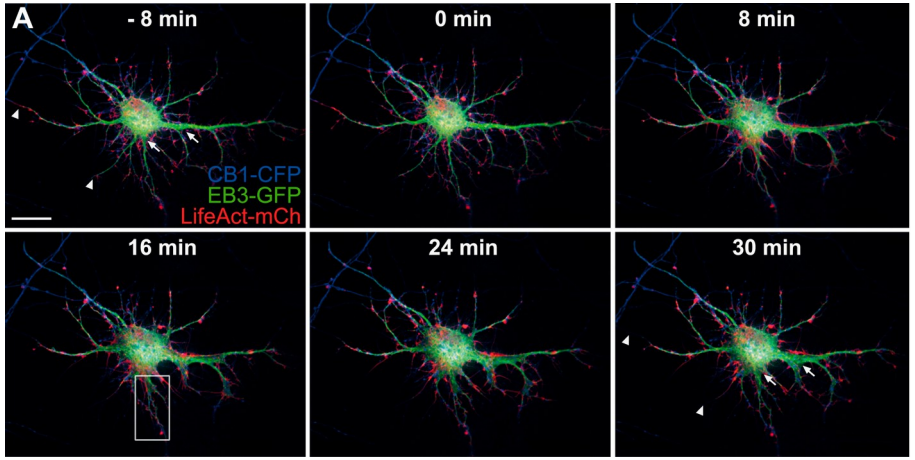
Jullié et al. (2022), Fig. 3 (A–B). Morphine-induced trafficking and presynaptic tolerance.

Cannabinoids reshape several neuronal compartments

Cannabinoids can change neuronal shape by activating **CB1 receptors**. Researchers examined cultured hippocampal neurons and followed changes in the cell body and **dendrites**, the branches that receive signals from other neurons. These structures form the **somatodendritic region**. After receptor activation, outer dendritic branches retracted while portions nearer the cell body broadened, changing the shape of this region within minutes.

These changes involve **actin**, a protein that forms part of the **cytoskeleton**, the internal framework that supports the cell and helps it change shape. CB1 activation reorganized actin into straighter bundles as nearby microtubules bent. Following the same neuron over several minutes revealed how this internal reorganization accompanied physical changes in its dendrites and the region around its cell body.

Brief activation changed shape within minutes, while overnight treatment reduced the number of developing dendrites. Researchers examined these outcomes separately because immediate contraction and longer-term growth are different processes. Blocking ROCK or myosin prevented the overnight reduction. The chemical changed neuronal structure through a pathway that regulated the cell's internal framework and the development of its branches over different timescales.



Roland et al. (2014), Fig. 7 (A). Somatic and dendritic changes during CB1 activation.



Roland et al. (2014), Video 8. Somatic and dendritic changes during CB1 activation.

Key takeaways

Exposure determines access. Absorption, distribution, metabolism, and excretion govern the amount of drug reaching a target. Membrane permeability and transporters also determine access to receptors inside cells.

Binding and response are separate properties. Affinity describes binding, potency describes the amount needed for an effect, and efficacy describes the response a drug can produce. Agonists activate receptors; antagonists prevent activation.

Synaptic drugs act at several steps. Synthesis, vesicle storage, release, receptor activation, reuptake, and enzymatic breakdown each provide targets for changing neurotransmission.

Repeated exposure changes drug responses. Tolerance reduces a response to the same exposure, whereas sensitization increases it. Receptor trafficking and cellular adaptations can change signaling during repeated treatment.

Transmitter identity does not specify one behavior. Glutamate, GABA, acetylcholine, monoamines, peptides, and lipid messengers act through distinct receptors and pathways. Receptor location and circuit target determine the resulting cellular and behavioral effects.

Experiments connect targets to behavior. Identify the manipulated molecule, the affected cell, and the measured response. Receptor imaging, transmitter sensors, pathway targeting, and behavioral recordings reveal different steps in a drug's action.

Carlson & Birkett (2022); cited primary studies